

WHO guideline on preventing early pregnancy and poor reproductive outcomes among adolescents in low- and middle-income countries

Web Annex. Evidence base



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Suggested citation. Web Annex. Evidence base. In: WHO guideline on preventing early pregnancy and poor reproductive outcomes among adolescents in low- and middle-income countries. Geneva: World Health Organization; 2025. <https://doi.org/10.2471/B09322>. Licence: CC BY-NC-SA 3.0 IGO.

Cataloguing-in-Publication (CIP) data. CIP data are available at <https://iris.who.int/>.

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This document is the web annex for: *WHO guideline on preventing early pregnancy and poor reproductive outcomes among adolescents in low- and middle-income countries*

Abbreviations

2SLS	two-stage least squares	IUD	intrauterine device
AHR	adjusted hazard ratio	LARC	long-acting reversible contraceptive
aOR	adjusted odds ratio	LMICs	low- and middle-income countries
ARR	adjusted risk ratio	MTBA	More Than Brides Alliance
ATT	average treatment effect on the treated	OLS	ordinary least squares
BPO	Business Process Outsourcing	OR	odds ratio
BRAC	Bangladesh Rural Advancement Committee	PICO	population, intervention, comparator, outcomes
CCT	conditional cash transfers	PRISMA	preferred reporting items for systematic reviews and meta-analyses
CEDAW	Convention on the Elimination of All Forms of Discrimination Against Women	RCT	randomized controlled trial
CI	confidence interval	RD	risk difference
CINAHL	Cumulative Index to Nursing and Allied Health Literature	RDD	regression discontinuity design
CRC	Convention on the Rights of the Child	RMA	Reaching Married Adolescents
CRCT	cluster randomized controlled trial	SD	standard deviation
DID	difference in differences	SE	standard error
DISHA	The Development Initiative Supporting Healthy Adolescents	SRH	sexual and reproductive health
DMPA-SC	subcutaneous depot medroxyprogesterone acetate	SRHR	sexual and reproductive health and rights
ELA	Employment and Livelihood for Adolescents	TPP	Tipping Point Programme (core) – components to enhance adolescent girls’ personal assets, intrinsic agency and instrumental agency
FSSSP	Female Secondary School Stipend Program	TPP+	Tipping Point Plus Programme (enhanced) – TPP components plus activities to enhance social-norms change by engaging community leaders and by facilitating girl-led community activities and social-norms-change events
GBV	gender-based violence		
GDG	Guideline Development Group	UCT	unconditional cash transfers
GE	Girl Empower – a programme delivering a life skills curriculum and individual savings start up to girls aged 13–14 in Liberia	VAWG	violence against women and girls
GE+	GE programme “plus” a cash incentive payment to caregivers for girls’ participation in weekly GE sessions	VE	violence prevention and education
GRADE	Grading of Recommendations Assessment, Development and Evaluation	VEH	violence prevention, education and health
ITT	intention to treat	VEHW	violence prevention, education, health and wealth creation

1. GRADE tables for PICO questions related to child marriage and related outcomes

PICO 1.1 Interventions to empower girls by building their protective assets and/or promoting positive gender socialization

GRADE Table WA.1.1(i) PICO question 1.1/outcome 1: Do interventions to empower girls by building their protective assets and/or promoting positive gender socialization (outcome 1)¹ **reduce child marriage?**

No. of studies/N	Study design	Certainty assessment					Intervention: girls' empowerment	Results	Certainty	Importance
		Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations				
11 studies (1-11) N≈44 572 girls and young women	RCTs	Very serious ^{a,b,c,d}	Not serious	Serious ^e	Not serious	Serious ^{f,g}	Building girls' protective assets (life skills, SRHR education, livelihoods skills/ financial literacy, and/or education support)	3 studies showed significant impact on reducing child marriage^{1,2,3} 1 multi-country study showed significant impact on reducing child marriage in some settings but not others⁴ 7 studies did not show significant impact on reducing child marriage⁵⁻¹¹	⊕○○○ VERY LOW	CRITICAL
3 studies (12-14) N≈6199 girls and young women	Quasi-experimental	Very serious ^{a,c,h,i,j}	Not serious	Serious ^k	Not serious	Not serious	Same as above	1 study found reduction in child marriage among younger adolescents, but increased prevalence of marriage among older adolescents¹² 2 studies found an impact on reducing child marriage (under 18)^{13,14}	⊕○○○ VERY LOW	CRITICAL

¹ On all of these GRADE tables, this labelling refers to the outcome as listed in Annex 4 in the main guideline.

No. of studies/N	Study design	Certainty assessment					Intervention: girls' empowerment	Results	Certainty	Importance
		Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations				
3 studies (1,10,11) N≈11 889 girls and young women	RCTs	Not serious	Not serious	Serious ^e	Serious ⁱ	Serious ^g	Promoting positive gender socialization	1 study found impact on reducing child marriage¹ 2 studies found no impact on reducing child marriage^{10,11}	⊕○○○ VERY LOW	CRITICAL
1 study (14) N≈949 girls and young women	Quasi-experimental	Serious ^{a,c,h}	Cannot assess	Not serious	Not serious	Not serious	Same as above	Impact on reducing child marriage¹⁴	⊕○○○ VERY LOW	CRITICAL

RCT: randomized controlled trial; SRHR: sexual and reproductive health and rights.

Explanations – certainty assessment

- ^a Risk of bias: Migration may have caused significant differences in populations captured in earlier and later rounds of data collection (2,4 [Mali],5,14).
- ^b Risk of bias: Appropriateness of the sample (2).
- ^c Risk of bias: Attrition bias related to selective dropout of participants who systematically differed from those who remained in the study (2,14).
- ^d Risk of bias: Lack of adherence to design (4 [Malawi]).
- ^e Indirectness: Exposure to all components of a multi-component intervention makes it difficult to ascertain which programme strategies or activities were most influential for achieving programme impact (2,4,5,11).
- ^f Other considerations: Study not able to address potential longer-term impacts/time to follow-up (8).
- ^g Other considerations: Potentially limited generalizability due to study being carried out in specific setting or with specific population (7,8,10).
- ^h Risk of bias: Absence of an adequate comparison group (14).
- ⁱ Risk of bias: Quality of counterfactual (12).
- ^j Risk of bias: Selection bias related to systematic differences between treatment and control groups (12).
- ^k Indirectness: Difficulty attributing impact to programme due to other changes/programmes operating in the area during the study period (12–14).
- ^l Imprecision: Uncertainty because confidence interval (CI) is large and low baseline prevalence rates may have resulted in incorrect power calculations (11).

Explanations (numbering matches superscripts in table AND references in list) – results

- Significant reduction in child marriage
 - The average programme effect in the community was significant with adjusted hazard ratio (AHR) 0.75 (95% CI: 0.60–0.92; $P = 0.007$) for education, 0.72 (95% CI: 0.59–0.88; $P = 0.001$) for gender, and 0.70 (95% CI: 0.56–0.87; $P = 0.001$) for livelihoods relative to those living in control arm villages.
 - BALIKA programme impact on reducing the risk of child marriage was of similar magnitude in all three intervention arms (25–30%) relative to control villages in an RCT, assessed at the end of a relatively short period of only 18 months.
- Significant reduction in child marriage
 - The midline intention-to-treat (ITT) estimate showed girls in treated communities to be 69 percentage points less likely to be married/cohabiting at follow-up, corresponding to 53% of the baseline mean. In control communities, marriage rates for adolescent girls rose naturally from 12% to 18% from baseline to follow-up. This divergence in trends continued to endline, at which point they were 8 percentage points less likely to be married or cohabiting.
 - Baseline prevalence: 0.129 (standard deviation [SD] = 0.335); change at endline: -0.08 (SD = 1.015).
- Significant impact on reducing child marriage
 - Beneficiaries of both programmes – Girl Empower (GE) and Girl Empower Plus (GE+) were 3.6 percentage points (or 46%) less likely to be married compared with the control group.
 - GE+ reduced the likelihood of marriage by more than 50%.

4. Programme-attributable reduction in child marriage in India, but not in Malawi, Mali or Niger
 - Intervention was successful at reducing the proportion of girls aged 12–19 years married in India (difference-in-differences [DID] coefficient -0.126, $P < 0.001$). Findings in the other countries did not show impact of the intervention on delaying marriage. In intervention areas in India, the observed declines in child marriage from baseline to midline were maintained from midline to endline, while in comparison areas child marriage prevalence declined from baseline to midline and slowed from midline to endline. Although data from the other three More Than Brides Alliance (MTBA) countries showed encouraging declines in child marriage from baseline to endline, these trends were not attributed to the MTBA programme.
5. No significant impact on reducing child marriage
 - The intervention was not found to have had an impact on reducing child marriage (DID coefficient 0.083, 95% CI: -0.20–0.37).
6. No significant impact on reducing child marriage
 - The empowerment arm was not found to have significantly reduced child marriage. For girls married by age 15 at endline, ordinary least squares (OLS) regression coefficient: 0.002 (Huber-White robust standard error [SE] = 0.012). For girls married between age 15 and 18, OLS regression coefficient: 0.006 (Huber-White robust SE = 0.008).
7. No significant impact on reducing child marriage
 - The clubs – both with the microfinance component (ITT estimate: 0.009 [SE = 0.012]) and without it (ITT estimate: 0.004 [SE = 0.009]) – were not found to have had a discernible impact on child marriage.
8. No significant impact on reducing child marriage
 - Results showed no change in the probability the child was married by endline. Beta coefficient: 0.042 (SE = 0.029).
9. No significant impact on reducing child marriage
 - No intervention impact was detected on child marriage in any of the arms (Tipping Point Programme [TPP] vs control: AHR 1.14, 95% CI: 0.79–1.63, $P = 0.47$; TPP “Plus” [TPP+] vs control: AHR 1.24, 95% CI: 0.89–1.71, $P = 0.19$; TPP vs TPP+: AHR 1.03, 95% CI: 0.72–1.47, $P = 0.87$).
10. No significant impact on reducing child marriage in the overall sample
 - Prevalence of child marriage within the last 12 months decreased from 35% at baseline to 32% at endline in the intervention arm (change not significant).
 - No difference in child marriage across treatment arms detected (adjusted odds ratio [AOR] 0.72, 95% CI: 0.46–1.15).
11. No significant impact on reducing child marriage
 - No significant change measured, as child marriage prevalence remained low at endline: 6.05% overall and < 6.33% in all study arms.
12. Delayed marriage for younger girls
 - Child marriage was reduced among younger adolescents, but prevalence of marriage among older adolescents increased. Among girls aged 10–14 years, those exposed to the programme were less likely to have ever been married (odds ratio [OR] 0.09, $P < 0.001$). However, among girls aged 15–19 years, those in the intervention area had an elevated likelihood of being married by the endline (OR 2.4, $P < 0.001$).
13. Impact on reducing child marriage
 - Age at marriage increased by nearly two years to just shy of 18 years. At baseline, the mean age at marriage was 15.9 years, and more than 50% of girls in the sample were married in the two years prior to the start of the Development Initiative Supporting Healthy Adolescents (DISHA), and nearly 60% were married before the age of 18 years. In contrast, only 40% of the 198 girls married during the two-year DISHA programme were younger than 18 years at marriage. By endline, the average age at marriage among girls surveyed had increased by nearly two years to 17.9 years.
14. Impact on reducing child marriage
 - According to the model that did not control for covariates, exposure to Employment and Livelihood for Adolescents (ELA) Centre significantly reduced the probability of marriage. However, when other relevant variables were considered, participation in ELA Centre became a weaker determinant of the probability of getting married. Age of the adolescent girls and education of their household heads were significant factors in the models that included them as covariates. When the interaction variable of centre participation and age was introduced, the interaction variable was positively associated with the probability of marriage and participation was negatively associated with the probability of marriage.

GRADE Table WA.1.1(ii) PICO question 1.1/outcomes 2 and 3: Do interventions to empower girls by building their protective assets and/or promoting positive gender socialization (outcome 2) **increase girls' knowledge and understanding about the negative effects/consequences of child marriage and/or (outcome 3) **increase girls' knowledge and understanding of adolescents' rights, and where they could seek help and support if needed?****

No. of studies/N	Certainty assessment						Intervention: girls' empowerment	Results	Certainty	Importance
	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations				
5 studies (2,4,5,7,11) N≈14 628 girls and young women	RCTs	Serious ^{a,b,c,d}	Not serious	Not serious	Not serious	Not serious	Building girls' protective assets (life skills, SRHR education, livelihoods skills/ financial literacy and/ or education support)	3 studies showed increased knowledge about SRHR^{2,5,11} 1 study showed impact on some SRHR knowledge indicators, but not others⁴ 1 study showed no significant impact on improving knowledge about SRHR⁷	⊕⊕⊕○ MODERATE	IMPORTANT
5 studies (13-17) N≈7750 girls and young women	Quasi-experimental	Serious ^{a,e,f}	Not serious	Not serious	Not serious	Not serious	Same as above	1 study showed increased knowledge about adolescents' rights¹⁵ 5 studies showed increased knowledge about SRHR¹³⁻¹⁷	⊕○○○ VERY LOW	IMPORTANT
1 study (11) N≈1124 girls and young women	RCT	Very serious ^c	Cannot assess	Not serious	Serious ^g	Not serious	Promoting positive gender socialization	Increased knowledge about SRHR¹¹	⊕○○○ VERY LOW	IMPORTANT
1 study (14) N≈949 girls and young women	Quasi-experimental	Very serious ^{a,b,d,h}	Cannot assess	Not serious	Not serious	Not serious	Same as above	Increased knowledge on health and sanitation¹⁴	⊕○○○ VERY LOW	IMPORTANT

RCT: randomized controlled trial; SRHR: sexual and reproductive health and rights.

Explanations – certainty assessment

- ^a Risk of bias: Migration may have caused significant differences in populations captured in earlier and later rounds of data collection (2,4 [Malawi],5,7,14,15).
- ^b Risk of bias: Appropriateness of the sample (5,14).
- ^c Risk of bias: Selection bias related to systematic differences between treatment and control groups (11).
- ^d Risk of bias: Attrition bias related to selective dropout of participants who systematically differed from those who remained in the study (2,14).
- ^e Risk of bias: Insufficient information provided on sampling methods and/or composition of comparison groups (13,15).
- ^f Risk of bias: Potential spillover effects (14).
- ^g Imprecision: Uncertainty because CI is large and low baseline prevalence rates may have resulted in incorrect power calculations (11).
- ^h Risk of bias: Absence of an adequate comparison group and potential spillover effects (14).

Explanations (numbering matches superscripts in table AND references in list) – results

2. Significantly increased SRHR knowledge
 - Programme shown to have significantly improved girls' pregnancy knowledge and HIV knowledge scores.
4. Impact found on some SRHR knowledge indicators, but not others
 - Significant increase in knowledge of HIV in India, but not in Malawi, Mali or Niger.
 - Significant increase in contraceptive knowledge (scale) in Niger, but not in India, Malawi or Mali.
5. Significantly increased SRHR knowledge
 - SRHR knowledge (as measured on the "fertile period and contraceptive methods knowledge scale" [0–11]) improved in intervention areas relative to control areas at endline and again two years later (DID coefficient 0.27, 95% CI: 0.06–0.48).
7. No significant impact on SRHR knowledge
 - The clubs – both with and without the microfinance component – were not found to have had a discernible impact on increasing knowledge of HIV.

11. Significantly increased SRHR knowledge
 - SRHR knowledge increased significantly among TPP and TPP+ versus control participants (OR 0.71, $P = 0.036$).
13. Impact on increasing SRHR knowledge and increasing knowledge related to child marriage
 - Knowledge of legal age at marriage for girls (AOR 1.78, $P < 0.05$).
 - Knowledge of contraceptive pills (AOR 6.32, $P < 0.05$).
 - Knowledge of condoms (AOR 2.43, $P < 0.05$).
 - Knowledge of source of contraceptive pills (AOR 6.91, $P < 0.05$).
14. Significantly increased SRHR knowledge
 - ELA Centre members who had received Adolescent Peer Organized Network training gained more knowledge on health and sanitation issues and reported becoming aware of hygienic menstruation practices.
 - Intervention had a significant impact on decreasing health superstition score (average treatment effect on the treated [ATT] -7.06, DID coefficient $P < 0.05$).
15. Significantly increased SRHR knowledge; increased knowledge about adolescents' rights (results not disaggregated by sex)
 - Awareness index for social and legal issues increased significantly (from a mean score of 0.49 to 0.63; DID coefficient $P < 0.01$).
 - Knowledge that legal action can be taken in the case of a child marriage increased from 77% to 95% in the intervention group (DID coefficient $P < 0.01$).
 - Knowledge that dowry giver and taker are eligible for punishment in the case of a child marriage increased from 78% to 92% in the intervention group (DID coefficient $P < 0.01$).
 - Knowledge about HIV/AIDs increased from 80% to 91% among intervention group (DID coefficient $P < 0.05$).
 - DID coefficient not significant on knowledge of child and women trafficking or knowledge of legal marriage age.
16. Significantly increased knowledge of harms of child marriage
 - Results showed a significant impact on increasing knowledge that early childbearing is injurious to the mother's health (OR 1.69, $P < 0.001$).
17. Significantly increased SRHR knowledge
 - Significant increase in knowledge about contraception (DID coefficient 0.189, $P < 0.001$).

GRADE Table WA.1.1(iii) PICO question 1.1/outcomes 4, 5 and 6: Do interventions to empower girls by building their protective assets and/or promoting positive gender socialization (outcome 4) **increase girls' self-esteem and agency to assert preferences and negotiate decisions** and/or (outcome 5) **help girls challenge internalized gender norms** and/or (outcome 6) **increase supportive social networks of peers and community members to whom they could turn for assistance?**

Certainty assessment							Intervention: girls' empowerment	Results	Certainty	Importance
No. of studies/N	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations				
6 studies (2-5,7,11) N≈15 844 girls and young women	RCTs	Very serious ^{a,b,c,d,e}	Not serious	Not serious	Serious ^f	Not serious	Building girls' protective assets (life skills, SRHR education, livelihoods skills/ financial literacy and/ or education support)	Self-esteem/agency: – 2 studies showed increased self-esteem/agency^{5,11} – 3 studies showed no significant impact on self-esteem/agency^{2,3,7} Helping girls challenge internalized gender norms: – 4 studies showed no significant impact on challenging internalized gender norms^{2,5,7,11} Supportive social networks: – 1 study showed increased supportive social networks⁵ – 1 multi-country study showed impact on supportive social networks in some settings but not in others⁴	⊕○○○ VERY LOW	IMPORTANT
4 studies (13-15,17) N≈5670 girls and young women	Quasi-experimental	Very serious ^{a,e,g,h,i}	Not serious	Not serious	Not serious	Not serious	Same as above	Self esteem/agency: – 1 study showed no significant impact on self esteem¹⁷ Helping girls challenge internalized gender norms: – 2 studies showed significant impact on challenging internalized gender norms^{13,15} Supportive social networks: – 1 study showed increased supportive social networks¹⁴	⊕○○○ VERY LOW	IMPORTANT
1 study (11) N≈1124 girls and young women	RCT	Very serious ^d	Cannot assess	Not serious,	Serious ^f	Not serious	Promoting positive gender socialization	Self-esteem/agency: – Increased self-esteem/agency¹¹ Helping girls challenge internalized gender norms: – Significant impact on challenging internalized gender norms¹¹	⊕○○○ VERY LOW	IMPORTANT

No. of studies/N	Study design	Certainty assessment					Intervention: girls' empowerment	Results	Certainty	Importance
		Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations				
1 study (15) N≈389 girls and young women	Quasi-experimental	Very serious ^{a,h}	Cannot assess	Not serious	Not serious	Not serious	Same as above	• Mixed results on helping challenge internalized gender norms ¹⁵	⊕○○○ VERY LOW	IMPORTANT

RCT: randomized controlled trial; SRHR: sexual and reproductive health and rights.

Explanations – certainty assessment

- ^a Risk of bias: Migration may have caused significant differences in populations captured in earlier and later rounds of data collection (2,4–6,14,15).
^b Risk of bias: Appropriateness of the sample (5).
^c Risk of bias: Social desirability bias related to self-reported attitudes towards child marriage (3).
^d Risk of bias: Selection bias related to systematic differences between treatment and control groups (11).
^e Risk of bias: Attrition bias related to selective dropout of participants who systematically differed from those who remained in the study (2,14).
^f Imprecision: Uncertainty because the CI is large and low baseline prevalence rates may have resulted in incorrect power calculations (11).
^g Risk of bias: Absence of an adequate comparison group (14).
^h Risk of bias: Insufficient information provided on sampling methods and/or composition of comparison groups (13,15).

Explanations (numbering matches superscripts in table AND references in list) – results

- No significant impact on self-esteem/agency and no significant impact on helping challenge internalized gender norms
 - Gender empowerment index (reflecting how girls perceive their role in tasks related to the labour market and household) scores increased significantly at midline, but by endline there was no statistical difference between adolescent girls in the intervention and control communities.
- No significant impact on self-esteem/agency
 - Small and statistically insignificant impact of both GE and GE+ compared with control on psychosocial well-being and protective factors ($\beta, \leq 0.11$ SD).
- Impact on increasing supportive social networks in some settings but not in others
 - Impact in increasing proportion of community-level sample reporting being part of a club or group in India, but not in Malawi, Mali or Niger.
- Impact on increasing self-esteem/agency and increasing supportive social networks, but no significant impact on helping challenge internalized gender norms
 - Self-efficacy levels increased over time among those in the intervention arm, from an average of 6–6.1 to 7.4–7.7 on a scale with a range of 0–10.
 - The percentage of respondents who reported no access to a safe space decreased over time, representing 45–48% of respondents in the intervention arms and 48% of respondents in the control arm two years after the endline. There was no effect on norms regarding gender equity or acceptability of intimate partner violence.

- No significant impact on self-esteem/agency or helping girls challenge gender norms
 - The adolescent girls' clubs (life skills, livelihoods and vocational training) – with and without the microfinance component – showed no significant impact on perceived gender roles or control over life indices.
- Impact on increasing self-esteem/agency, but no significant impact on helping to challenge internalized gender norms
 - Significant increases in gender attitudes and self-efficacy scores were found, based on a scale of 0–22 (scores at baseline: TPP mean 12.58 [SD = 0.52], TPP+ mean 12.95 [SD = 0.47]; scores at follow-up: TPP mean 12.73 [SD = 0.62]; TPP+ mean 13.73 [SD = 0.65]).
 - For measures of instrumental agency, significant increases were observed in mean scores for girls' mobility and freedom of movement, communication and negotiation with parents, and participation in decision-making.
 - DID coefficient analyses (with covariate adjustments) of outcomes related to social norms and perceptions about gender discrimination in the family showed no significant programme effect.
- Impact on helping girls change internalized gender norms
 - Ideal age at marriage for girls is 18 or older (AOR 5.89, $P < 0.05$).
 - Contraception should be available to married youth (AOR 4.34, $P < 0.05$).
 - Able to talk to elders about timing of marriage (AOR 1.81, $P < 0.05$).
 - Able to travel unaccompanied outside the village for health services (AOR 1.65, $P < 0.05$).
- Impact on increasing girls' supportive social networks and mobility
 - Significant increase in girls' self-reported sociability was found (ATT 4.07, $P < 0.01$).
 - Significant increase in girls' mobility score was found (ATT 2.42, $P < 0.01$).
- Impact found on some indicators related to helping adolescents challenge internalized gender norms but not on other indicators (results not disaggregated by sex)
 - The proportion of adolescents who agreed that everyone in a family should abide by the husband decreased from 96% to 94% in the intervention group (DID coefficient $P < 0.01$).
 - The proportion of adolescents who agreed that it is usually not possible for a woman to be involved in earning money if she has a child decreased from 52% to 43% in the intervention group (DID coefficient $P < 0.05$).
 - The proportion of adolescents who agreed that if a woman gets involved in earning, it becomes difficult for her to maintain a marital relationship with her husband decreased from 49% to 45% in the intervention group (DID coefficient $P < 0.01$).
 - DID coefficient was not statistically significant on agreement that there should be a difference between sons and daughters in enjoying facilities at home.
 - DID coefficient was not statistically significant on agreement that a husband must endorse a decision to sell land.
- No significant impact on increasing self-esteem/agency
 - DID coefficient estimates of the impact of the Ishraq programme on self-esteem were not statistically significant (DID coefficient $P = 0.021$).

GRADE Table WA.1.1(iv) PICO question 1.1/outcome 7: Do interventions to empower girls by building their protective assets and/or promoting positive gender socialization (outcome 7) result in harms and/or unintended consequences?

No. of studies/N	Certainty assessment						Intervention: girls' empowerment	Results	Certainty	Importance
	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations				
1 study (6) N = 15 576 girls aged 15-17 at programme start	RCT	Not serious	Cannot assess	Not serious	Not serious	Not serious	Peer-facilitated empowerment curriculum including education support and social competency training	• Possible indirect impacts on the marriage market, incentivizing a subset of girls within the community with specific characteristics (in this case, more conservative girls) to marry earlier, offsetting programme impact on delaying marriage for girls	⊕⊕⊕○ MODERATE	IMPORTANT

RCT: randomized controlled trial.

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PICO 1.2 Interventions to (a) build awareness of and change attitudes and norms about child marriage, the rights of girls, and gender equality among parents/guardians and/or the broader community, and (b) build gender-equitable attitudes, norms and behaviours among boys and/or men

Table WA.1.2a(i) PICO question 1.2a/outcome 1: Do interventions to build awareness of and change attitudes and norms about child marriage, the rights of girls, and gender equality among parents/guardians and/or the broader community (outcome 1) **reduce child marriage?**

No. of studies/N	Study design	Certainty assessment					Interventions	Results	Certainty	Importance
		Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations				
4 studies (1-4) N≈13 148 girls	RCTs	Not serious	Not serious	Not serious	Serious ^a	Not serious	Interventions to build awareness of and change attitudes and norms about child marriage, the rights of girls, and gender equality among parents/guardians and the broader community	1 study showed significant reductions in child marriage in the intervention arms¹ 3 studies showed non-significant reductions in child marriage²⁻⁴	⊕⊕⊕○ MODERATE	CRITICAL
2 studies (5,6) N≈6823 youth; N≈1562 adults	Quasi-experimental	Serious ^{b,c,d}	Not serious	Not serious	Not serious	Not serious	Same as above	1 study showed significant reductions in child marriage in 2 settings and non-significant reduction in the 3rd setting⁵ 1 study showed non-significant reduction in child marriage⁶	⊕⊕○○ LOW	CRITICAL

RCT: randomized controlled trial.

Explanations – certainty assessment

- ^a Imprecision: Uncertainty because the CI is large and/or low baseline prevalence rates may have resulted in incorrect power calculations (2-4).
^b Risk of bias: Selection bias related to systematic differences between treatment and control group (5).
^c Risk of bias: Quality of counter-factual (5).
^d Risk of bias: Insufficient information provided on sampling methods and/or composition of comparison groups (balance table not presented) (6).

Explanations (numbering matches superscripts in table AND references in list) – results

- Significant reductions in child marriage
 - Overall, the probability of child marriage (age at marriage < 18) declined significantly by about a third in all intervention villages relative to control villages. The probability of child marriage was 31% lower for the education and gender-rights awareness interventions and 23% lower for the livelihoods

- intervention relative to control villages. The decline in child marriage is attributable to the 18-month-long intervention in all three skill-building strategies. While the change in intervention villages is significantly greater than in control villages, the difference between villages with different intervention strategies is not statistically significant.
- Proportion married before age 18 increased in all arms from baseline to endline, but increased significantly more in the control arm (from 17% to 29%).
 - In the education arm, marriage before age 18 increased from 18% to 24% (OR 0.693, $P \leq 0.001$).
 - In the gender awareness arm, marriage before age 18 increased from 19% to 25% (OR 0.695, $P \leq 0.001$).
 - In the livelihoods skills arm, marriage before age 18 increased from 16% to 23% (OR 0.767, $P \leq 0.01$).
 - Girls who were unmarried at the beginning of the study were one fourth less likely to be married by the end of the study (0.76–0.78 relative odds adjusted for age, religion and family wealth status).
2. Non-significant reduction in child marriage
 - No intervention impact was detected on child marriage in any of the arms (TPP vs control: AHR 1.14 [95% CI: 0.79–1.63], $P = 0.47$; TPP+ vs control: AHR 1.24 [95% CI: 0.89–1.71], $P = 0.19$; TPP vs TPP+: AHR 1.03 [95% CI: 0.72–1.47], $P = 0.87$).
 3. Non-significant reduction in child marriage
 - Prevalence of child marriage within the last 12 months decreased from 35% at baseline to 32% at endline in intervention areas (change not significant).
 - No difference in child marriage detected at endline across intervention arms: AOR 0.72 (95% CI: 0.46–1.15).
 4. Non-significant reduction in child marriage
 - No significant change measured, as child marriage prevalence remained low at endline: 6.05% overall and < 6.33% in all study arms.
 5. Results differed by setting (multi-country study)
 - United Republic of Tanzania: No significant impact on child marriage among younger or older girls in any arm: Adjusted risk ratios (ARRs) 1.13 (95% CI: 0.48–2.68) for girls aged 12–14 and 0.74 (95% CI: 0.43–1.27) for girls aged 15–17.
 - Burkina Faso: Significant impact on child marriage found for older girls aged 15–17 in the community dialogue arm: ARR 0.33 (95% CI: 0.19–0.60), $P < 0.001$. No significant impact found for older girls in the education or comprehensive model arms. No results shown for girls aged 12–14.
 - Ethiopia (compared against baseline data due to inadequate control sample): Child marriage significantly declined in the community dialogue arm for younger girls aged 12–14: ARR 0.42 (95% CI: 0.25–0.70), $P < 0.001$. No significant decline among girls aged 15–19: ARR 0.81 (95% CI: 0.60–1.09).
 6. Non-significant reduction in child marriage
 - Age at marriage increased by nearly two years to just shy of 18. However, a significant programme effect was not found on child marriage prevalence.
 - At baseline, the mean age at marriage was 15.9, and more than 50% of girls in the sample were married in the two years prior to the start of DISHA, nearly 60% before the age of 18. In contrast, only 40% of the 198 girls married during the two-year DISHA programme were younger than 18 at marriage. By endline, the average age at marriage among girls surveyed had increased by nearly two years to 17.8 years.

Table WA.1.2a(ii) PICO question 1.2a/outcome 2: Do interventions to build awareness of and change attitudes and norms about child marriage, the rights of girls, and gender equality among parents/guardians and/or the broader community (outcome 2) increase knowledge and understanding of the negative effects/consequences of child marriage?

No. of studies/N	Study design	Certainty assessment					Interventions	Results	Certainty	Importance
		Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations				
1 study (6) N = 4323 youth; N = 1562 adults	Quasi-experimental	Serious ^a	Cannot assess	Serious ^{b,c}	Not serious	Not serious	Interventions to build awareness of and change attitudes and norms about child marriage, the rights of girls, and gender equality among parents/guardians and the broader community	• Significant improvements on knowledge related to early marriage among adults in intervention villages (analysis of comparison vs control group was not shown) ⁶	⊕○○○ VERY LOW	IMPORTANT

Explanations – certainty assessment

- ^a Risk of bias: Insufficient information provided on sampling methods and/or composition of comparison groups (balance table not presented) (6).
- ^b Indirectness: Exposure to all components of a multi-component intervention makes it difficult to ascertain which programme strategies or activities were most influential for achieving programme impact (6).
- ^c Indirectness: Analysis presented does not directly respond to PICO question – results on this indicator presented for intervention/control comparison at endline only (6).

Explanations (numbering matches superscripts in table AND references in list) – results

6. Significant improvements on knowledge related to early marriage among adults in intervention villages (analysis of comparison vs control group was not shown)
- Knowledge of legal age at marriage for girls increased significantly among adults in intervention areas from baseline (26%) to endline (57%) ($P < 0.05$).
 - Proportion of adults reporting that the ideal age at marriage for girls is 18 or older increased significantly in the intervention group from baseline (48%) to endline (83%) ($P < 0.05$).

Table WA.1.2a(iii) PICO question 1.2a/outcomes 3 and 4: Do interventions to build awareness of and change attitudes and norms about child marriage, the rights of girls, and gender equality among parents/guardians and/or the broader community (outcome 3) **produce more equitable attitudes towards girls and boys, women and men, in terms of their roles, responsibilities, relations and sexuality** and/or (outcome 4) **enhance supportive social norms regarding rejection of child marriage?**

No. of studies/N	Certainty assessment						Interventions	Results	Certainty	Importance
	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations				
2 studies (1,4) N≈11 106 girls	RCTs	Serious ^{a,b}	Not serious	Not serious	Not serious	Serious ^c	Interventions to build awareness of and change attitudes and norms about child marriage, the rights of girls, and gender equality among parents/guardians and the broader community	1 study showed significant impact in producing more gender-equitable attitudes, behaviours and/or norms¹ 1 study did not find a statistically significant impact in producing more gender-equitable attitudes, behaviours and/or norms⁴	⊕⊕○○ LOW	IMPORTANT
3 studies (6-8) N≈14 235 youth; N≈5967 adults	Quasi-experimental	Very serious ^{a,d,e,f}	Not serious	Not serious	Not serious	Not serious	Same as above	1 study showed a significant positive impact among parents on gender attitudes index⁷ 1 study found significant positive impact on gender knowledge and attitudes among adults in intervention areas, but analysis of comparison vs control group was not shown⁵ 1 study found non-significant impact on increasing gender-equitable attitudes among girls' mothers and/or brothers⁸	⊕○○○ VERY LOW	IMPORTANT

RCT: randomized controlled trial.

Explanations – certainty assessment

- ^a Risk of bias: Attrition bias related to selective dropout of participants who systematically differed from those who remained in the study (1,8).
- ^b Risk of bias: Selection bias related to systematic differences between intervention and control groups (4).
- ^c Other considerations: Potentially limited generalizability due to study being carried out in a specific setting or with a specific population. Target population's caste composition has relevant implications for TPP+ approach (4).
- ^d Risk of bias: Migration may have caused significant differences in populations captured in earlier and later rounds of data collection (7).
- ^e Risk of bias: Insufficient information provided on sampling methods and/or composition of comparison groups (6,7).
- ^f Risk of bias: Potential spillover effects (8).

Explanations (numbering matches superscripts in table AND references in list) – results

- Significant impact on producing more gender-equitable attitudes, behaviours, and/or norms
 - Proportion of girls married between baseline and endline reporting that parents asked for their consent before marriage significantly increased in education arm (OR 0.665, $P = 0.05$) and gender awareness arm (OR 0.685, $P \leq 0.05$), but not in the livelihoods arm (OR 0.831).
- No significant impact on gender-equitable attitudes, behaviours and/or norms
 - For TPP+ vs control group, regression-based DID coefficient models showed no significant programme effects on outcomes related to network social norms and perceptions about gender discrimination in the family.
 - Detailed results not shown.

Table WA.1.2(iii) PICO question 1.2a/outcomes 3 and 4 (continued)

6. Significant improvements on knowledge and attitudes among adults related to early marriage in intervention villages
 - Proportion of adults reporting that the ideal age at marriage for girls is 18 or older increased significantly in the intervention group from baseline (48%) to endline (83%) ($P < 0.05$).
 - Analysis comparing change in intervention areas relative to change in control areas was not shown.
7. Significant positive impact found among parents on gender attitudes index
 - An index was constructed for parents' awareness on various gender-related attitudes. It was found that the mean score on the index for the comparison group increased slightly by 0.03, while that of the intervention group increased significantly, from 0.62 at baseline to 0.73 at endline. The DID coefficient was found to be significant at 1% level, indicating that the programme had a positive impact on awareness among parents.
8. Non-significant impact on increasing gender-equitable attitudes among girls' mothers and/or brothers
 - Non-significant impact of Ishraq programme found on proportion of mothers and/or brothers who report that the appropriate age at marriage for girls is 18 or older.
 - Non-significant impact of Ishraq programme found among mothers and/or brothers on support for girls playing sports, support for girls' participation in decision-making, or agreement that girls should have at least a secondary education.

PICO question 1.2a/outcome 5: Do interventions to build awareness of and change attitudes and norms about child marriage, the rights of girls, and gender equality among parents/guardians and/or the broader community (outcome 5) result in harms and/or unintended consequences?

No articles were identified that met the PICO inclusion criteria for this outcome.

Table WA.1.2b(i) PICO question 1.2b/outcome 1: Do interventions to build gender-equitable attitudes, norms and behaviours among boys (as peers and/or partners) and/or men (as partners or fathers) (outcome 1) **reduce child marriage?**

No. of studies/N	Study design	Certainty assessment					Interventions	Results	Certainty	Importance
		Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations				
1 study (6) N = 4323 youth; N = 1562 adults	Quasi-experimental	Serious ^a	Cannot assess	Serious ^{b,c}	Not serious	Not serious	Interventions to build gender-equitable attitudes, norms and behaviours among boys and/or men	Reduction in child marriage, but no intervention-attributable effect ⁵	⊕○○○ VERY LOW	CRITICAL

Explanations – certainty assessment

- ^a Risk of bias: Insufficient information provided on sampling methods and/or composition of comparison groups (balance table not presented) (6).
- ^b Indirectness: Exposure to all components of a multi-component intervention makes it difficult to ascertain which programme strategies or activities were most influential for achieving programme impact (6).
- ^c Indirectness: Analysis presented does not directly respond to PICO question—results on this indicator presented for intervention/control comparison at endline only (6).

Explanations (numbering matches superscripts in table AND references in list) – results

6. Reduction in child marriage, but no intervention-attributable effect
- Average age at marriage increased by nearly two years to just shy of 18; however, a significant programme effect was not found on child marriage prevalence.
 - At baseline, the mean age at marriage was 15.9, and more than 50% of girls in the sample were married in the two years prior to the start of DISHA, nearly 60% before the age of 18. In contrast, only 40% of the 198 girls married during the two-year DISHA programme were younger than 18 at marriage. By endline, the average age at marriage among girls surveyed had increased by nearly two years to 17.8 years.

Table WA.1.2b(ii) PICO question 1.2b/outcome 2: Do interventions to build gender-equitable attitudes, norms and behaviours among boys (as peers and/or partners) and/or men (as partners or fathers) (outcome 2) **increase knowledge and understanding of the negative effects/consequences of child marriage?**

No. of studies/N	Study design	Certainty assessment					Interventions	Results	Certainty	Importance
		Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations				
1 study (6) N = 4323 youth; N = 1562 adults	Quasi-experimental	Serious ^a	Cannot assess	Not serious	Not serious	Not serious	Interventions to build gender-equitable attitudes, norms and behaviours among boys and/or men	• Significantly higher knowledge related to early marriage among boys in intervention areas relative to control areas at endline (analysis of change since baseline was not shown) ⁶	⊕○○○ VERY LOW	IMPORTANT

Explanations – certainty assessment

^a Risk of bias: Insufficient information provided on sampling methods and/or composition of comparison groups (balance table not presented) (6).

Explanations (numbering matches superscripts in table AND references in list) – results

- Significant positive results reported on knowledge and attitudes related to early marriage among boys in intervention villages relative to control areas at endline (analysis of change since baseline was not shown)
 - Knowledge of legal age at marriage for girls was significantly higher among boys in the intervention group relative to control group at endline (AOR 2.79, $P < 0.05$).

Table WA.1.2b(iii) PICO question 1.2b/outcomes 3, 4 and 5: Do interventions to build gender-equitable attitudes, norms and behaviours among boys (as peers and/or partners) and/or men (as partners or fathers) (outcome 3) **produce more equitable attitudes towards girls and boys, women and men, in terms of their roles, responsibilities, relations and sexuality, (outcome 4) **help men challenge internalized masculine norms** and/or (outcome 5) **produce more gender equitable behaviours, including but not limited to division of domestic labour and shared decision-making?****

No. of studies/N	Study design	Certainty assessment					Interventions	Results	Certainty	Importance
		Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations				
2 studies (6,8) N≈13 438 youth; N≈5170 adults	Quasi-experimental	Serious ^{a,b,c}	Not serious	Serious ^{d,e}	Not serious	Not serious	Interventions to build gender-equitable attitudes, norms and behaviours among boys and/or men	1 study found significant differences in attitudes toward contraceptive access and ideal age at marriage among boys in intervention areas relative to control areas at endline (analysis of change since baseline was not shown)⁶ 1 study did not find a statistically significant impact on increasing gender-equitable attitudes among adolescent girls' brothers⁸	⊕○○○ VERY LOW	IMPORTANT

Explanations – certainty assessment

- ^a Risk of bias: Attrition bias related to selective dropout of participants who systematically differed from those who remained in the study (8).
- ^b Risk of bias: Insufficient information provided on sampling methods and/or composition of comparison groups (balance table not presented) (6).
- ^c Risk of bias: Potential spillover effects (8).
- ^d Indirectness: Exposure to all components of a multi-component intervention makes it difficult to ascertain which programme strategies or activities were most influential for achieving programme impact (6,8).
- ^e Indirectness: Analysis presented does not directly respond to PICO question—results on this indicator presented for intervention/control comparison at endline only (6).

Explanations (numbering matches superscripts in table AND references in list) – results

6. Significant positive results reported on gender-equitable attitudes among boys in intervention areas relative to control areas at endline (analysis of change since baseline was not shown)
- Proportion of adolescent boys agreeing that contraception should be available to married youth was significantly higher among boys in the intervention group at endline (AOR 7.14, $P < 0.05$).
 - Proportion of adolescent boys reporting that the ideal age at marriage for girls is 18 or older was significantly higher among boys in the intervention group at endline (AOR 2.30, $P < 0.05$).
8. No statistically significant impact on increasing gender-equitable attitudes among adolescent girls' brothers
- Non-significant impact found among brothers on support for girls playing sports, support for girls' participation in decision-making, or agreement that girls should have at least a secondary education.

PICO question 1.2b/outcome 6: Do interventions to build gender-equitable attitudes, norms and behaviours among boys (as peers and/or partners) and/or men (as partners or fathers) (outcome 6) result in harms and/or unintended consequences?

No articles were identified that met the PICO inclusion criteria for this outcome.

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PICO 1.3 Incentives: conditional financial incentives, unconditional financial incentives and non-financial incentives

Table WA.1.3(i) PICO question 1.3/outcome 1: Do the following types of incentives – conditional financial incentives, unconditional financial incentives (labelled or otherwise) and/or non-financial incentives – (outcome 1) **reduce child marriage?**

No. of studies/N	Study design	Certainty assessment					Intervention: Type of incentive offered	Results	Certainty	Importance
		Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations				
5 studies (1-5) N = 27 383	RCTs or CRCTs	Serious ^{a,b,c,d}	Not serious	Not serious	Not serious	Not serious	Conditional financial incentives	Mixed results: 1 study found no significant impact on reducing child marriage¹ 3 studies found significant reductions in child marriage²⁻⁴ 1 study found significant reduction in staying unmarried but no impact on child marriage⁵	⊕⊕⊕○ MODERATE	CRITICAL
4 studies (6-9) N = 865	Quasi-experimental	Serious ^{d,e,f}	Serious ^g	Serious ^h	Not serious	Serious ⁱ	Same as above	1 study found significant impact on reducing child marriage⁶ Mixed results: 1 study found impact on reducing child marriage for younger adolescents but increase in child marriage for older adolescents and overall⁷ 1 study found non-significant impact on overall child marriage in 1 site, and significant impact on child marriage in older adolescent girls only in 2 sites⁸ 1 study found ambiguous intervention impact on child marriage⁹	⊕○○○ VERY LOW	CRITICAL
2 studies (10,11) N > 4000	Observational	Serious ^{d,f}	Not serious	Not serious	Serious ^j	Not serious	Same as above	1 study found impact on delaying marriage for girls, but not on reducing child marriage¹⁰ 1 study found impact on reducing child marriage¹¹	⊕○○○ VERY LOW	CRITICAL
4 studies (1,3,12,13) N~10 345	RCTs	Not serious	Not serious	Not serious	Not serious	Serious ^k	Unconditional financial incentives	1 study showed significant reductions in child marriage¹ 3 studies showed no significant impact on child marriage^{3,12,13}	⊕⊕⊕○ MODERATE	CRITICAL

Table WA.1.3(i) PICO question 1.3/outcome 1 (continued)

No. of studies/N	Study design	Certainty assessment					Intervention: Type of incentive offered	Results	Certainty	Importance
		Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations				
1 study (11) Village population N~4000	Observational/ natural experiment	Serious ^d	Cannot assess	Serious ^h	Not serious	Not serious	Non-financial incentives	Impact on reducing child marriage¹¹	⊕○○○ VERY LOW	CRITICAL

CRCT: cluster randomized controlled trial; RCT: randomized controlled trial.

Explanations – certainty assessment

- ^a Risk of bias: Lack of adherence to design (4).
^b Risk of bias: Attrition bias related to selective dropout of participants who systematically differed from those who remained in the study (1).
^c Risk of bias: Issues related to timing of study (time to follow-up, longer-term impacts) (3).
^d Risk of bias: Quality of counter-factual/isolation of exposure (2,8,11).
^e Risk of bias: Migration may have caused significant differences in populations captured in earlier and later rounds of data collection (7).
^f Risk of bias: Selection bias related to systematic differences between intervention and control groups (8,10,11).
^g Inconsistency: Across settings for multi-country studies (8).
^h Indirectness: Exposure to all components of a multi-component intervention makes it difficult to ascertain which programme strategies or activities were most influential for achieving programme impact (7,11).
ⁱ Other consideration: Study does not report results for aggregate adolescent sample (8).
^j Imprecision: Difficulty attributing impact to programme due to time since exposure (10).
^k Other considerations: Potentially limited generalizability due to study being carried out in a specific setting or with a specific population (12,13).

Explanations (numbering matches superscripts in table AND references in list) – results

- No significant impact on reducing child marriage in the conditional cash transfers (CCT) arm; significant impact on reducing child marriage in the unconditional cash transfers (UCT) arm
 - The impact on the proportion ever-married was small and not significant in the CCT arm, where there was a reduction by 1.2 percentage points (-0.012 [SE 0.024]).
 - The likelihood of ever being married was 7.9 percentage points (44%) lower in the UCT arm than in the control group at the end of the two-year intervention (0.079 [SE 0.022], $P < 0.05$).

- Significant reduction in child marriage
 - The estimated ITT effects of the violence prevention and education (VE) intervention relative to violence prevention only (using extended controls including baseline schooling to improve precision and account for potential initial imbalances): -0.062 (95% CI: -0.11, -0.02), $P < 0.01$.
 - The pooled estimate (i.e. all three study arms with education pooled into a single treatment variable) with extended controls as above: -0.041 (95% CI: -0.8, 0.00), $P < 0.05$.
- Sustained significant impact on reducing child marriage in the CCT arm; no significant impact on reducing child marriage in the UCT arm two years after the end of the programme
 - CCTs caused sustained reduction in incidence of child marriage.
 - No statistically significant impact of UCTs on marriage two years after the end of the programme.
- Significant reduction in child marriage
 - 4.5 years after programme completion, girls eligible for the incentive for at least two years were 24% (-8.9 percentage points, $P < 0.01$) less likely to be married before age 18 years.
- Child marriage before age 18 years declined; significant impact on remaining unmarried
 - Beneficiaries were 3.6 percentage points (46%) less likely to be married.
- Significant reduction in child marriage
 - Coefficient for basic model for marriage: -20.7307 (SE = 1%).
 - No significant impact was found on the child marriage programme treatment indicator; the effect was insignificant across all models examining impact on child marriage (OR 0.06, $P = 0.21$).
- Impact on reducing child marriage among younger adolescents, but increased prevalence of marriage among older adolescents
 - Among girls aged 10–14 years, those exposed to the programme were less likely to have ever been married (OR 0.1, $P < 0.01$). However, among girls aged 15–19 years, those in the intervention area had an elevated likelihood of having gotten married by the endline (OR 2.4, $P < 0.01$).
- Results differed by country, age group and intervention
 - United Republic of Tanzania: Child marriage impact was non-significant by arm and age group, except for the asset transfer only arm, which showed a significant impact (RR 0.52, $P < 0.05$).
 - Ethiopia: Among girls aged 15–17 years, those residing in the conditional asset transfer area and the comprehensive model had roughly half the risk of being married at endline compared with their counterparts at baseline (risk ratio [RR] 0.57 [$P < 0.05$] for conditional asset transfer; RR 0.38 [$P < 0.001$] for the comprehensive model).

Table WA.1.3(i) PICO question 1.3/outcome 1 (continued)

- Burkina Faso: Dropped from analysis due to significant differences between the intervention and control sites at both baseline and endline, which were impossible to control for in the multivariate models.
9. No statistically significant impact on child marriage
 - No effect on marriage before age 18. Proportion marrying during 18th year of age higher for beneficiaries (59%) than non-beneficiaries (45%), leading authors to conclude that the intervention may have incentivized misreporting of age or age at marriage, or waiting until age 18 to marry in order to cash in on incentives for dowry payment.
 10. Impact on delaying marriage for girls, no significant impact on reducing child marriage or probability of being married (aged 15–19 years)
 - Overall, estimates from DID and regression discontinuity design (RDD) models do not show any impacts of the programme on the probability of girls aged 15–19 years being legally married. However, when looking at age at marriage, the analysis identified that girls in stipend districts are likely to marry about 1.2–1.5 years after the programme started. Results are statistically significant and consistent across model specifications and subsamples with different age cohorts.
 11. Impact on reducing child marriage
 - Proportion of married girls declined from 29% to 14% for 13–15 year-olds, and from 72% to 64% for 16–19 year-olds.
 12. No significant impact on child marriage
 - The cash transfer programmes had no significant impact on early marriage in either Malawi (AOR –0.00428 [SE 0.0176], not significant) or Zambia (AOR 0.0117 [SE 0.0141], not significant).
 13. Unclear impact on reducing child marriage.

Table WA.1.3(ii) PICO question 1.3/outcome 2: Do the following types of incentives – conditional financial incentives, unconditional financial incentives (labelled or otherwise) and/or non-financial incentives – (outcome 2) **increase school enrolment, attendance and/or completion?**

No. of studies/N	Study design	Certainty assessment					Intervention: Type of incentive offered	Results	Certainty	Importance
		Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations				
5 studies (2–5,12) N ~ 27 383	RCTs or CRCTs	Not serious	Not serious	Serious ^a	Not serious	Not serious	Conditional financial incentives	4 studies found improved education outcomes (enrolment and completion rates)^{2–4,12} 1 study found no significant impact on probability of being in school⁵	⊕⊕⊕○ MODERATE	IMPORTANT
3 studies (7–9) N ~ 8650	Quasi-experimental	Serious ^{b,c,d}	Serious ^e	Serious ^f	Not serious	Serious ^g	Same as above	2 studies found improved education outcomes^{7,9} 1 multi-country study found impact on education outcomes differed by setting⁸	⊕○○○ VERY LOW	IMPORTANT
2 studies (10,11) N > 4000	Observational	Serious ^{c,d}	Not serious	Serious ^f	Not serious	Not serious	Same as above	2 studies found improved education outcomes (attendance and completion rates)^{10,11}	⊕○○○ VERY LOW	IMPORTANT
2 studies (3,4) N = 6703	RCTs	Not serious	Not serious	Not serious	Not serious	Serious ^{h,i}	Unconditional financial incentives	Improved education outcomes (dropout rate) in the short term,³ but impact not sustained⁴	⊕⊕⊕○ MODERATE	IMPORTANT
1 study (11) N ~ 4000	Observational/natural experiment	Serious ^c	Cannot assess	Serious ^f	Not serious	Not serious	Non-financial incentives (food)	Impact on primary school enrolment rates¹¹	⊕○○○ VERY LOW	IMPORTANT

CRCT: cluster randomized controlled trial; RCT: randomized controlled trial.

Explanations – certainty assessment

- ^a Indirectness: Difficulty attributing impact to programme due to time since exposure (4).
^b Risk of bias: Migration may have caused significant differences in populations captured in earlier and later rounds of data collection (7).
^c Risk of bias: Quality of counterfactual/isolation of exposure (8,11).
^d Risk of bias: Selection bias related to systematic differences between intervention and control groups (8).
^e Inconsistency: Across settings for multi-country studies (8).

- ^f Indirectness: Exposure to all components of a multi-component intervention makes it difficult to ascertain which programme strategies or activities were most influential for achieving programme impact (7,10,11).
^g Other considerations: Study does not report results for aggregate adolescent sample (8).
^h Other considerations: Study not able to address potential longer-term impacts/time to follow-up (4).
ⁱ Other considerations: Potentially limited generalizability due to study being carried out in a specific setting or with a specific population (4).

Explanations (numbering matches superscripts in table AND references in list) – results

2. Impact on improving education outcomes
 - All study arms (VE: violence prevention and education; VEH: Violence prevention, education and health; VEHW: violence prevention, education, health and wealth creation) increased educational outcomes between 0.1 and 0.2 SD, although VEH was only marginally significant.
 - VE-VEH-VEHW pooled estimate: 0.134 (95% CI: 0.05, 0.22), $P < 0.01$.
 - VE estimate: extended controls: 0.199 (95% CI: 0.10, 0.30), $P < 0.01$.
 - VEH estimate: extended controls: 0.103 (95% CI: -0.01, -0.22), $P < 0.1$ (marginally significant).
 - VEHW estimate: extended controls: 0.110 (95% CI: 0.01, 0.21), $P < 0.05$.
 - Pooled estimate: extended controls: 0.137 (95% CI: 0.06, 0.22), $P < 0.01$.
3. Impact on improving education outcomes; effects not sustained for UCTs
 - Short-term improvements in the UCT arm observed during the programme were not sustained.
 - CCTs caused sustained effects on school attainment, but only among the stratum of adolescent females who had already dropped out of school at baseline and were all assigned to CCTs.
4. Impact on improving education outcomes
 - 4.5 years after programme completion, girls eligible for the incentive for at least two years were 15% (-4.8%, $P < 0.05$) less likely to have given birth before age 20, and 25% (7.0%, $P < 0.01$) more likely to be in school at age 22 years.
5. No significant impact on child schooling
 - At 24 months, the standardized effects of both GE and GE+ on schooling, sexual violence, psychosocial well-being and protective factors were small compared with the control group (≤ 0.11 SD) and not significant at the 95% level of confidence.
7. Impact on improving education outcomes
 - Among girls aged 10–14 years, at the endline survey, those exposed to the programme were more likely to be in school (OR 3.0) and less likely to have ever been married (OR 0.1) than those in the control area.
8. Impact on improving education outcomes differed by setting
 - United Republic of Tanzania: All the interventions tested were associated with increased probability of girls aged 12–14 years being in school.
 - Ethiopia: No impact of the interventions on schooling was detected.
 - Burkina Faso: Girls aged 12–14 years residing in the community dialogue site had a greater chance of being in school (RR 1.2) compared with their counterparts in the control area.
9. Impact on improving education outcomes
 - Beneficiary girls had a 12% higher probability of completing 8th grade and a 19% higher probability of aspiring to study beyond 12th grade.
10. Impact on improving education outcomes
 - Participant girls were 3–6% more likely to complete middle school.
11. Impact on improving education outcomes
 - The two primary school incentive programmes — the Bangladesh Rural Advancement Committee (BRAC) school in Village A and the wheat-for-education programme in Village B — were available only to poor families, and the principal criterion on which poverty was assessed was landlessness.
 - In Village B, primary school enrolment levels improved much more among the children of poor families than among children from the village as a whole, although the two groups had similar enrolment levels in 1992. Similarly in Village A, the increase in girls' schooling was greater among poor families than in the population as a whole.
 - In Village B, girls' primary school enrolment rates increased sharply, from 44% to 59%.
 - Increased primary school attendance resulted in higher rates of completion as well as increased secondary school attendance by 1996.
12. Significantly improved education outcomes (probability of attending school)
 - In Malawi, cash transfers led to a positive impact on the proportion of male and female youth currently attending school, in the range of 8–11 percentage points ($P < 0.01$).
 - In Zambia, a positive impact was also found but it was significant only for the male youth (for female youth it was insignificant at $P < 0.1$) and was smaller in magnitude, around 5–6 percentage points.

Table WA.1.3(iii) PICO question 1.3/outcome 3: Do the following types of incentives – conditional financial incentives, unconditional financial incentives (labelled or otherwise) and/or non-financial incentives – (outcome 3) **delay first pregnancy and/or birth?**

No. of studies/N	Study design	Certainty assessment					Intervention: Type of incentive offered	Results	Certainty	Importance
		Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations				
5 studies (1-5) N = 27 383	CRCTs or RCTs	Not serious	Not serious	Not serious	Not serious	Not serious	Conditional financial incentives	4 studies found improved SRHR outcomes^{2,3-5} 1 study found no improvement in SRHR outcomes¹	⊕⊕⊕○ MODERATE	IMPORTANT
3 studies (6-8) N = 4706	Quasi-experimental	Serious ^{a,b,c}	Not serious	Serious ^d	Not serious	Serious ^e	Same as above	1 study showed impact on first births differed by setting⁸ 2 studies showed improved SRHR outcomes^{6,7}	⊕○○○ VERY LOW	IMPORTANT
1 study (10) N = 4000	Observational/natural experiment	Serious ^{a,c}	Cannot assess	Serious ^f	Not serious	Not serious	Same as above	- The study did not assess direct impact on SRHR - The impact was on improved schooling and more schooling is associated indirectly with improved contraceptive knowledge¹⁰	⊕○○○ VERY LOW	IMPORTANT
4 studies (1,3,12,13) N = 10 345	RCTs	Not serious	Not serious	Not serious	Not serious	Not serious	Unconditional financial incentives	2 studies found improved SRHR outcomes^{1,13} 1 study found improved SRHR outcomes in the short term,⁴ but impact not sustained³ 1 study found no significant impact on SRHR outcomes¹²	⊕⊕⊕⊕ HIGH	IMPORTANT

CRCT: cluster randomized controlled trial; RCT: randomized controlled trial; SRHR: sexual and reproductive health and rights.

Explanations – certainty assessment

- ^a Risk of bias: Quality of counter-factual/isolation of exposure (8).
- ^b Risk of bias: Migration may have caused significant differences in populations captured in earlier and later rounds of data collection (8,10).
- ^c Risk of bias: Selection bias related to systematic differences between intervention and control groups (8,10).
- ^d Indirectness: Exposure to all components of a multi-component intervention makes it difficult to ascertain which programme strategies or activities were most influential for achieving programme impact (7).
- ^e Other considerations: Study does not report results for the aggregate adolescent sample (8).
- ^f Indirectness: Difficulty attributing impact to the programme due to time since exposure (10).

Explanations (numbering matches superscripts in table AND references in list) – results

1. Impact on improving SRHR outcomes for UCT arm compared with control and CCT arm
 - Likelihood of ever being pregnant was 27% and 44% lower in the UCT arm than in the control group at the end of the two-year intervention and two years after the end of the intervention, respectively. Pregnancy rates were significantly lower in the UCT than the CCT arm whereas programme impacts on these two outcomes were small and significant in the CCT arm. The difference in programme impacts between the two treatment arms at the end of two years and two years after the end are both significant at the 95% confidence level. Pregnancy and marriage rates were substantially lower in the UCT than the CCT arm, due to the impact of UCTs on these outcomes among girls who dropped out of school.
2. Impact on improving SRHR outcomes
 - Reduction of 0.11 SD in fertility summary, with point estimates for individual arms ranging between -0.08 and -0.16 SD.
 - VE estimate: extended controls: -0.16 (95% CI: -0.28, -0.03), $P < 0.05$.
 - Pooled estimate: extended controls: -0.113 (95% CI: -0.22, -0.01), $P < 0.05$.
3. Impact on improving SRHR outcomes for CCTs; effects not sustained for UCTs
 - Short-term improvements in the UCT arm observed during the programme failed to translate into increased welfare in the longer term.
 - CCTs caused sustained effects on pregnancy, age at first birth, total number of births and desired fertility, but only among the stratum of adolescent females who had already dropped out of school at baseline and were all assigned to CCTs.

4. Impact on improving SRHR outcomes
 - 4.5 years after programme completion, girls eligible for the incentive for at least two years were 15% (-4.8 percentage points, $P < 0.05$) less likely to have given birth before age 20 and 25% (7.0 percentage points, $P < 0.01$) more likely to be in school at age 22 years.
5. Impact on improving SRHR outcomes
 - Programme beneficiaries had a lower number of sexual partners (0.58 fewer partners, 43% lower), and were more likely to practise safer sex compared with the control group.
 - GE+ reduced the likelihood of marriage and the number of sexual partners in the past 12 months by more than 50%.
6. Impact on improving SRHR outcomes
 - Adolescent girls who participated in the programme experienced a delay in first premarital sexual intercourse, first and second births. Coefficients for model 3: premarital sex -0.7611 (0.3585), first birth 0.6771 (0.1974), second birth -0.0155 (0.3036).
7. Impact on improving SRHR outcomes
 - Improved SRHR knowledge and behaviour outcomes among girls in the intervention group: awareness of contraceptive method, 91.1% ($P < 0.01$); ever used injectable contraceptives, 66.4% ($P < 0.001$); ever used any contraceptive method, 74.1% ($P < 0.001$).
8. Impact on improving SRHR outcomes (experiencing a birth) differed by setting
 - The study attempted to assess impact of the programme on the proportion of girls who experienced a birth, but the authors were unable to present comparable results because of differences in the samples by arm and across settings. A more detailed probe in Ethiopia showed the programme impacted SRHR knowledge on a range of indicators.
10. Impact on SRHR significant but indirectly mediated through schooling
 - Participant girls had improved schooling and more schooling was shown to be associated indirectly with improved contraceptive knowledge.
12. No significant impact on SRHR outcomes
 - The cash transfer programmes had no significant impact on fertility outcomes in either country.
13. Impact on improving SRHR outcomes
 - Programme reduced likelihood of pregnancy by 5%.

PICO question 1.3/outcome 4: Do the following types of incentives – conditional financial incentives, unconditional financial incentives (labelled or otherwise) and/or non-financial incentives – (outcome 4) **result in harms and/or unintended consequences?**

No articles were identified that met the PICO inclusion criteria for this outcome.

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PICO 1.4 Interventions to improve the availability and/or quality of educational opportunities for girls and young women

Table WA.1.4(i) PICO question 1.4/outcome 1: Do interventions to improve the availability and/or quality of educational opportunities for girls and young women (outcome 1) **reduce child marriage?**

No. of studies/N	Study design	Certainty assessment					Interventions to improve the availability and/or quality of educational opportunities for girls and young women	Results	Certainty	Importance
		Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations				
4 studies (1-4) N≈12 562 girls	RCTs	Not serious ^a	Not serious	Not serious	Not serious	Not serious	Life skills curriculums implemented by “social mobilizer” or provision of support to stay in school, including fees, uniforms, school supplies and school-based helpers	1 study found that results differed by intervention arm¹ 1 study found no significant impact on child marriage prevalence, however the endline sample was skewed towards younger ages² 2 studies showed impact on reducing child marriage^{3,4}	⊕⊕⊕⊕ HIGH	CRITICAL
1 study (5) N = 2500 girls aged 12–17	Quasi-experimental	Serious ^a	Cannot assess	Not serious	Not serious	Not serious	Provision of school supplies to encourage retention in school	No significant impact on reducing child marriage⁵	⊕○○○ VERY LOW	CRITICAL
12 studies (6-17) N > 1 million respondents	Observational/natural experiment	Serious ^{a,b,c}	Not serious	Not serious	Not serious	Not serious	Improved availability of primary, secondary and tertiary educational opportunities through school construction, tuition fee waivers, scholarships and stipend programmes	2 studies showed no significant impact on reducing child marriage^{6,7} 2 studies showed marginally significant impact on reducing child marriage^{8,9} 7 studies showed impact on reducing child marriage¹⁰⁻¹⁶ - In a multi-country study, impact on child marriage differed by country¹⁷	⊕○○○ VERY LOW	CRITICAL

RCT: randomized controlled trial.

Explanations – certainty assessment

- ^a Risk of bias: Quality of counterfactual/isolation of exposure (5–8,11,15–17).
- ^b Risk of bias: Selection bias related to systematic differences between intervention and control groups (13).
- ^c Risk of bias: Potential spillover effects (6).

Explanations (numbering matches superscripts in table AND references in list) – results

1. Impact on reducing child marriage differed by study arm
 - Girls in the “subsidy plus” programme were 2.6 percentage points (20%) less likely to be married relative to those in the standalone “education subsidy” programme.
2. No effect found on probability of marriage
 - No change in the probability that the child is married by endline (approximately age 14); coefficient 0.042 (SE = 0.029).
3. Impact on reducing child marriage
 - Fewer girls among the comprehensive intervention group reported marriage (n = 18, 11.2%) compared with those in the delayed partial intervention group (n = 29, 23.6%).
4. Impact on reducing child marriage
 - The intervention reduced marriage by 63% after two years. Compared with intervention participants, control participants had nearly three times higher odds of getting married (OR 2.92; *P* = 0.02).
5. No significant impact on reducing child marriage
 - United Republic of Tanzania: Education promotion arm, girls aged 12–14 years: ARR 0.70 (95% CI: 0.30, 1.65), girls aged 15–17: ARR 0.99 (95% CI: 0.64, 1.53).
 - Burkina Faso: No significant impact was found for older girls aged 15–17 years in the education arm: ARR 0.99 (95% CI: 0.66, 1.49). No results were shown for girls aged 12–14.
 - Ethiopia: No results were shown for the education arm.
6. No significant impact on reducing child marriage
 - There is no effect of exposure to school construction on the age at first marriage. Coefficients are small (0.050) and statistically non-significant.
7. No statistically significant impact on child marriage
 - No statistically significant change in age at marriage; however, impact measure on child marriage is an indirect estimation (study assesses impact of increased educational attainment on age at marriage).
8. Marginally significant impact found on reducing marriage and cohabitation
 - Lottery winners were less likely to be married or cohabiting (coefficient in probit model with basic controls for female sample: -0.0114 [SD = 0.0068]).
9. Marginally significant reduction in age at first marriage
 - Each additional year of schooling reduced the age of first marriage by 0.34 years, although this association was only marginally significant (*P* = 0.083).
10. Significantly decreased child marriage rates
 - Marriage rates for adolescent girls decreased from 36% to 32%.
11. Significantly reduced probability of child marriage (before age 15 and 18 years)
 - While the tuition-free primary education policy led to significant changes in the probability of marriage before age 15 or 18 years and in the probability of childbirth before age 18, the extension of tuition-free policy to secondary education led to significantly greater reductions in all four outcomes.
 - Abolition of tuition fees at the primary level reduced the probability of marriage before age 15 years by 0.8 percentage points (-0.75, 95% CI: -1.51, -0.07), and reduced the probability of marriage before age 18 years by 2.5 percentage points (-2.5, 95% CI: -4.19, -0.85) on average.
 - Abolition of tuition fees at the secondary level reduced the probability of marriage before age 15 years by 4.1 percentage points (-4.05, 95% CI: -4.96, -3.20), and reduced the probability of marriage before age 18 years by 5.5 percentage points (-5.47, 95% CI: -7.81, -3.23) on average.
 - Tuition-free secondary education reduced the probability of marriage before 15 by 3.3 percentage points on average (-3.30, 95% CI: -4.06, -2.54) and that of marriage before 18 by 3.0 percentage points on average (-2.97, 95% CI: -5.16, -0.79) compared with tuition-free primary education alone.
12. Significantly reduced probability of child marriage
 - At the ages of 20 and 21, each additional year of schooling reduced the likelihood of marriage by 6.4 and 5.3 percentage points, respectively.
13. Significant impact on delaying marriage for girls
 - Increased education led to delayed age at marriage.
 - Impact of expanded access to maternal education on marriage: 0.015 (clustered SE = 0.012).
14. Significant impact on reducing the probability of child marriage
 - Primary school reduced the probability of marriage by age 18 by 46 percentage points.
15. Significant impact on delaying marriage age for girls
 - Exposure to the Female Secondary School Stipend Program (FSSSP) delayed age at first marriage by 0.57 years (3.6%) or, on average, 0.11 years for each year of exposure for those in Cohort 1, who received the stipend for five years.
 - For girls in Cohort 2, who received the stipend for two years, exposure to the FSSSP increased age at first marriage by 0.34 years (2.1%), or 0.17 years for each year of exposure.
 - Girls eligible for the stipend got married on average between 0.11 and 0.17 years later for each year of exposure.
16. Significant increase in probability of not being married
 - Both reduced form (ITT effect) and the two-stage least squares (2SLS, effect of treatment on the treated [ETT]) estimates indicate that higher education increased the probability of never getting married by 5–6 percentage points (about 34–40%).
17. Impact on child marriage prevalence differed by country (multi-county study)
 - All countries: marriage before 18, risk difference (RD) -0.03 (95% CI: -0.09, 0.03).
 - Ethiopia: marriage before 18: RD -0.10 (95% CI: -0.13, -0.08).
 - Ghana: marriage before 18: RD -0.06 (95% CI: -0.08, -0.04).
 - Rwanda: marriage before 18: RD -0.15 (95% CI: -0.17, -0.13).
 - Uganda: marriage before 18: RD -0.07 (95% CI: -0.11, -0.04).
 - Zambia: marriage before 18: RD -0.05 (95% CI: -0.06, -0.03).
 - Cameroon: no evidence that removal of tuition fees had an impact on child marriage rates.
 - Malawi: no evidence that removal of tuition fees had an impact on child marriage rates.

Table WA.1.4(ii) PICO question 1.4/outcome 2: Do interventions to improve the availability and/or quality of educational opportunities for girls and young women (outcome 2) **increase school enrolment, attendance and/or completion (i.e. improve education outcomes)?**

No. of studies/N	Study design	Certainty assessment					Interventions to improve the availability and/or quality of educational opportunities for girls and young women	Results	Certainty	Importance
		Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations				
4 studies (1-4) N≈12,562 girls	RCTs	Not serious ^a	Not serious	Not serious	Not serious	Not serious	Life skills curriculums implemented by “social mobilizer” or provision of support to stay in school, including fees, uniforms, school supplies and school-based helpers	1 study found an impact on increasing educational attainment¹ 3 studies found an impact on reducing school dropout rates²⁻⁴	⊕⊕⊕⊕ HIGH	IMPORTANT
10 studies (6-8,10-15,17) N > 1 million respondents	Observational/natural experiment	Serious ^{a,b,c}	Not serious	Not serious	Not serious	Not serious	Improved availability of primary, secondary and tertiary educational opportunities through school construction, tuition fee waivers, scholarships and stipend programmes	8 studies found an impact on increasing years of schooling/ educational attainment^{6,9,10,12-16} 1 study found a non-significant increase in number of years of schooling⁸ 1 multi-country study found that impact on primary school completion differed by country¹⁷ 1 study found impact on decreasing probability of grade repetition⁸	⊕○○○ VERY LOW	IMPORTANT

RCT: randomized controlled trial.

Explanations – certainty assessment

^a Risk of bias: Quality of counterfactual/isolation of exposure (3,6,8,17).

^b Risk of bias: Selection bias related to systematic differences between intervention and control groups (6,8).

^c Risk of bias: Potential spillover effects (6).

Explanations (numbering matches superscripts in table AND references in list) – results

1. Stand-alone education subsidy increased educational attainment more than the HIV education programme and joint programme
 - School dropout rates declined significantly in the stand-alone education subsidy arm (-0.031 [SE 0.012]) but not in the HIV education arm (0.003 [SE 0.011]) or the joint programme offering subsidy and HIV education (-0.016 [SE 0.012]).
2. Decline in school dropout rates
 - The intervention caused a 25% decline in school dropout that persisted from 7th grade through the transition to high school.
 - No statistically significant impact on test scores.
3. Reduced school dropout and increased years of schooling
 - Those in the comprehensive intervention group (5 years) were less likely to drop out of school; school dropout: OR 0.24, 95% CI: 0.10–0.55, $P < 0.001$.
 - Those in the comprehensive intervention group received almost one additional year of schooling compared with the delayed partial intervention group; years of schooling: coefficient 0.68, 95% CI: 0.41–0.98, $P < 0.001$.
4. Reduced school dropout
 - The intervention reduced school dropout by 82%.
 - Control participants had six times the odds of dropping out of school (OR 8.5; $P \leq 0.001$).
6. Significant increase in children's education outcomes as a result of school construction in parents' districts. An increase in the mean number of schools in a mother's birth district led to an increase in the likelihood that her child completes tertiary education, relative to average tertiary education levels.
 - Some effects are SD increase from the mean in programme villages.
 - Mean years of schooling: father 7.967 (SD = 4.34), mother 8.854 (SD = 4.278).
 - Effect estimate: father 0.097 (SE = 0.032), mother 0.169 (SE = 0.045).
 - Aggregate score: father 0.030 (SE = 0.013), mother 0.056 (SE = 0.016).
8. Non-significant increase in years of schooling. Decreased probability of grade repetition.
 - Lottery winners in the 1995 Bogotá sample completed an additional 0.12–0.16 years (0.8 years in the full sample, not statistically significant).
 - The probability of grade repetition was reduced by 5–6 percentage points for lottery winners.
9. Increase in schooling attainment
 - Schooling attainment increased rapidly following the implementation of Malawi's education policies. The median number of completed school years was 1 grade higher for the cohort interviewed in 2000 relative to the oldest cohort.
10. Increased school completion and progression
 - The proportion of girls aged 11–19 years who had completed fifth grade increased from 22% to 36% between 1992 and 1995.
 - The proportion of these primary-school graduates who opted to continue to secondary school also increased from 48% to 62%.
12. Increase in schooling
 - For each additional year of free schooling made available, years of schooling increased by over one tenth of a year. This relationship was statistically significant at the 95% confidence level for all three samples, and at the 99% CI when the census data are included.
13. Increased probability of attending secondary school
 - The discontinuity in education is associated with an increase in education of 1.68 years, and an increased probability of attending some secondary school by 25.2 percentage points, after controlling for other variables.
14. Increase in primary school completion
 - The estimate of the increase in primary school completion due to the compulsory schooling law was 0.006 ($P < 0.001$).
15. Increased years of schooling
 - The stipend increased years of education for eligible girls by 25% (1.2 years of additional schooling).
16. Increased years of schooling due to higher education reform (RDD assessing relative impact on affected cohorts)
 - Years of schooling for ages 15–24 increased by 0.67 (SE = 0.132).
17. Impact on primary school completion differed by country (multi-country study)
 - No additional gains in primary school completion in Malawi or Uganda.
 - Modest effects on completion rates in Ethiopia and Ghana.
 - Probability of completing primary school rose by 19 percentage points in Rwanda and 6 percentage points in Zambia.

Table WA.1.4(iii) PICO question 1.4/outcome 3: Do interventions to improve the availability and/or quality of educational opportunities for girls and young women (outcome 3) **increase self-efficacy and autonomy?**

No. of studies/N	Study design	Certainty assessment					Interventions to improve the availability and/or quality of educational opportunities for girls and young women	Results	Certainty	Importance
		Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations				
1 study (2) N = 2459 adolescent girls	RCT	Not serious	Cannot assess	Not serious	Not serious	Not serious	Life skills curriculum implemented by “social mobilizer”	• No statistically significant impact on parental perceptions of girls’ self-efficacy ²	⊕⊕⊕○ MODERATE	IMPORTANT
1 study (15) N = 24 329 ever-married women	Observational/ natural experiment	Serious ^a	Cannot assess	Not serious	Not serious	Not serious	Secondary school stipend programme	• Impact on increasing girls’ self-efficacy ¹⁵	⊕○○○ VERY LOW	IMPORTANT

RCT: randomized controlled trial.

Explanations – certainty assessment

^a Risk of bias: Quality of counterfactual/isolation of exposure (15).

Explanations (numbering matches superscripts in table AND references in list) – results

2. No statistically significant treatment effect found for parental perception of girls’ self-efficacy
15. Impact found on increasing girls’ autonomy among cohorts of women who benefited from the national scholarship programme
 - Eligible girls experienced relatively greater autonomy and were able to make their own decisions about their health care, visit relatives outside of the home and make their own purchases of household goods (0.039 [SE = 0.021], $P < 0.1$).

PICO question 1.4/outcome 4: Do interventions to improve the availability and/or quality of educational opportunities for girls and young women (outcome 4) result in harms and/or unintended consequences?

No articles were identified that met the PICO inclusion criteria for this outcome.

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PICO 1.5 Interventions to improve the economic empowerment of girls through savings, bundled services, demand-driven job services, childcare services, rural electrification, land rights, microcredit, business management training, networks and mentors and/or integrated services for farming

Table WA.1.5(i) PICO question 1.5/outcome 1: Do interventions to improve the economic empowerment of girls through savings, bundled services, demand-driven job services, childcare services, rural electrification, land rights, microcredit, business management training, networks and mentors and/or integrated services for farming (outcome 1) **reduce child marriage?**

No. of studies/N	Study design	Certainty assessment					Interventions to improve the economic empowerment of girls	Results	Certainty	Importance
		Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations				
1 study (1) N = 3080 girls	RCT	Serious ^{a,b}	Cannot assess	Not serious	Not serious	Not serious	Financial supports and services interventions – girls' clubs in safe spaces with provision of information and resources, including savings accounts, life skills and livelihood training, and microfinance access	1 study showed no significant impact on reducing child marriage¹	⊕○○○ VERY LOW	CRITICAL
1 study (2) N = 949 girls	Quasi-experimental	Very serious ^{a,c,d,e,f}	Not serious	Not serious	Not serious	Not serious	Financial supports and services interventions – safe spaces and other resources provided for socialization and cognitive development; combined with income-generation skills and microfinance access	1 study showed impact on reducing child marriage²	⊕○○○ VERY LOW	CRITICAL
4 studies (3-6) N≈13 320 girls	RCTs	Serious ^{a,d}	Not serious	Not serious	Not serious	Not serious	Human capital development interventions (financial literacy/business management training)	3 studies showed impact on reducing child marriage³⁻⁵ 1 study showed no impact on child marriage⁶	⊕⊕⊕○ MODERATE	CRITICAL

Table WA.1.5(i) PICO question 1.5/outcome 1 (continued)

No. of studies/N	Study design	Certainty assessment					Interventions to improve the economic empowerment of girls	Results	Certainty	Importance
		Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations				
3 studies (2,7,8) N≈7118 youth	Quasi-experimental	Very serious ^{a,c,d,g}	Not serious	Serious ^{h,i}	Not serious	Not serious	Same as above	1 study showed impact on delaying marriage for girls, but not reducing marriage before age 18⁷ 2 studies showed impact on reducing child marriage^{2,8}	⊕○○○ VERY LOW	CRITICAL
3 studies (1,5,9) N≈6507 girls	RCTs	Serious ^{a,b}	Not serious	Not serious	Not serious	Not serious	Social infrastructure interventions (childcare, networks and mentors, job services, career resources)	2 studies showed impact on reducing child marriage^{5,9} 1 study showed no significant impact on reducing child marriage¹	⊕⊕⊕○ MODERATE	CRITICAL
3 studies (2,7,8) N≈6199 youth	Quasi-experimental	Very serious ^{a,c,d,e,f,g}	Not serious	Serious ^{h,i}	Not serious	Not serious	Same as above	3 studies showed impact on reducing child marriage^{2,7,8} 1 study showed that reduction was limited to younger adolescents⁷	⊕○○○ VERY LOW	CRITICAL
1 study (10) N = 5003 women	Observational/natural experiment	Serious ^j	Cannot assess	Serious ^k	Not serious	Not serious	Same as above	1 study showed impact on reducing child marriage¹⁰	⊕○○○ VERY LOW	CRITICAL

RCT: randomized controlled trial.

Explanations – certainty assessment

- ^a Risk of bias: Migration may have caused significant differences in populations captured in earlier and later rounds of data collection (1,2,4,7).
- ^b Risk of bias: Appropriateness of the sample (1).
- ^c Risk of bias: Absence of an adequate comparison group (2).
- ^d Risk of bias: Attrition bias related to selective dropout of participants who systematically differed from those who remained in the study (2,4).
- ^e Risk of bias: Timing of baseline evaluation (2).
- ^f Risk of bias: Potential spillover effects (2).

- ^g Risk of bias: Insufficient information provided on sampling methods and/or composition of comparison groups (balance table not presented) (8).
- ^h Indirectness: Exposure to all components of a multi-component intervention makes it difficult to ascertain which programme strategies or activities were most influential for achieving programme impact (4,7,8).
- ⁱ Indirectness: Analysis presented did not directly respond to the PICO question – results on this indicator were presented for intervention/control comparison at endline only (8).
- ^j Risk of bias: Quality of counterfactual/isolation of exposure (10).
- ^k Indirectness: Difficulty attributing impact to programme due to other changes/programmes operating in the area during the study period (10).

Explanations (numbering matches superscripts in table AND references in list) – results

1. No impact found on reducing child marriage
 - The intervention did not have an impact on reducing child marriage (DID coefficient: 0.083, 95% CI: -0.200-0.366).
2. Impact on reducing child marriage
 - According to the model that did not control for covariates, exposure to ELA Centre significantly reduced the probability of marriage. However, when other relevant variables were considered, participation in ELA Centre became a weaker determinant of the probability of getting married. Age of the adolescent girls and education of their household heads were significant factors in models that included them as covariates. When the interaction variable of centre participation and age was introduced, the interaction variable was positively associated with the probability of marriage and participation was negatively associated with the probability of marriage.
 - According to the model that did not control for covariates, exposure to ELA Centre significantly reduces the probability of marriage (robustness z-statistic 2.30, significant at 5%). However, when other relevant variables were considered, participation in ELA Centre became a weaker determinant of the probability of getting married (1.80, significant at 10%; 1.97, significant at 5%). Age of the adolescent girls and education of their household heads were significant factors in models that included them as covariates. When the interaction variable of centre participation and age was introduced, the interaction variable was positively associated with the probability of marriage and participation was negatively associated with the probability of marriage.
3. Impact on reducing child marriage
 - The programme reduced child marriage (< 18 years) significantly in all arms relative to control (AHR 0.75; 95% CI: 0.60–0.92, $P = 0.007$), including the livelihoods arm (AHR 0.70; 95% CI: 0.56–0.87; $P = 0.001$).
4. Impact on reducing child marriage
 - The midline ITT estimate shows girls in treated communities to be 69 percentage points less likely to be married/cohabiting at follow-up, corresponding to 53% of the baseline mean. In control communities, marriage rates for adolescent girls rose naturally from 12% to 18% from baseline to follow-up, and the evidence suggests this is almost entirely prevented from happening by the programme in intervention communities. This divergence in trends continues to endline, at which point they are 8 percentage points less likely to be married or cohabiting.
 - Baseline prevalence = 0.129 (SD = 0.335); change at endline = -0.08 (SD = 1.015).
5. Impact on reducing child marriage
 - Beneficiaries of both programmes were 3.6 percentage points (or 46%) less likely to be married compared with the control group (AOR 0.036 [SD = 0.014]).
 - GE + (including CCTs, in addition to livelihoods training) reduced the likelihood of marriage by more than 50% (AOR 0.041 [SD = 0.014]).
6. No statistically significant impact on reducing child marriage in the overall sample
 - Prevalence of child marriage within the last 12 months decreased from 35% at baseline to 32% at endline in the intervention arm (change not significant).
 - No difference in child marriage detected across intervention arms (AOR 0.72, 95% CI: 0.46–1.15).
7. Impact on delaying marriage for girls
 - Prevalence of child marriage reduced among younger adolescents, but increased among older adolescents: among girls aged 10–14 years, those exposed to the programme were less likely to have ever been married (OR 0.09, $P < 0.001$). However, among girls aged 15–19 years, those in the intervention area had an elevated likelihood of being married by the endline (OR 2.4, $P < 0.001$).
8. Impact on reducing child marriage
 - At baseline, the mean age at marriage was 15.9 years, and more than 50% of girls in the sample were married in the two years prior to the start of DISHA, nearly 60% before the age of 18. In contrast, only 40% of the 198 girls married during the two-year DISHA programme were younger than 18 years at marriage. By endline, the average age at marriage among girls surveyed had increased by nearly two years to 17.9 years.
9. Decreased probability of marriage among girls
 - Girls and young women aged 15–21 years at baseline from treatment villages were 5.1 percentage points less likely to get married during the three year period of the study.
10. Decreased probability of marriage among girls
 - Girls living in villages within commuting distance of garment factories, which had been operating for 6.4 years (which was the average exposure in the garment-proximate villages across all units of observation in the regression) had a ($6.4 \times 0.00048 =$) 0.3 percentage point lower probability of getting married by that year relative to girls living in control villages in the same district that were not within commuting distance to garment factories.

Table WA.1.5(ii) PICO question 1.5/outcomes 2, 3 and 4: Do interventions that improve the economic empowerment of girls through savings, bundled services, demand-driven job services, childcare services, rural electrification, land rights, microcredit, business management training, networks and mentors and/or integrated services for farming (outcome 2) **increase initiation and/or continuation of gainful employment (including self-employment)**, (outcome 3) **increase girls' access to income and assets** and (outcome 4) **increase girls' control of and benefit from economic gains**?

No. of studies/N	Study design	Certainty assessment					Interventions to improve the economic empowerment of girls	Results	Certainty	Importance
		Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations				
3 studies (1,4,9) N≈9765 girls and young women	RCTs	Serious ^{a,b}	Not serious	Not serious	Not serious	Not serious	Financial literacy, layering of microfinance and information about employment opportunities	1 study showed impact on increasing economic assets (financial literacy, saved money)¹ 2 studies showed impact on increasing income-generating activities/paid employment among girls^{4,9}	⊕⊕⊕○ MODERATE	IMPORTANT
1 study (10) N≈5003 women	Observational/natural experiment	Serious ^c	Cannot assess	Serious ^d	Not serious	Not serious	New employment opportunities in the garment industry	1 study showed impact on increasing female labour force participation¹⁰	⊕○○○ VERY LOW	IMPORTANT

RCT: randomized controlled trial.

Explanations – certainty assessment

- ^a Risk of bias: Migration may have caused significant differences in populations captured in earlier and later rounds of data collection (1,4).
- ^b Risk of bias: Appropriateness of the sample (1).
- ^c Risk of bias: Attrition bias related to selective dropout of participants who systematically differed from those who remained in the study (4).
- ^d Indirectness: Difficulty attributing impact to programme due to other changes/programmes operating in the area during the study period (10).

Explanations (numbering matches superscripts in table AND references in list) – results

- Increased economic assets (financial literacy, saved money)
 - Financial literacy score saw a modest increase of 0.24 points out of 9 (95% CI: 0.01–0.48), and girls who had saved money in the past year increased by 6.7 percentage points (95% CI: 1.8–11.5). Two years after the end of the intervention, improved outcomes relative to the control were still observed in saving money (DID coefficient 6.7, 95% CI: 2.0–11.3).
- Increase in income-generating activities among girls
 - Eligible girls were more likely to be engaged in income-generating activities by 6.8 percentage points (67%, SD = 0.16) at midline and 4.9 percentage points (48%, SD = 0.02) at endline.
- Increase in paid employment
 - In intervention villages three years after baseline, women aged 18–24 years were 4.6 percentage points more likely to be employed in the Business Process Outsourcing (BPO) sector than similarly aged women in control villages (for whom BPO employment was close to 0).
- Increased female labour force participation
 - Girls who were exposed to factory jobs when they were aged 10–23 years (which is the critical age group at risk for early marriage in Bangladesh) were 17 percentage points more likely to have done wage work outside the home (79% increase over the control group).

Table WA.1.5(iii) PICO question 1.5/outcomes 5 and 6: Do interventions that improve the economic empowerment of girls through savings, bundled services, demand-driven job services, childcare services, rural electrification, land rights, microcredit, business management training, networks and mentors and/or integrated services for farming (outcome 5) **influence social and gender norms and attitudes about the potential for girls to contribute financially to the household through the attainment of education and employment** and/or (outcome 6) **increase autonomy of girls (e.g. their influence on the timing of their marriage, their choice of partner, their economic autonomy, financial decision-making)?**

No. of studies/N	Study design	Certainty assessment					Interventions to improve the economic empowerment of girls	Results	Certainty	Importance
		Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations				
1 study (1) N = 3080 girls	RCTs	Serious ^{a,b}	Cannot assess	Not serious	Not serious	Not serious	Financial supports and services interventions – Girls' clubs in safe spaces with provision of information and resources, including savings accounts, life skills and livelihood training, and microfinance access	• 1 study showed impact in increasing girls' self-efficacy ¹	⊕⊕○○ LOW	IMPORTANT
1 study (11) N = 1852 adolescent girls	Observational/natural experiment	Not serious	Cannot assess	Not serious	Not serious	Not serious	Gender quota in leadership policy for elected positions on village councils as a means of increasing the aspirations of women and girls	• 1 study showed impact on increasing girls' aspirations ¹¹	⊕⊕○○ LOW	IMPORTANT

RCT: randomized controlled trial.

Explanations – certainty assessment

^a Risk of bias: Migration may have caused significant differences in populations captured in earlier and later rounds of data collection (1).

^b Risk of bias: Appropriateness of the sample (1).

Explanations (numbering matches superscripts in table AND references in list) – results

1. Sustained impact on increased self-efficacy
 - Positive but modest programme impacts emerged in self-efficacy with a 0.31 higher score out of 10 (95% CI: 0.02–0.6).
11. Increase in girls' aspirations
 - Adolescent girls in villages with a female leader for two election cycles were more likely to not want to be a housewife or have their occupation determined by their in-laws (8.3 percentage points; $P = 0.053$).

PICO question 1.5/outcome 7: Do interventions that improve the economic empowerment of girls through savings, bundled services, demand-driven job services, childcare services, rural electrification, land rights, microcredit, business management training, networks and mentors and/or integrated services for farming (outcome 7) **result in harms and/or unintended consequences?**

No articles were identified that met the PICO inclusion criteria for this outcome.

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PICO 1.6 Child marriage prevention laws in line with those recommended by international treaty bodies, such as the United Nations Convention on the Rights of the Child (CRC) and the Convention on the Elimination of All Forms of Discrimination against Women (CEDAW)

Table WA.1.6(i) PICO question 1.6/outcome 1: Does the existence of child marriage prevention laws in line with those recommended by international treaty bodies, such as the CRC and CEDAW (outcome 1) **reduce child marriage?**

No. of studies	Study design	Certainty assessment					Results	Certainty	Importance
		Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations			
3 studies (1-3) N≈276 486	Observational/ natural experiment	Not serious	Not serious	Serious ^a	Not serious	Not serious	Results showed that in some cases, the implementation of laws on minimum age at marriage brought benefits in reducing the prevalence of child marriage, but that the observed impacts appear to have only been among subsets of the population or were not sustained over time. 1 multi-country study found reductions in child marriage rates in two countries during the period immediately following introduction of child marriage laws, but not in later cohorts; no reductions were observed in the other four countries¹ 1 study found a significant reduction in child marriage following introduction of a law on age at marriage, but only in the younger subset of the population (i.e. marriage under age 16)² 1 study found no impact of introduction of laws on age at marriage on the prevalence of child marriage in the overall sample, but noted significant declines in child marriage in urban areas³	⊕○○○ VERY LOW	CRITICAL

Explanations – certainty assessment

^a Indirectness: Difficulty attributing impact to the programme due to other changes/programmes operating in the area during the study period (1,2,3).

Explanations (numbering matches superscripts in table AND references in list) – results

1. Child marriage rates declined in two countries during the period immediately following introduction of child marriage laws, but not in later cohorts; no declines in the rate of child marriage were observed in other four countries included in the study following introduction of age at marriage laws in those settings

- Results suggest that changes in laws on the minimum age at marriage were not effective in curbing early marriage in Benin, Bhutan, Kazakhstan and Mauritania, where child marriage showed little evidence of decline across cohorts. Significant reductions in early marriage following implementation of laws were observed in Nepal and Tajikistan, yet their effectiveness depended on the model specification (i.e. no controls, additional continuous variable of women's age at the time of the law implementation in months, or additional categorical variables for women's age, place and region of residence) and window (i.e. width of the interval that determines which women around the cut-off defining exposure to the law are included in the analysis) adopted.
- 2. Significant reduction in child marriage following the introduction of the marriage age law, but only in the younger subset of the population (marriage under age 16)
- Marriages of girls under 16 were delayed by about 17% (6.8 percentage points) following implementation of a law on the minimum age at marriage in areas where early marriage was more common prior to the reform. However, the effect of the reform, though larger, was statistically

Table WA.1.6(i) PICO question 1.6/outcome 1 (continued)

- insignificant among women belonging to ethnic groups with the strongest norms favouring child marriage.
3. Introduction of laws on age at marriage had no impact on the prevalence of child marriage in the overall sample, but significant declines in child marriage were seen in urban areas following the introduction of the law, suggesting that laws may have differential impact among subsets of the population
 - Impact observed for high “ban intensity” areas. “Ban intensity” refers to the difference between the age up to which child marriage is banned and the mean age at marriage at the time of the ban in that area – the further the mean age at marriage is from the age limit specified in the law, the higher the “ban intensity”.
 - Rural and urban difference observed: bans were more effective in urban areas; this was interpreted as a result of stronger enforcement of child marriage laws in urban areas.

PICO question 1.6/outcomes 2 and 3: Does the existence of child marriage prevention laws in line with those recommended by international treaty bodies, such as the CRC and CEDAW (outcome 2) **increase awareness of and/or support for child marriage prevention laws** and/or (outcome 3) **result in harms and/or unintended consequences?**

No articles were identified that met the PICO inclusion criteria for these outcomes.

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PICO 1.7 Interventions to mobilize political, governmental, religious, traditional and/or other leaders

Table WA.1.7(i) PICO question 1.7/outcome 1: Do interventions to mobilize political, governmental, religious, traditional, and/or other leaders (outcome 1) **reduce child marriage**?

No. of studies/N	Study design	Certainty assessment					Intervention to mobilize political, governmental, religious, traditional and/or other influential leaders	Results	Certainty	Importance
		Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations				
1 study (1) N = 1124 girls; N = 531 adults	Longitudinal CRCT	Very serious ^a	Cannot assess	Serious ^b	Not serious	Not serious	Engaging religious leaders, teachers, and community leaders through community sensitizations to raise awareness about harms of child marriage	1 study found no significant impact on child marriage prevalence¹	⊕○○○ VERY LOW	CRITICAL
1 study (2) N = 2500 girls	Quasi-experimental	Serious ^a	Serious ^{c,d}	Serious ^e	Not serious	Serious ^f	Engaging local leaders in community dialogues to raise awareness about harms of child marriage	1 study found a significant decrease in child marriage in the community dialogue arm in 1 of 3 settings²	⊕○○○ VERY LOW	CRITICAL
1 study (3) N = 18 089 women	Observational/ natural experiment	Not serious	Cannot assess	Serious ^e	Serious ^{g,h}	Serious ^{i,j}	Exposure to female political leadership	1 study found a significant reduction in child marriage, increased age at first marriage, and delayed gauna ceremony³	⊕○○○ VERY LOW	CRITICAL

CRCT: cluster randomized controlled trial.

Explanations – certainty assessment

- ^a Risk of bias: Selection bias related to systematic differences between intervention and control groups (1,2).
- ^b Indirectness: Leadership engagement component not isolated (1).
- ^c Inconsistency: Intervention results varied across settings (2).
- ^d Inconsistency: Results varied across age groups for single-country studies (2).
- ^e Indirectness: Interventions did not involve direct mobilization of leaders for the purpose of impacting child marriage/other PICO outcomes (2,3).
- ^f Other considerations: Study does not report results for aggregate adolescent sample (2).
- ^g Imprecision: Difficulty attributing impact to programme due to other changes/programmes operating in the area during the study period (3).
- ^h Imprecision: Potential spillover effects (3).
- ⁱ Other considerations: Study not able to address potential longer-term impacts/time to follow-up (3).
- ^j Other considerations: Difficulty measuring causality due to the inherent nature of the topic (e.g. complexity of social norms and factors influencing norms change) (3).

Explanations (numbering matches superscripts in table AND references in list) – results

1. No significant change was found, as child marriage prevalence remained low at endline: overall (6.05%) and in all study arms (< 6.33%).
2. Results differed by setting
 - Burkina Faso: The impact on marriage age among younger girls aged 12–14 years was not examined, as too few girls were married at both baseline (12 married girls) and endline (5 married girls). Girls aged 15–17 years residing in the community dialogue arm had over two thirds less risk (RR = 0.33) of being married compared to those in the control site.
 - Ethiopia: Among girls aged 12–14 years, those residing in the community dialogue site at endline had over half the risk of being married compared to their counterparts at baseline (RR = 0.42, 95% CI: 0.25–0.70).
 - United Republic of Tanzania: The comprehensive model was significantly associated with a delay in marriage among girls aged 12–14 years. Girls aged 12–14 years residing in the comprehensive site had two thirds less risk of being married compared to girls in the control site (RR = 0.33).
3. On average, women who married after the first election with seats reserved for female candidates were significantly less likely to be married at age <18 years relative to women who married prior to the policy in the intervention districts. Women married 2.3–2.8 years older and delayed their gauna ceremony by 1.6–2 years. Women who married after the first woman Pradhan was elected were on average 19–24 percentage points less likely to be married at age <18 years. While slightly smaller in magnitude, the results were robust to controlling for women's age rank within the household, household size, schooling of each spouse, household income, assets, caste and religion.

PICO question 1.7/outcomes 2 and 3: Do interventions to mobilize political, governmental, religious, traditional, and/or other leaders (outcome 2) increase development of supportive laws and policies and/or (outcome 3) increase implementation of supportive laws and policies?

No articles were identified that met the PICO inclusion criteria for these outcomes.

Table WA.1.7(ii) PICO question 1.7/outcome 4: Do interventions to mobilize political, governmental, religious, traditional, and/or other leaders (outcome 4) **increase supportive social norms related to the value of preventing child marriage and commitment to acting to prevent child marriage?**

No. of studies/N	Study design	Certainty assessment					Intervention to mobilize political, governmental, religious, traditional and other influential leaders	Results	Certainty	Importance
		Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations				
1 study (1) N = 1124 girls; N = 531 adults	Longitudinal CRCT	Very serious ^a	Cannot assess	Serious ^b	Not serious	Not serious	Engaging religious leaders, teachers and community leaders through community sensitization to raise awareness about harms of child marriage	Significant increases in girls' agency, social norms and perceptions of gender discrimination¹	⊕○○○ VERY LOW	IMPORTANT
2 studies (3,4) N = 1852 girls; N = 4773 adults; N = 18 089 women	Observational/ natural experiment	Not serious	Not serious	Serious ^c	Serious ^{d,e}	Serious ^{f,g}	Exposure to female leadership	1 study found impact in increasing the likelihood of wanting to marry after age 18⁴ 1 study found impact in decreasing the gender gap in aspirations among parents and adolescents³	⊕○○○ VERY LOW	IMPORTANT

CRCT: cluster randomized controlled trial.

Explanations – certainty assessment

- ^a Risk of bias: Selection bias related to systematic differences between intervention and control groups (1).
^b Indirectness: Leadership engagement component not isolated (1).
^c Indirectness: Interventions did not involve direct mobilization of leaders for the purpose of impacting child marriage/other PICO outcomes (3,4).
^d Imprecision: Difficulty attributing impact to programme due to other changes/programmes operating in the area during the study period (3).
^e Imprecision: Potential spillover effects (3).
^f Other considerations: Study not able to address potential longer-term impacts/time to follow-up (3,4).
^g Other considerations: Nature of topic inherently difficult to measure (e.g. complexity of social norms and factors influencing norms change) (3,4).

Explanations (numbering matches superscripts in table AND references in list) – results

- Significant increases in 10 of 15 outcomes related to girls' agency, social norms and perceptions of gender discrimination.
- Gender gap in aspirations closed by 25% among parents and 32% among adolescents in villages assigned to a female leader.
- Girls in intervention areas were more likely to want to marry after 18 years (8.8 percentage points; $P = 0.053$).

PICO question 1.7/outcome 5: Do interventions to mobilize political, governmental, religious, traditional, and/or other leaders (outcome 5) **result in harms and/or unintended consequences?**

No articles were identified that met the PICO inclusion criteria for this outcome.

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PICO 1.8 Interventions to improve the health and social well-being of married girls

Table WA.1.8(i) PICO question 1.8/outcome 1: Do interventions to improve the health and social well-being of married girls (outcome 1) **mitigate the impact of child marriage on health outcomes and mental health?**

No. of studies/N	Study design	Certainty assessment					Intervention to improve the health of married girls	Results	Certainty	Importance
		Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations				
1 study (1) N = 2670 married adolescent girls and young women aged 15–29	RCT	Serious ^a	Cannot assess	Serious ^b	Not serious	Not serious	Interactive group sessions and awareness-raising campaigns addressing sexual harassment, rape, domestic violence and other forms of violence, and information about how to access health and legal services	1 study showed significant impact on reported physical violence¹	⊕○○○ VERY LOW	CRITICAL
4 studies (2-5) N≈7208 married adolescent girls	Quasi-experimental	Serious ^{a,c}	Not serious	Serious ^b	Not serious	Not serious	Community awareness-raising, mass media and education interventions aiming to increase knowledge of SRHR, gender-based violence and related topics among married adolescents and others in the community	SRHR behaviours: 3 studies showed significant impact on improving SRHR behaviours^{2,4} SRHR knowledge and attitudes 2 studies showed significant impact on increasing SRHR knowledge^{2,3} 2 studies showed a significant impact on improving SRHR attitudes^{2,5}	⊕○○○ VERY LOW	CRITICAL

RCT: randomized controlled trial; SRHR: sexual and reproductive health and rights.

Explanations – certainty assessment

^a Risk of bias: Potential spillover effects (1,2,5).

^b Indirectness: Analysis presented does not directly respond to PICO question; outcomes relate primarily to knowledge and attitudes, rather than direct health outcome (1,5).

^c Risk of bias: Absence of an adequate comparison group (5).

Explanations (numbering matches superscripts in table AND references in list) – results

- Significant impact on reducing reported physical violence
 - SRHR and violence against women and girls (VAWG) awareness sessions had significant impact among young women aged 15–19 years, reducing reported physical violence, aRR 0.79 (95% CI: 0.62–0.99; $P < 0.05$).

Table WA.1.8(i) PICO question 1.8/outcome 1 (continued)

2. Significant impact on improving SRHR knowledge, attitudes and behaviours
 - Impact on SRHR knowledge: Increased knowledge of fertile days of the menstrual cycle (OR 2.97, $P < 0.001$).
 - Impact on SRHR attitudes: Increase in agreement that early childbearing is injurious to mother's health (OR 2.61, $P < 0.01$) and agreement that contraceptive use is safe and necessary for delaying first birth (OR 1.57, $P < 0.01$).
 - Impact on contraceptive behaviour: Increased contraceptive demand (OR 1.49, $P < 0.01$) and use (OR 3.84, $P < 0.001$).
3. Significant impact on improving SRHR knowledge and behaviours
 - Results showed that participation in either intervention arm (SRH only or integrated SRH/economic empowerment) significantly improved four of the five SRH outcomes, with the largest gains in the single-focus arm (SRH only).
 - In the SRH arm, significant impact was shown in increasing current use of contraceptives: DID estimate 0.21 (95% CI: 0.14–0.28; $P < 0.001$); proportion who had visited a family planning clinic in the last six months: 0.19 (95% CI: 0.14–0.25; $P < 0.001$); proportion who had discussed family planning with their husbands: 0.04 (95% CI: 0.01–0.08; $P < 0.1$), knowledge of STIs: 0.20 (95% CI: 0.13–0.27; $P < 0.001$); and proportion who had ever been tested for HIV: 0.14 (95% CI: 0.06–0.21; $P < 0.1$).
4. Significant impact on increasing use of family planning
 - Individuals named by participants of the Reaching Married Adolescents (RMA) intervention as important in their lives ("alters") were more likely to have reported ever use of family planning as compared to alters of control respondents (AOR 2.29, 95% CI: 1.07–4.93), but this association was lost when the alter's knowledge of the RMA programme was included in the model.
5. Significant impact on improving attitudes about gender-based violence (GBV), decreasing GBV, and increasing equitable SRH attitudes
 - Inequitable attitudes about GBV scale: –4.7 (95% CI: –9.8 to –0.3; $P < 0.05$).
 - Percent who reacted violently to their partner: –15.7 (95% CI: –27.1 to –4.4; $P < 0.05$).
 - Inequitable SRH attitudes scale: –12.9 (95% CI: –17.3 to –8.5; $P < 0.05$).

Table WA.1.8(ii) PICO question 1.8/outcome 2: Do interventions to improve the health and social well-being of married girls (outcome 2) mitigate the impact of child marriage on social outcomes?

No. of studies/N	Study design	Certainty assessment					Intervention to improve the social well-being of married girls	Results	Certainty	Importance
		Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations				
1 study (6) N = 150 married adolescent girls	RCT	Not serious	Cannot assess	Serious ^a	Not serious	Not serious	Functional analytic psychotherapy with enhanced cognitive therapy sessions focused on sexual efficacy, satisfaction and communication	• 1 study found significant improvement in sexual quality of life among married adolescents ⁶	⊕⊕○○ LOW	CRITICAL
1 study (5) N = 307 newly married/newly parenting girls aged 15–19	Quasi-experimental	Serious ^b	Cannot assess	Not serious	Not serious	Not serious	Multi-component, age-tailored, narrative-based mass media intervention including community mobilization activities, radio and community discussions about gender, violence and SRHR	• 1 study showed significant impact in producing more equitable gender roles ⁵	⊕○○○ VERY LOW	CRITICAL

RCT: randomized controlled trial; SRHR: sexual and reproductive health and rights.

Explanations – certainty assessment

- ^a Indirectness: Analysis presented does not directly respond to PICO question; insufficient information provided on components of outcome indices (6).
- ^b Risk of bias: Potential spillover effects (5).

Explanations (numbering matches superscripts in table AND references in list) – results

- Significant impact shown in producing more equitable gender roles
 - Among married adolescents, intervention effects include shifts on the “inequitable household roles scale” score: -6.5 (95% CI: -10.8 to -2.2 ; $P < 0.05$).
- Significant improvement in sexual quality of life among married adolescent women
 - The paired t-test showed a significant difference between the mean score of sexual quality of life before (52.33 ± 23.09) and after (88.08 ± 10.51) counselling in the intervention group ($P < 0.0001$).
 - According to the analysis of covariance, there was a significant difference between the score on sexual quality after counselling between the intervention (88.08 ± 10.51) and control (60.32 ± 23.73) groups ($P < 0.0001$).

PICO question 1.8/outcomes 3 and 4: Do interventions to improve the health and social well-being of married girls (outcome 3) improve the health, education and/or well-being of the children of the girl/young woman and/or (outcome 4) result in harms and/or unintended consequences?

No articles were identified that met the PICO inclusion criteria for these outcomes.

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PICO 1.9 Meaningful engagement of adolescents in the design, implementation and/or monitoring and evaluation of programmes

Table WA.1.9(i) PICO question 1.9/outcome 1: Does meaningful engagement of adolescents in the design, implementation and/or monitoring and evaluation of programmes (outcome 1) **reduce child marriage?**

No. of studies/N	Study design	Certainty assessment					Intervention: Meaningful engagement of adolescents in the design, implementation and/or monitoring and evaluation of programmes	Results	Certainty	Importance
		Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations				
2 studies (1,2) N = 2247 girls; N = 531 adults	RCTs	Serious ^a	Not serious	Not serious	Not serious	Not serious	Adolescents involved in implementation of programme: girl movement-building; girl-led community activities and social-norms-change events	• 2 studies found non-significant impact on reducing child marriage ^{1,2}	⊕⊕⊕○ MODERATE	CRITICAL
1 study (3) N = 15 576 girls aged 15–17	RCT	Serious ^b	Cannot assess	Serious ^{c,d}	Not serious	Not serious	Adolescents involved in implementation of programme: peer education	• 1 study found a non-significant reduction in child marriage in the arm including a peer education component ³	⊕○○○ VERY LOW	CRITICAL
1 study (4) N = 4323 youth; N = 1562 adults	Quasi-experimental	Serious ^e	Cannot assess	Serious ^{c,f,g}	Not serious	Not serious	Adolescents involved in implementation of programme: peer education	• 1 study found reduction in child marriage, but no intervention-attributable effect ⁴	⊕○○○ VERY LOW	CRITICAL

RCT: randomized controlled trial.

Explanations – certainty assessment

^a Risk of bias: Selection bias related to systematic differences between intervention and control groups (1,2).

^b Risk of bias: Lack of adherence to design; results not presented for study arms as described in the trial registry (AEA RCT Registry (socialscienceregistry.org); results not shown for livelihood or savings components (3).

^c Indirectness: Youth engagement component not isolated (3,4).

^d Indirectness: Difficulty attributing impact to programme due to time since exposure (3).

^e Risk of bias: Insufficient information provided on sampling methods and/or composition of comparison groups (balance table not presented) (4).

^f Indirectness: Exposure to all components of a multi-component intervention makes it difficult to ascertain which programme strategies or activities were most influential for achieving programme impact (4).

^g Indirectness: Analysis presented does not directly respond to the PICO question; results on this indicator were presented for intervention/control comparison at endpoint only (4).

Explanations (numbering matches superscripts in table AND references in list) – results

1. Non-significant impact on reducing child marriage
 - No impact on child marriage was detected in any of the arms of the trial (TPP vs control: AHR 1.14, 95% CI: 0.79–1.63, $P = 0.47$; TPP+ vs control: AHR 1.24, 95% CI: 0.89–1.71, $P = 0.19$; TPP vs TPP+: AHR 1.03, 95% CI: 0.72–1.47, $P = 0.87$).
2. Non-significant impact on reducing child marriage
 - No significant change was found and child marriage prevalence remained low at endline: overall (6.05%) and in all study arms (< 6.33%).
3. Non-significant reduction in child marriage in the intervention arm including a peer education component
 - The empowerment arm was not found to have significantly reduced child marriage. For girls married by age 15 at endline: OLS regression coefficient 0.002 (Huber-White robust SE = 0.012). For girls married between ages 15 and 18: OLS regression coefficient was 0.006 (Huber-White robust SE = 0.008).
4. Reduction in child marriage, but no intervention-attributable effect
 - Age at marriage increased by nearly two years to just shy of 18; however, a significant programme effect was not found on child marriage prevalence.
 - At baseline, the mean age at marriage was 15.9, and more than 50% of girls in the sample were married in the two years prior to the start of DISHA, nearly 60% before the age of 18. In contrast, only 40% of the 198 girls married during the two-year DISHA programme were younger than 18 at marriage. By endline, the average age at marriage among girls surveyed had increased by nearly two years to 17.8 years.

PICO question 1.9/outcomes 2, 3 and 4: Does meaningful engagement of adolescents in the design, implementation and/or monitoring and evaluation of programmes (outcome 2) **increase community engagement in efforts to prevent child marriage**, (outcome 3) **increase acceptability and appropriateness of programmes** and/or (outcome 4) **result in harms and/or unintended consequences?**

No articles were identified that met the PICO inclusion criteria for these outcomes.

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2. GRADE tables for PICO questions related to adolescents' use of contraception and related outcomes

PICO 2.1 Interventions to (a) empower girls and change their perceptions, (b) shift social norms and behaviours among parents/guardians/the broader community and (c) shift social norms and behaviours among boys (peers/partners) and/or men (fathers/partners)

Table WA.2.1(i) PICO questions 2.1a, b and c/outcome 1: Do interventions to (2.1a) empower girls to make decisions about their fertility and contraceptive use, and to change their perceptions regarding the role and status of women and girls, or to (2.1b and c) shift social norms and behaviours regarding adolescent fertility and contraceptive decision-making, as well as wider social norms (such as the role and status of women and girls), among (2.1b) parents/guardians/the broader community or (2.1c) boys (as peers and/or partners) and/or men (as partners or fathers) (outcome 1) **increase adolescents' new or continued use of a modern contraceptive method, their contraceptive method of choice, and/or dual methods and/or (for 2.1c only) male methods?**

Certainty assessment							No. of patients		Effect		Certainty	Importance
No. of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Interventions to empower girls and to change social norms and behaviours	No intervention	Relative (95% CI)	Absolute (95% CI)		
Use of a modern contraceptive method or contraceptive method of choice (assessed with: any modern contraceptive use)												
1 (1) ^a	RCT	Not serious	Serious ^b	Not serious ^c	Not serious	None	NR/81	NR/158	OR 4.53 (1.60 to 12.87)		⊕⊕⊕○ MODERATE	CRITICAL

Table WA.2.1(i) PICO questions 2.1a, b and c/outcome 1 (continued)

Certainty assessment							No. of patients		Effect		Certainty	Importance
No. of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Interventions to empower girls and to change social norms and behaviours	No intervention	Relative (95% CI)	Absolute (95% CI)		
Use of a modern contraceptive method or contraceptive method of choice (assessed with: consistent condom use)												
1 (2) ^d	RCT	Serious ^e	Serious ^b	Not serious ^c	Not serious	None	26/135 (19.3%)	25/181 (13.8%)	OR 0.93 (0.83 to 1.03)	8 fewer per 1000 (from 21 fewer to 4 more)	⊕⊕○○ LOW	CRITICAL
Use of a modern contraceptive method or contraceptive method of choice (assessed with: any use of modern contraceptives in the last three months [female or male condom, hormonal pills, injections, implants or IUDs], among those ever sexually active)												
1 (3)	RCT	Not serious	Serious ^b	Not serious ^c	Not serious	None	138/545 (25.3%)	141/566 (24.9%)	RR 1.00 (0.81 to 1.24)	0 fewer per 1000 (from 47 fewer to 60 more)	⊕⊕⊕○ MODERATE	CRITICAL
Use of a modern contraceptive method or contraceptive method of choice (assessed with: any use of modern contraceptives at last sex [female or male condom, hormonal pills, injections, implants or IUDs], among those sexually active in the past four weeks)												
1 (3)	RCT	Not serious	Serious ^b	Not serious ^c	Not serious	None	92/143 (64.3%)	87/166 (52.4%)	RR 1.26 (1.06 to 1.50)	136 more per 1000 (from 31 more to 262 more)	⊕⊕⊕○ MODERATE	CRITICAL

CI: confidence interval; IUD: intrauterine devices; NR: not reported; OR: odds ratio; RCT: randomized controlled trial; RR: risk ratio.

Explanations – certainty assessment and/or results

- ^a Results for combined intervention compared with control: Additional increase in any modern contraceptive use was 17.9% (95% CI: 5.3%–30.5%), $P = 0.005$. For Intervention 1 (home visits alone), it was 11.5% (95% CI: –0.4%–23.4%), AOR 4.11 (95% CI: 1.42–11.88), $P = 0.009$. For Intervention 2 (small group discussions alone), it was 35.6% (95% CI: 22.2%–49.1%), AOR 7.94 (95% CI: 2.96–21.29), $P < 0.001$. All AORs were statistically significant.
- ^b Inconsistency: Cannot assess, single study.
- ^c Indirectness: Though the intervention was multi-component and results cannot be linked to specific PICO-related intervention components, we did not downgrade for indirectness because social norms interventions by nature need to work through multiple ways.
- ^d Condom use was also reported for last vaginal penetration (AOR 1.12, 95% CI: 1.01–1.25, $P = 0.02$), last anal penetration (AOR 1.09, 95% CI: 0.88–1.35, $P = 0.40$), last oral sex episode (AOR 1.08, 95% CI: 0.96–1.21, $P = 0.19$), and percentage condom use (AOR 7.52, 95% CI: 0.41–16.62, $P = 0.03$).
- ^e Risk of bias: Downgraded once because the total sample size for the six-month follow-up condom use outcome was not reported, but it appears to have been substantially smaller than the total sample size reported (even for sexually active participants).

PICO questions 2.1a, b and c/outcome 2: Do interventions to (2.1a) empower girls to make decisions about their fertility and contraceptive use, and to change their perceptions regarding the role and status of women and girls, or to (2.1b) shift social norms and behaviours regarding adolescent fertility and contraceptive decision-making, as well as wider social norms (such as the role and status of women and girls), among (2.1b) parents/guardians/the broader community or (2.1c) boys (as peers and/or partners) and/or men (as partners or fathers) (outcome 2) **increase access to any modern method of contraception or to a range of contraceptive methods?**

No articles were identified that met the PICO inclusion criteria for this outcome.

Table WA.2.1(ii) PICO questions 2.1a/outcome 3: Do interventions to empower girls to make decisions about their fertility and contraceptive use, and to change their perceptions regarding the role and status of women and girls (outcome 3) **increase agency and/or autonomy to use contraception and make decisions about contraceptive use?**

Certainty assessment							No. of patients		Effect		Certainty	Importance
No. of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Interventions to empower girls	No intervention	Relative (95% CI)	Absolute (95% CI)		
Agency/autonomy (assessed with: control beliefs regarding obtaining contraceptives)												
1 (3)	RCT	Not serious	Serious ^a	Not serious ^b	Not serious	None	1207/1663 (72.6%)	1184/1723 (68.7%)	RR 1.04 (0.98 to 1.11)	27 more per 1000 (from 14 fewer to 76 more)	⊕⊕⊕○ MODERATE	IMPORTANT
Agency/autonomy (assessed with: control beliefs regarding condom use)												
1 (3)	RCT	Not serious	Serious ^a	Not serious ^b	Not serious	None	1380/1657 (83.3%)	1400/1720 (81.4%)	RR 1.02 (0.98 to 1.06)	16 more per 1000 (from 16 fewer to 49 more)	⊕⊕⊕○ MODERATE	IMPORTANT

CI: confidence interval; RCT: randomized controlled trial; RR: risk ratio.

Explanations – certainty assessment

^a Inconsistency: Cannot assess consistency, since it is a single study.

^b Indirectness: Though the intervention was multi-component and results cannot be linked to specific PICO-related intervention components, we did not downgrade for indirectness because social norms interventions by nature need to work through multiple ways.

Table WA.2.1(iii) PICO questions 2.1b and c/outcome 3 for 2.1b and outcome 2 for 2.1c: Do interventions to shift social norms and behaviours regarding adolescent fertility and contraceptive decision-making, as well as wider social norms (such as the role and status of women and girls), among (2.1b) parents/guardians/the broader community or (2.1c) boys (as peers and/or partners) and/or men (as partners or fathers) (outcome 3 for 2.1b) **increase supportive social norms and/or beliefs regarding contraceptive use and/or contraceptive decision-making among adolescents or (outcome 2 for 2.1c) **increase knowledge, attitudes and/or supportive beliefs and/or social norms of boys/men regarding contraceptive use among adolescents, shared contraceptive decision-making and consent?****

Certainty assessment							No. of patients		Effect		Certainty	Importance
No. of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Interventions to change social norms and behaviours	No intervention	Relative (95% CI)	Absolute (95% CI)		
Supportive norms (assessed with: percentage of first-time mothers aged 15–19 who strongly agree that most people expect them to use postpartum contraception in the six weeks following childbirth)												
1 (4)	Observational study	Not serious	Serious ^a	Not serious ^b	Not serious	None	77/487 (15.8%)	54/439 (12.3%)	RR 1.28 (0.93 to 1.77)	34 more per 1000 (from 9 fewer to 95 more)	⊕○○○ VERY LOW	IMPORTANT
Supportive norms (assessed with: percentage of first-time mothers aged 15–19 who strongly agree that most people expect them to discuss use of postpartum contraception in the six weeks following childbirth with their husband/partner before the baby is born)												
1 (4)	Observational study	Not serious	Serious ^a	Not serious ^b	Not serious	None	106/487 (21.8%)	60/439 (13.7%)	RR 1.59 (1.19 to 2.13)	81 more per 1000 (from 26 more to 154 more)	⊕○○○ VERY LOW	IMPORTANT
Supportive norms (assessed with: percentage of first-time mothers aged 15–19 who believe that most [more than half or all] new mothers in the community use postpartum contraception within the six weeks following childbirth)												
1 (4)	Observational study	Not serious	Serious ^a	Not serious ^b	Not serious	None	103/487 (21.1%)	70/439 (15.9%)	RR 1.32 (1.01 to 1.75)	51 more per 1000 (from 2 more to 120 more)	⊕○○○ VERY LOW	IMPORTANT
Supportive norms (assessed with: percentage of first-time mothers aged 15–19 who believe that most [more than half or all] new mothers in the community discuss postpartum contraception with their husband/partner before the baby is born)												
1 (4)	Observational study	Not serious	Serious ^a	Not serious ^b	Not serious	None	100/487 (20.5%)	86/439 (19.6%)	RR 1.05 (0.81 to 1.36)	10 more per 1000 (from 37 fewer to 71 more)	⊕○○○ VERY LOW	IMPORTANT

CI: confidence interval; RR: risk ratio.

Explanations – certainty assessment

- ^a Inconsistency: Cannot assess consistency, since it is a single study.
- ^b Indirectness: Though the intervention was multi-component and results cannot be linked to specific PICO-related intervention components, we did not downgrade for indirectness because social norms interventions by nature need to work through multiple ways.

PICO questions 2.1b and c/outcome 4: Do interventions to shift social norms and behaviours regarding adolescent fertility and contraceptive decision-making, as well as wider social norms (such as the role and status of women and girls), among (2.1b) parents/guardians/the broader community or (2.1c) boys (as peers and/or partners) and/or men (as partners or fathers) (outcome 4) **decrease reproductive coercion?**

No articles were identified that met the PICO inclusion criteria for this outcome.

Table WA.2.1(iv) PICO question 2.1c/outcome 3: Do interventions to shift social norms and behaviours regarding adolescent fertility and contraceptive decision-making, as well as wider social norms (such as the role and status of women and girls) among boys (as peers and/or partners) and/or men (as partners or fathers) (outcome 3) **increase agency/self-efficacy of girls to negotiate with partner, obtain contraception and/or refuse sex?**

Certainty assessment							No. of patients		Effect		Certainty	Importance
No. of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Interventions to change social norms and behaviours	No intervention	Relative (95% CI)	Absolute (95% CI)		
Agency/self-efficacy (assessed with: contraceptive self-efficacy – unmarried adolescents)												
1 (5)	Observational study	Serious ^a	Serious ^b	Not serious ^c	Not serious	None	Observed: 48.4% Counter-factual (using propensity score matching to baseline): 37.8% Intervention effect: 10.6% (95% CI: 5.8 to 15.4, <i>P</i> < 0.05)				⊕○○○ VERY LOW	IMPORTANT
Agency/self-efficacy (assessed with: contraceptive self-efficacy – newly married/newly parenting adolescents)												
1 (5)	Observational study	Serious ^a	Serious ^b	Not serious ^c	Not serious	None	Observed: 67.5% Counter-factual (using propensity score matching to baseline): 59.2% Intervention effect: 8.3% (95% CI: 4.2 to 12.4, <i>P</i> < 0.05)				⊕○○○ VERY LOW	IMPORTANT
Agency/self-efficacy (assessed with: embarrassed to get condoms – out-of-school adolescents)												
2 (6,7)	Observational studies	Not serious	Serious ^b	Not serious ^c	Not serious	None	Intervention: 61.27%, Control: 64.75%, not statistically significant				⊕○○○ VERY LOW	IMPORTANT
Agency/self-efficacy (assessed with: embarrassed to get condoms – in-school adolescents)												
2 (6,7)	Observational studies	Not serious	Serious ^b	Not serious ^c	Not serious	None	Intervention: 75.20%, Control: 73.33%, not statistically significant				⊕○○○ VERY LOW	IMPORTANT
Agency/self-efficacy (assessed with: embarrassed to get contraception – out-of-school adolescent girls)												
2 (6,7)	Observational studies	Not serious	Serious ^b	Not serious ^c	Not serious	None	Intervention: 48.10%, Control: 65.79%, OR 0.36 (95% CI: 0.16 to 0.86)				⊕○○○ VERY LOW	IMPORTANT
Agency/self-efficacy (assessed with: embarrassed to get contraception – in-school adolescent girls)												
2 (6,7)	Observational studies	Not serious	Serious ^b	Not serious ^c	Not serious	None	Intervention: 51.44%, Control: 51.83%, not statistically significant				⊕○○○ VERY LOW	IMPORTANT
Agency/self-efficacy (assessed with: percentage of first-time mothers aged 15–19 who believe that they have total control over the decision to use a postpartum contraception method)												
1 (4)	Observational study	Not serious	Serious ^b	Not serious ^c	Not serious	None	201/487 (41.3%)	159/439 (36.2%)	RR 1.14 (0.97 to 1.34)	51 more per 1000 (from 11 fewer to 123 more)	⊕○○○ VERY LOW	IMPORTANT
Agency/self-efficacy (assessed with: mean postpartum contraception self-efficacy score among first-time mothers aged 15–19 [score range: 7 to 28])												
1 (4)	Observational study	Not serious	Serious ^b	Not serious ^c	Not serious	None	Intervention: 19.3, Control: 18.6, MD = 0.9				⊕○○○ VERY LOW	IMPORTANT

Table WA.2.1(iv) PICO question 2.1c/outcome 3 (continued)

			Certainty assessment				No. of patients		Effect		Certainty	Importance
No. of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Interventions to change social norms and behaviours	No intervention	Relative (95% CI)	Absolute (95% CI)		
Agency/self-efficacy (assessed with: percentage of first-time mothers aged 15–19 who would use postpartum contraception even if all five named referents did not want them to)												
1 (4)	Observational study	Not serious	Serious ^b	Not serious ^c	Not serious	None	189/487 (38.8%)	160/439 (36.4%)	RR 1.06 (0.90 to 1.26)	22 more per 1000 (from 36 fewer to 95 more)	⊕○○○ VERY LOW	IMPORTANT

CI: confidence interval; MD: mean difference; OR: odds ratio; RR: risk ratio.

Explanations – certainty assessment

- ^a Risk of bias: Downgraded because the original design included an intervention and matched control group, but contamination (high levels of intervention exposure) in the control villages meant the authors changed their design to create a propensity-score matched “counterfactual” comparison group from baseline data.
- ^b Inconsistency: Cannot assess consistency, since it is a single study.
- ^c Indirectness: Though the intervention was multi-component and results cannot be linked to specific PICO-related intervention components, we did not downgrade for indirectness because social norms interventions by nature need to work through multiple ways.

PICO questions 2.1a/outcome 4 and 2.1b and c/outcome 5: Do interventions to (2.1a) empower girls to make decisions about their fertility and contraceptive use, and to change their perceptions regarding the role and status of women and girls, or to (2.1b) shift social norms and behaviours regarding adolescent fertility and contraceptive decision-making, as well as wider social norms (such as the role and status of women and girls), among (2.1b) parents/guardians/the broader community or (2.1c) boys (as peers and/or partners) and/or men (as partners or fathers) (outcome 4 or 5) **result in harms and/or unintended consequences?**

No articles were identified that met the PICO inclusion criteria for these outcomes.

References for all tables for PICO 2.1

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7. Impact of Growing Up GREAT! on the lives of very young adolescents: one and two years after program implementation: research brief. La Jolla, CA: Center on Gender Equity and Health at the University of California San Diego for the Bill & Melinda Gates Foundation; 2022 (<https://resourcecentre.savethechildren.net/document/impact-of-growing-up-great-on-the-lives-of-very-young-adolescents-one-and-two-years-after-program-implementation/>).

PICO 2.2 Interventions to expand opportunities for self-care, including availability of hormonal contraceptives over the counter, DMPA-SC and contraceptives direct to consumer

PICO question 2.2/outcomes 1,2,3 and 4: Do the following interventions to expand opportunities for self-care – availability of (i) hormonal contraceptives (including emergency contraception) over the counter (i.e. without prescription), (ii) subcutaneous depot medroxyprogesterone acetate (DMPA-SC) and training on self-injection, and (iii) contraceptives direct to consumer – (outcome 1) **increase adolescents' new or continued use of a modern contraceptive method or their contraceptive method of choice**, (outcome 2) **increase access to any modern method of contraception or a range of contraceptive methods**, (outcome 3) **improve the quality of care (as perceived by adolescents) and/or** (outcome 4) **result in harms and/or unintended consequences?**

No articles were identified that met the PICO inclusion criteria for these outcomes.

PICO 2.3 Interventions to address financial barriers to using contraceptives

Table WA.2.3(i) PICO question 2.3/outcome 1: Do the following interventions to address financial barriers to contraceptives – availability of vouchers, free and/or reduced cost of contraceptives, social marketing and/or social franchising, and insurance-based programmes – (outcome 1) **increase adolescents' new or continued use of a modern contraceptive method or their contraceptive method of choice?**

Certainty assessment							No. of patients		Effect		Certainty	Importance
No. of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Interventions to address financial barriers to contraceptives	Standard practice	Relative (95% CI)	Absolute (95% CI)		
New or continued use of a modern contraceptive method (assessed with: use of long-acting reversible contraceptives [LARCs])												
1 (1)	Observational study	Not serious ^a	Serious ^b	Not serious	Serious ^c	None	0/24 (0.0%)	0/24 (0.0%)	Not estimable		⊕○○○ VERY LOW	CRITICAL
New or continued use of a contraceptive method of choice (assessed with: patient left with a contraceptive method)												
1 (2)	Observational study	Not serious ^a	Serious ^b	Not serious	Serious ^d	None	16/17 (94.1%)	8/16 (50.0%)	RR 1.88 (1.14 to 3.12)	440 more per 1000 (from 70 more to 1000 more)	⊕○○○ VERY LOW	CRITICAL
New or continued use of a contraceptive method of choice (assessed with: self-reported use of modern contraceptive methods [IUD, sterilization, oral or injectable contraceptives] among girls who are sexually active, not pregnant, and do not want to be pregnant in the coming year)												
1 (3)	Observational study	Not serious ^a	Serious ^b	Not serious	Not serious	None	Intervention: 50%, Control: 50%, AOR 1.33 (95% CI: 0.77 to 2.29)				⊕○○○ VERY LOW	CRITICAL
New or continued use of a contraceptive method of choice (assessed with: condom use at last sexual contact among girls who are sexually active, not pregnant, and do not want to be pregnant in the coming year)												
1 (3)	Observational study	Not serious ^a	Serious ^b	Not serious	Not serious	None	Intervention: 23%, Control: 20%, AOR 1.84 (95% CI: 1.11 to 3.03)				⊕○○○ VERY LOW	CRITICAL

AOR: adjusted odds ratio; CI: confidence interval; IUD: intrauterine device; LARC: long-acting reversible contraceptive; OR: odds ratio; RR: risk ratio.

Explanations – certainty assessment

^a Risk of bias: Not downgraded for detection bias. Blinding was not possible given the nature of the intervention. Detection bias was unlikely as the outcome was judged unlikely to have been affected by lack of blinding.

^b Inconsistency: Cannot assess consistency, since it is a single study.

^c Imprecision: Small sample size of adolescents; no outcomes in either study arm.

^d Imprecision: Small sample size; unit of analysis is clinics, with different standardized patients² visiting clinics before and during the voucher programme.

² Standardized patients are individuals trained to act as real patients in order to simulate a set of symptoms or problems for education, training or monitoring/evaluation purposes. They are also sometimes called simulated patients, sample patients or patient instructors.

Table WA.2.3(ii) PICO question 2.3/outcome 3: Do the following interventions to address financial barriers to contraceptives – availability of vouchers, free and/or reduced cost of contraceptives, social marketing and/or social franchising, and insurance-based programmes – (outcome 3) **improve the quality of care (as perceived by adolescents)?**

Certainty assessment							No. of patients		Effect		Certainty	Importance
No. of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Interventions to address financial barriers to contraceptives	Standard practice	Relative (95% CI)	Absolute (95% CI)		
Quality of care (assessed with: four domains of satisfaction measured with 21 individual items) ^a												
1 (2)	Observational study	Not serious ^b	Serious ^c	Not serious	Serious ^d	None	A. Contraception, choice of method: Before: 5.4; After: 7.4 B. Contraception, continuity of use: Before: 5.6; After: 7.9 C. STI/HIV prevention: Before: 4.5; After: 6.0 D. Organization of the clinic: Before: 7.9; After: 8.1 Statistical comparisons all > 0.05 except for A4. “Decides on best method with the girl” (P = 0.02) and B1 “Patient left with a contraceptive method” (P < 0.01)				⊕○○○ VERY LOW	IMPORTANT
Quality of care (assessed with: overall satisfaction [would consult same doctor again])												
1 (4) ^e	Observational study	Serious ^f	Serious ^c	Not serious	Not serious	None	199/221 (90.0%)	394/479 (82.3%)	OR 2.23 (1.24 to 4.02)	89 more per 1000 (from 29 more to 127 more)	⊕○○○ VERY LOW	IMPORTANT
Quality of care (assessed with: doctors’ explanations were clear)												
1 (4) ^e	Observational study	Serious ^f	Serious ^c	Not serious	Not serious	None	181/221 (81.9%)	370/479 (77.2%)	OR 1.37 (0.87 to 2.17)	51 more per 1000 (from 25 fewer to 108 more)	⊕○○○ VERY LOW	IMPORTANT
Quality of care – All adolescents (assessed with: satisfied with the reception)												
1 (4) ^g	Observational study	Serious ^f	Serious ^c	Not serious	Not serious	None	190/221 (86.0%)	374/479 (78.1%)	OR 1.99 (1.22 to 3.26)	96 more per 1000 (from 32 more to 140 more)	⊕○○○ VERY LOW	IMPORTANT
Quality of care – Adolescents who have not had sexual debut (assessed with: satisfied with the reception)												
1 (4) ^e	Observational study	Very serious ^{f,h}	Serious ^c	Not serious	Not serious	None	NR/62	NR/106	OR 2.22 (0.87 to 5.63)		⊕○○○ VERY LOW	IMPORTANT
Quality of care – Adolescents with sexual debut who are not pregnant or mothers (assessed with: satisfied with the reception)												
1 (4) ^e	Observational study	Very serious ^{f,h}	Serious ^c	Not serious	Not serious	None	NR/65	NR/99	OR 6.93 (1.51 to 31.84)		⊕○○○ VERY LOW	IMPORTANT

Table WA.2.3(ii) PICO question 2.3/outcome 3 (continued)

			Certainty assessment				No. of patients		Effect		Certainty	Importance
No. of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Interventions to address financial barriers to contraceptives	Standard practice	Relative (95% CI)	Absolute (95% CI)		
Quality of care – Adolescents who are pregnant or mothers (assessed with: satisfied with the reception)												
1 (4) ^e	Observational studies	Very serious ^{fh}	Serious ^c	Not serious	Not serious	None	NR/91	NR/271	OR 1.13 (0.58 to 2.20)		⊕○○○ VERY LOW	IMPORTANT

CI: confidence interval; OR: odds ratio; RR: risk ratio.

Explanations – certainty assessment and results

- ^a Measured with 21 individual items in the following categories: (1) Contraception, choice of method (five items); (2) Contraception, continuity in use (five items); (3) STI/HIV prevention (six items); (4) Organization of clinic (five items).
- ^b Risk of bias: Not downgraded for detection bias. Blinding was not possible given the nature of the intervention. Detection bias was unlikely as the outcome was judged unlikely to have been affected by lack of blinding.
- ^c Inconsistency: Cannot assess consistency, since it is a single study.
- ^d Imprecision: Small sample size; unit of analysis is clinics, with different standardized patients visiting clinics before and during the voucher programme.

- ^e Odds ratio (OR) adjusted for category of childbearing experience, age group, school attendance, level of educational attainment, refrigerator in the house, number of people per bedroom, socioeconomic classification of the survey site, survey site, self-filled or completed by an interviewer.
- ^f Risk of bias: Downgraded once for detection bias. Blinding was not possible given the nature of the intervention. Detection bias was possible as data were self-reported and may have been affected by lack of blinding.
- ^g OR is unadjusted (no adjusted data were presented).
- ^h Risk of bias: Downgraded once for potential selective reporting bias. Disaggregation by population subgroups was only conducted for this outcome without explanation.

PICO question 2.3/outcomes 2 and 4: Do the following interventions to address financial barriers to contraceptives – availability of vouchers, free and/or reduced cost of contraceptives, social marketing and/or social franchising, and insurance-based programmes – (outcome 2) **increase access to any modern method of contraception or to a range of contraceptive methods,** and/or (outcome 4) **result in harms and/or unintended consequences?**

No articles were identified that met the PICO inclusion criteria for these outcomes.

References for all tables for PICO 2.3

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PICO 2.4 Digital interventions for clients, including targeted and untargeted communication to persons, person-to-person communication, and on-demand communication with persons

Table WA.2.4(i) PICO question 2.4/outcome 1: Do digital targeted and untargeted communication to persons (outcome 1) increase adolescents' new or continued use of a modern contraceptive method or their contraceptive method choice?

Certainty assessment			No. of patients				Effect		Certainty	Importance		
No. of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Targeted and untargeted communication to persons	No intervention	Relative (95% CI)	Absolute (95% CI)		
Use of a modern contraceptive method or a contraceptive method of choice (assessed with: current use [or partner’s use] of effective contraception [oral contraceptive pills, IUDs, injectables and implants]) ^a												
1 (1)	RCT	Serious ^b	Serious ^c	Not serious	Very serious ^d	None	2/128 (1.6%)	1/128 (0.8%)	RR 2.00 (0.18 to 21.78)	8 more per 1000 (from 6 fewer to 162 more)	⊕○○○ VERY LOW	CRITICAL
Knowledge (assessed with: knowledge of effective contraception)												
1 (1)	RCT	Serious ^b	Serious ^c	Not serious	Serious ^e	None	4.08	4		MD 0.08 score higher (0.29 lower to 0.44 higher)	⊕○○○ VERY LOW	IMPORTANT
Knowledge (assessed with: contraception myths and misconceptions)												
1 (2)	RCT	Serious ^f	Serious ^c	Not serious	Not serious	None	Adjusted mean difference in scores between baseline and endline comparing push notifications to control arm: -3.68% (95% CI: -6.14% to -1.22%), <i>P</i> = 0.003				⊕⊕○○ LOW	IMPORTANT
Agency (assessed with: personal agency in using contraception [women only])												
1 (1)	RCT	Serious ^b	Serious ^c	Not serious	Serious ^g	None	30/68 (44.1%)	24/56 (42.9%)	RR 1.03 (0.69 to 1.54)	13 more per 1000 (from 133 fewer to 231 more)	⊕○○○ VERY LOW	IMPORTANT
Agency (assessed with: personal agency in communicating with partners about contraception)												
1 (1)	RCT	Serious ^b	Serious ^c	Not serious	Not serious	None	82/127 (64.6%)	83/126 (65.9%)	RR 0.98 (0.82 to 1.17)	13 fewer per 1000 (from 119 fewer to 112 more)	⊕⊕○○ LOW	IMPORTANT

CI: confidence interval; IUD: intrauterine device; MD: mean difference; RCT: randomized controlled trial; RR: risk ratio.

Explanations – certainty assessment and results

- ^a This outcome was also measured as use in the four-month study period vs current use; one fewer participant in the intervention arm reported contraceptive use in this measure, and the total sample size was 127 in the intervention and 126 in the control group, creating an RR of approximately 1.0.
- ^b Risk of bias: Downgraded once because the control group experienced contamination by being able to see intervention content on the app, contrary to what was planned in the trial protocol.
- ^c Inconsistency: Cannot assess consistency, since it is a single study.

- ^d Imprecision: Downgraded twice because there were extremely few events in either arm and RR is > 1.25 or < 0.8, which includes both appreciable benefit and harm.
- ^e Imprecision: Downgraded once because mean difference includes both appreciable benefit and harm.
- ^f Risk of bias: Downgraded once for detection bias. Blinding was not possible given the nature of the intervention. Detection bias was possible as data were self-reported and may have been affected by lack of blinding.
- ^g Imprecision: Downgraded once because RR is > 1.25 or < 0.8, which includes both appreciable benefit and harm.

PICO question 2.4/outcomes 2, 3 and 4: Do digital targeted and untargeted communication to persons (outcome 2) **increase correct knowledge about methods of contraception and their side-effects/return to fertility or about sources of contraception**, (outcome 3) **increase agency to negotiate with partner, obtain contraception or manage side-effects**, and/or (outcome 4) **result in harms and/or unintended consequences?**

No articles were identified that met the PICO inclusion criteria for these outcomes.

PICO question 2.4/outcomes 1–4: Do digital person-to-person communications (outcome 1) **increase adolescents' new or continued use of a modern contraceptive method or their contraceptive method choice**, (outcome 2) **increase correct knowledge about methods of contraception and their side-effects/return to fertility or about sources of contraception**, (outcome 3) **increase agency to negotiate with partner, obtain contraception or manage side-effects**, and/or (outcome 4) **result in harms and/or unintended consequences?**

No articles were identified that met the PICO inclusion criteria for these outcomes.

Table WA.2.4(ii) PICO question 2.4/outcome 1: Do digital on-demand information services to clients (outcome 1) increase adolescents' new or continued use of a modern contraceptive method or their contraceptive method choice?

Certainty assessment							No. of patients		Effect		Certainty	Importance
No. of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	On-demand information services to clients	No intervention	Relative (95% CI)	Absolute (95% CI)		
Knowledge (assessed with: knowledge of contraceptives)												
1 (3)	RCT	Not serious	Serious ^a	Not serious	Not serious	None	Mean difference in scores from baseline to endline comparing intervention to control group, <i>P</i> = 0.06.			⊕⊕⊕○ MODERATE	IMPORTANT	
Knowledge (assessed with: contraception myths and misconceptions)												
1 (2)	RCT	Serious ^b	Serious ^a	Not serious	Not serious	None	Adjusted mean difference in scores between baseline and endline comparing on-demand information arm to control arm: -1.40% (95% CI: -3.80% to 1.00%), <i>P</i> = 0.254			⊕⊕○○ LOW	IMPORTANT	
Agency (assessed with: if someone your age wanted to buy condoms, how hard would it be for him or her to get a condom at a low cost? Percent very/somewhat easy)												
1 (4)	Observational study	Not serious	Serious ^a	Not serious	Not serious	None	In Nairobi, Kenya: Intervention: 70.33%, Control: 47.31% In Rio de Janeiro, Brazil: Intervention: 61.15%, Control: 57.89% No tests of statistical significance presented			⊕○○○ VERY LOW	IMPORTANT	

CI: confidence interval; RCT: randomized controlled trial.

Explanations – certainty assessment

^a Inconsistency: Cannot assess consistency, since it is a single study.

^b Risk of bias: Downgraded once for detection bias. Blinding was not possible given the nature of the intervention. Detection bias was possible as data were self-reported and may have been affected by lack of blinding.

PICO question 2.4/outcomes 2, 3 and 4: Do digital on-demand information services to clients (outcome 2) **increase correct knowledge about methods of contraception and their side-effects/return to fertility or about sources of contraception**, (outcome 3) **increase agency to negotiate with partner, obtain contraception or manage side-effects**, and/or (outcome 4) **result in harms and/or unintended consequences?**

No articles were identified that met the PICO inclusion criteria for these outcomes.

References for all tables for PICO 2.4

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3. Macharia P, Pérez-Navarro A, Sambai B, Inwani I, Kinuthia J, Nduati R et al. An unstructured supplementary service data-based mHealth app providing on-demand sexual reproductive health information for adolescents in Kibra, Kenya: randomized controlled trial. *JMIR mHealth and uHealth*. 2022;10(4):e31233 (<https://doi.org/10.2196/31233>).
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PICO 2.5 Interventions to mobilize political, religious, traditional and/or other leaders

PICO question 2.5: Do interventions directed at political, religious, and traditional leaders, as well as other influential leaders and groups in the community (outcome 1) **increase adolescents' new or continued use of a modern contraceptive method, their contraceptive method of choice, and/or dual methods**, (outcome 2) **increase supportive beliefs and/or social norms regarding contraceptive use among adolescents** and/or (outcome 3) **result in harms and/or unintended consequences?**

No articles were identified that met the PICO inclusion criteria for these outcomes.

PICO 2.6 Interventions to improve the following aspects of quality of services: appropriate package of services; provider competencies and bias; facility characteristics; equity and non-discrimination; and data and quality improvement

Table WA.2.6 PICO question 2.6: Do interventions to improve the following aspects of quality of services (drawn from WHO's *Global standards for quality health-care services for adolescents*)³ – appropriate package of services; provider competencies and bias; facility characteristics; equity and non-discrimination; and data and quality improvement – (outcome 1) **increase adolescents' new or continued use of a modern contraceptive method or their contraceptive method of choice?**

Certainty assessment							No. of patients		Effect		Certainty	Importance
No. of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Interventions to improve service quality	Standard practice	Relative (95% CI)	Absolute (95% CI)		
Use of a modern contraceptive method or a contraceptive method of choice (assessed with: condom use in last 12 months)												
1 (1)	Observational study	Serious ^a	Serious ^b	Serious ^c	Not serious	None	31/47 (66.0%)		RR 0.96 (0.73 to 1.27)	27 fewer per 1000 (from 185 fewer to 185 more)	⊕○○○ VERY LOW	CRITICAL
							Results					
Quality of care (assessed with: composite scores of overall facility performance using WHO quality assessment survey ⁴ [%])												
1 (2)	Observational study	Not serious	Serious ^b	Not serious	Not serious	None	Baseline: 27% Endline 5 years later: 83%				⊕○○○ VERY LOW	IMPORTANT
Quality of care (assessed with: composite scores by standard of service provision using WHO quality assessment survey [%])												
1 (2)	Observational study	Not serious	Serious ^b	Not serious	Not serious	None	Baseline: 28% Endline 5 years later: 81%				⊕○○○ VERY LOW	IMPORTANT
Use of a modern contraceptive method or a contraceptive method of choice (assessed with: monthly average of number of contraception clients aged 15–19 years)												
1 (3)	RCT	Not serious	Serious ^b	Not serious	Not serious	None	• United Republic of Tanzania: Intervention: 22.988, Control: 23.693, MD = -0.704 • Pakistan: Intervention: 4.856, Control: 2.665, MD = 2.192 (<i>P</i> < 0.01) • Burkina Faso: Intervention: 18.288, Control: 14.923, MD = 3.365 • Pooled difference: 1.614				⊕⊕⊕○ MODERATE	CRITICAL

³ World Health Organization, UNAIDS. Global standards for quality health-care services for adolescents: a guide to implement a standards-driven approach to improve the quality of health care services for adolescents. Geneva: World Health Organization; 2015 (<https://iris.who.int/handle/10665/183935>).

⁴ A survey developed by the authors of the study with the WHO Regional Office for South-East Asia (unpublished resource).

Table WA.2.6 PICO question 2.6 (continued)

Certainty assessment							No. of patients	Effect		Certainty	Importance	
No. of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Interventions to improve service quality	Standard practice	Relative (95% CI)			Absolute (95% CI)
							Results					
Use of a modern contraceptive method or a contraceptive method of choice (assessed with: number of clients using a short- or long-acting contraceptive method)												
1 (4)	Observational study	Serious ^d	Serious ^b	Not serious	Not serious	None	Number of contraception clients aged 15–19 using short- or long-acting methods increased from 40 to 153 per quarter during the intervention period, then decreased to 102 after activities ended.			⊕○○○ VERY LOW	CRITICAL	
Use of a modern contraceptive method or a contraceptive method of choice (assessed with: monthly average of contraception client visits among adolescents aged 10–14 years)												
1 (5)	Observational study	Not serious	Serious ^b	Not serious	Not serious	None	• Core+ group: increased number of monthly clients by 0.02 compared to baseline • Core group: increased number of monthly clients by 0.02 compared to baseline			⊕○○○ VERY LOW	CRITICAL	
Use of a modern contraceptive method or a contraceptive method of choice (assessed with: monthly average of contraception client visits among adolescents aged 15–19 years)												
1 (5)	Observational study	Not serious	Serious ^b	Not serious	Not serious	None	• Core+ group: increased average number of monthly clients by 8.6 compared to baseline • Core group: increased average number of monthly clients by 3.5 compared to baseline			⊕○○○ VERY LOW	CRITICAL	
Quality of care (assessed with: 5 items)												
1 (6)	Observational study	Not serious	Serious ^b	Serious ^c	Not serious	None	• Comfortable enough to clarify all the discomforts: Intervention: 91.2%*, Control: 84.7% • Enough auditory privacy during consultation: Intervention: 85.4%***, Control: 48.9% • Enough visual privacy during consultation: Intervention: 85.7%***, Control: 44.4% • Felt confident about disclosure of services received: Intervention: 84.7%***, Control: 59.8% • Had to wait long before getting any services: Intervention: 27.6%***, Control: 54.2%			⊕○○○ VERY LOW	IMPORTANT	
Use of a modern contraceptive method or a contraceptive method of choice (assessed with: average number of services delivered to adolescent clients per week per centre [combining IUD, short-term methods and implants])												
1 (7)	Observational study	Not serious	Serious ^b	Serious ^e	Not serious	None	• Pre-intervention: 3 • During intervention: 11 • Post-intervention: 16			⊕○○○ VERY LOW	CRITICAL	

Table WA.2.6 PICO question 2.6 (continued)

			Certainty assessment				No. of patients		Effect		Certainty	Importance
No. of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Interventions to improve service quality	Standard practice	Relative (95% CI)	Absolute (95% CI)		
							Results					
Use of a modern contraceptive method or a contraceptive method of choice (assessed with: number of women receiving contraceptive services [monthly average])												
1 (8)	Observational study	Not serious	Serious ^b	Serious ^f	Not serious	None	• Baseline: Intervention: 209, Control: 124 • Follow-up: Intervention: 70, Control: 41 ^g				⊕○○○ VERY LOW	CRITICAL

CI: confidence interval; MD: mean difference; RCT: randomized controlled trial; RR: risk ratio.

Key: * = $P < 0.05$, ** = $P < 0.01$

Explanations – certainty assessment and results

- ^a Risk of bias: Downgraded for potential selective reporting bias as the outcome measure described in the methods (condom use in the last six months) is different from the outcome measure reported in the results (condom use in the last 12 months).
- ^b Inconsistency: Cannot assess consistency, since it is a single study.
- ^c Indirectness: Interventions to improve service quality were just one part of a multi-component intervention.
- ^d Risk of bias: Downgraded for potential temporal bias as the total number of clients changed over time but the outcome reported was just the number of clients selecting these contraceptive methods (no total number of clients was presented); also, the start and end dates of the intervention were not clear.

- ^e Indirectness: Intervention to improve the service quality (facility characteristics) was one of many other intervention components so the researchers could not tease out specific effects.
- ^f Indirectness: Downgraded because outcome of number of contraceptive service delivery clients is not the same as contraceptive use.
- ^g A reduction in the average number of women seen for contraceptive services in both intervention and comparison clinics was attributed to COVID-19 and a provider strike.

PICO question 2.6: Do interventions to improve the following aspects of quality of services (drawn from *WHO's Global standards for quality health-care services for adolescents*) – appropriate package of services; provider competencies and bias; facility characteristics; equity and non-discrimination; and data and quality improvement – (outcome 2) **increase access to any modern method of contraception or to a range of contraceptive methods**, (outcome 3) **improve the quality of care (as objectively assessed and as perceived by adolescents)** and/or (outcome 4) **result in harms and/or unintended consequences?**

No articles were identified that met the PICO inclusion criteria for these outcomes.

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PICO 2.7 Enabling laws and policies on the following issues related to sexual activity, access to sexual and reproductive health (SRH) services, and access to specific contraceptive methods: age, marital status, and consent procedures

Table WA.2.7 PICO question 2.7/outcomes 1 and 2: Does the existence and/or proper application and implementation of enabling laws and policies on the following issues related to sexual activity, access to SRH services, and access to specific contraceptive methods – age, marital status and consent procedures – (outcomes 1 and 2) **increase adolescents’ new or continued use of a modern contraceptive method or their contraceptive method of choice?**

Certainty assessment							Results	Certainty	Importance
No. of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations			
Use of modern contraceptive method or contraceptive method of choice (assessed with: contraceptive use among married adolescent girls aged 15–19)									
1 (1)	Observational study	Serious ^a	Serious ^b	Serious ^c	Not serious	None	3.0% in 2000 vs 36.4% in 2019	⊕○○○ VERY LOW	CRITICAL
Use of modern contraceptive method or contraceptive method of choice (assessed with: postpartum contraceptive use among married adolescent girls aged 15–19)									
1 (1)	Observational study	Serious ^a	Serious ^b	Serious ^c	Not serious	None	8.6% in 2005 vs 46.3% in 2016	⊕○○○ VERY LOW	CRITICAL

CI: confidence interval.

Explanations – certainty assessment

^a Risk of bias: Downgraded because of many potential confounding variables in the time period assessed; effects are unlikely to be attributed to the laws and policies alone.

^b Inconsistency: Cannot assess consistency, since it is a single study.

^c Indirectness: Age of consent (law/policy of interest) was one part of a multi-component law.

PICO question 2.7/outcomes 3,4 and 5: Does the existence and/or proper application and implementation of enabling laws and policies on the following issues related to sexual activity, access to SRH services, and access to specific contraceptive methods – age, marital status and consent procedures – (outcome 3) **increase access to any modern method of contraception or to a range of contraceptive methods**, (outcome 4) **increase agency and/or autonomy to use contraception and make decisions about contraceptive use** and/or (outcome 5) **result in harms and/or unintended consequences?**

No articles were identified that met the PICO inclusion criteria for these outcomes.

References for PICO 2.7

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PICO 2.8 Interventions to meaningful engagement of adolescents in the design, implementation, and/or monitoring of programmes

PICO question 2.8: Does meaningful engagement of adolescents in the design, implementation and/or monitoring of programmes (outcome 1) **increase adolescents' new or continued use of a modern contraceptive method and/or their contraceptive method of choice**, (outcome 2) **increase access to any modern method of contraception or to a range of contraceptive methods**, (outcome 3) **improve the quality of care (as perceived by adolescents)** and/or (outcome 4) **result in harms and/or unintended consequences?**

No articles were identified that met the PICO inclusion criteria for these outcomes.

3. Evidence-To-Decision tables

Table WA.3.1a Preventing child marriage and responding to the needs and rights of married girls

Recommendation	Recommendation 1.1	Recommendation 1.2	Recommendation 1.3	Recommendation 1.4	Recommendation 1.5
Intervention	Interventions to empower girls by building their knowledge, skills, assets and social networks	Engagement with parents/guardians, boys and men, and the broader community to create and sustain a gender-equitable and enabling environment	Conditional incentives to increase educational attainment and reduce child marriage as a part of social protection interventions for girls at highest risk of child marriage	Interventions to remove gender-related barriers to education and ensure girls' completion of 12 years of quality education	Interventions aimed at the economic empowerment of girls to improve their financial literacy, access to savings, employment skills and prospects, and expand alternatives to marriage before age 18
Problem a priority?	☒ Yes ☐ No ☐ Uncertain ☐ Varies	☒ Yes ☐ No ☐ Uncertain ☐ Varies	☒ Yes ☐ No ☐ Uncertain ☐ Varies	☒ Yes ☐ No ☐ Uncertain ☐ Varies	☒ Yes ☐ No ☐ Uncertain ☐ Varies
Certainty of the evidence	☐ High ☐ Moderate ☒ Low ☐ Very low ☐ Nil	☐ High ☐ Moderate ☒ Low ☐ Very low ☐ Nil	☐ High ☒ Moderate ☐ Low ☐ Very low ☐ Nil	☐ High ☒ Moderate ☐ Low ☐ Very low ☐ Nil	☐ High ☒ Moderate ☐ Low ☐ Very low ☐ Nil
Balance of benefits and harms	☐ Favours intervention ☒ Probably favours intervention ☐ Against intervention ☐ Varies ☐ Uncertain	☐ Favours intervention ☒ Probably favours intervention ☐ Against intervention ☐ Varies ☐ Uncertain	☒ Favours intervention ☐ Probably favours intervention ☐ Against intervention ☐ Varies ☐ Uncertain	☐ Favours intervention ☒ Probably favours intervention ☐ Against intervention ☐ Varies ☐ Uncertain	☐ Favours intervention ☒ Probably favours intervention ☐ Against intervention ☐ Varies ☐ Uncertain

Table WA.3.1a Preventing child marriage and responding to the needs and rights of married girls (continued)

Recommendation	Recommendation 1.1	Recommendation 1.2	Recommendation 1.3	Recommendation 1.4	Recommendation 1.5
Intervention	Interventions to empower girls by building their knowledge, skills, assets and social networks	Engagement with parents/guardians, boys and men, and the broader community to create and sustain a gender-equitable and enabling environment	Conditional incentives to increase educational attainment and reduce child marriage as a part of social protection interventions for girls at highest risk of child marriage	Interventions to remove gender-related barriers to education and ensure girls' completion of 12 years of quality education	Interventions aimed at the economic empowerment of girls to improve their financial literacy, access to savings, employment skills and prospects, and expand alternatives to marriage before age 18
Resource requirements (costs) and cost-effectiveness	☐ Large costs ☐ Moderate costs ☐ Negligible costs and savings ☐ Moderate savings ☐ Large savings ☐ Varies ☒ Uncertain	☐ Large costs ☐ Moderate costs ☐ Negligible costs and savings ☐ Moderate savings ☐ Large savings ☐ Varies ☒ Uncertain	☐ Large costs ☐ Moderate costs ☐ Negligible costs and savings ☐ Moderate savings ☐ Large savings ☐ Varies ☒ Uncertain	☐ Large costs ☐ Moderate costs ☐ Negligible costs and savings ☐ Moderate savings ☐ Large savings ☐ Varies ☒ Uncertain	☐ Large costs ☐ Moderate costs ☐ Negligible costs and savings ☐ Moderate savings ☐ Large savings ☐ Varies ☒ Uncertain
Equity	☐ Increased ☒ Probably increased ☐ Reduced ☐ Varies ☐ Uncertain	☐ Increased ☒ Probably increased ☐ Reduced ☐ Varies ☐ Uncertain	☐ Increased ☒ Probably increased ☐ Reduced ☐ Varies ☐ Uncertain	☐ Increased ☒ Probably increased ☐ Reduced ☐ Varies ☐ Uncertain	☐ Increased ☒ Probably increased ☐ Reduced ☐ Varies ☐ Uncertain
Acceptability	☐ Yes ☒ Probably yes ☐ No ☐ Varies ☐ Uncertain	☐ Yes ☐ Probably yes ☐ No ☒ Varies ☐ Uncertain	☐ Yes ☒ Probably yes ☐ No ☐ Varies ☐ Uncertain	☐ Yes ☒ Probably yes ☐ No ☐ Varies ☐ Uncertain	☐ Yes ☒ Probably yes ☐ No ☐ Varies ☐ Uncertain
Feasibility	☒ Yes ☐ Probably yes ☐ No ☐ Varies ☐ Uncertain	☐ Yes ☒ Probably yes ☐ No ☐ Varies ☐ Uncertain	☐ Yes ☒ Probably yes ☐ No ☐ Varies ☐ Uncertain	☐ Yes ☒ Probably yes ☐ No ☐ Varies ☐ Uncertain	☐ Yes ☒ Probably yes ☐ No ☐ Varies ☐ Uncertain
Enough evidence to make a recommendation?	☒ Yes ☐ No	☒ Yes ☐ No	☒ Yes ☐ No	☒ Yes ☐ No	☒ Yes ☐ No
Recommendation in favour or against the intervention	☒ In favour ☐ Against	☒ In favour ☐ Against	☒ In favour ☐ Against	☒ In favour ☐ Against	☒ In favour ☐ Against
Type of recommendation	☐ Strong recommendation ☒ Conditional recommendation ☐ Good practice statement	☐ Strong recommendation ☒ Conditional recommendation ☐ Good practice statement	☐ Strong recommendation ☒ Conditional recommendation ☐ Good practice statement	☒ Strong recommendation ☐ Conditional recommendation ☐ Good practice statement	☒ Strong recommendation ☐ Conditional recommendation ☐ Good practice statement

Table WA.3.1a (continued)

Recommendation	Recommendation 1.6	Good practice statement 1.1	Good practice statement 1.2	Good practice statement 1.3
Intervention	Formulation and implementation of laws that restrict marriage before age 18	Mobilization of political, governmental, religious, traditional and other influential leaders to support the prevention of child marriage and promotion of girls' rights	Recognizing and addressing the specific needs and rights of ever-married girls and those in formal or informal unions	Meaningful engagement of adolescents, including those who are ever married or in formal or informal unions, in the design, implementation, monitoring and evaluation efforts to address their needs and rights
Problem a priority?	☒ Yes ☐ No ☐ Uncertain ☐ Varies	☒ Yes ☐ No ☐ Uncertain ☐ Varies	☒ Yes ☐ No ☐ Uncertain ☐ Varies	☒ Yes ☐ No ☐ Uncertain ☐ Varies
Certainty of the evidence	☐ High ☐ Moderate ☐ Low ☒ Very low ☐ Nil	☐ High ☐ Moderate ☐ Low ☒ Very low ☐ Nil	☐ High ☐ Moderate ☐ Low ☒ Very low ☐ Nil	☐ High ☐ Moderate ☐ Low ☒ Very low ☐ Nil
Balance of benefits and harms	☐ Favours intervention ☒ Probably favours intervention ☐ Against intervention ☐ Varies ☐ Uncertain	☐ Favours intervention ☒ Probably favours intervention ☐ Against intervention ☐ Varies ☐ Uncertain	☐ Favours intervention ☒ Probably favours intervention ☐ Against intervention ☐ Varies ☐ Uncertain	☐ Favours intervention ☒ Probably favours intervention ☐ Against intervention ☐ Varies ☐ Uncertain
Resource requirements (costs) and cost-effectiveness	☐ Large costs ☐ Moderate costs ☐ Negligible costs and savings ☐ Moderate savings ☐ Large savings ☒ Varies ☐ Uncertain	☐ Large costs ☐ Moderate costs ☐ Negligible costs and savings ☐ Moderate savings ☐ Large savings ☒ Varies ☐ Uncertain	☐ Large costs ☐ Moderate costs ☐ Negligible costs and savings ☐ Moderate savings ☐ Large savings ☒ Varies ☐ Uncertain	☐ Large costs ☐ Moderate costs ☐ Negligible costs and savings ☐ Moderate savings ☐ Large savings ☒ Varies ☐ Uncertain
Equity	☐ Increased ☒ Probably increased ☐ Reduced ☐ Varies ☐ Uncertain	☐ Increased ☒ Probably increased ☐ Reduced ☐ Varies ☐ Uncertain	☐ Increased ☒ Probably increased ☐ Reduced ☐ Varies ☐ Uncertain	☐ Increased ☒ Probably increased ☐ Reduced ☐ Varies ☐ Uncertain

Table WA.3.1a Preventing child marriage and responding to the needs and rights of married girls (continued)

Recommendation	Recommendation 1.6	Good practice statement 1.1	Good practice statement 1.2	Good practice statement 1.3
Intervention	Formulation and implementation of laws that restrict marriage before age 18	Mobilization of political, governmental, religious, traditional and other influential leaders to support the prevention of child marriage and promotion of girls' rights	Recognizing and addressing the specific needs and rights of ever-married girls and those in formal or informal unions	Meaningful engagement of adolescents, including those who are ever married or in formal or informal unions, in the design, implementation, monitoring and evaluation efforts to address their needs and rights
Acceptability	☐ Yes ☒ Probably yes ☐ No ☐ Varies ☐ Uncertain	☐ Yes ☒ Probably yes ☐ No ☐ Varies ☐ Uncertain	☐ Yes ☒ Probably yes ☐ No ☐ Varies ☐ Uncertain	☐ Yes ☒ Probably yes ☐ No ☐ Varies ☐ Uncertain
Feasibility	☐ Yes ☒ Probably yes ☐ No ☐ Varies ☐ Uncertain	☐ Yes ☒ Probably yes ☐ No ☐ Varies ☐ Uncertain	☐ Yes ☒ Probably yes ☐ No ☐ Varies ☐ Uncertain	☐ Yes ☒ Probably yes ☐ No ☐ Varies ☐ Uncertain
Enough evidence to make a recommendation?	☒ Yes ☐ No	☒ Yes ☐ No	☒ Yes ☐ No	☒ Yes ☐ No
Recommendation in favour or against the intervention	☒ In favour ☐ Against	☒ In favour ☐ Against	☒ In favour ☐ Against	☒ In favour ☐ Against
Type of recommendation	☐ Strong recommendation ☒ Conditional recommendation ☐ Good practice statement	☐ Strong recommendation ☐ Conditional recommendation ☒ Good practice statement	☐ Strong recommendation ☐ Conditional recommendation ☒ Good practice statement	☐ Strong recommendation ☐ Conditional recommendation ☒ Good practice statement

Table WA.3.1b Improving access to, uptake of, and continued use of contraception among adolescents

Recommendation	Recommendation 2.1a	Recommendation 2.1b	Recommendation 2.2	Recommendation 2.3	Recommendation 2.4
Intervention	Gender-transformative behaviour change interventions with adolescents to strengthen contraceptive decision-making	Interventions to shift gender and other social norms to support contraceptive decision-making and use among adolescents	Self-care interventions relevant to contraceptive use	Interventions to reduce financial barriers related to contraceptive use among adolescents	Accurate and safe digital health interventions for adolescents as part of sexual and reproductive health (SRH) programming
Problem a priority?	☒ Yes ☐ No ☐ Uncertain ☐ Varies	☒ Yes ☐ No ☐ Uncertain ☐ Varies	☒ Yes ☐ No ☐ Uncertain ☐ Varies	☒ Yes ☐ No ☐ Uncertain ☐ Varies	☒ Yes ☐ No ☐ Uncertain ☐ Varies
Certainty of the evidence	☐ High ☒ Moderate ☐ Low ☐ Very low ☐ Nil	☐ High ☒ Moderate ☐ Low ☐ Very low ☐ Nil	☐ High ☐ Moderate ☐ Low ☐ Very low ☒ Nil	☐ High ☐ Moderate ☐ Low ☒ Very low ☐ Nil	☐ High ☐ Moderate ☒ Low ☐ Very low ☐ Nil
Balance of benefits and harms	☐ Favours intervention ☒ Probably favours intervention ☐ Against intervention ☐ Varies ☐ Uncertain	☐ Favours intervention ☒ Probably favours intervention ☐ Against intervention ☐ Varies ☐ Uncertain	☒ Favours intervention ☐ Probably favours intervention ☐ Against intervention ☐ Varies ☐ Uncertain	☐ Favours intervention ☒ Probably favours intervention ☐ Against intervention ☐ Varies ☐ Uncertain	☐ Favours intervention ☐ Probably favours intervention ☐ Against intervention ☐ Varies ☒ Uncertain
Resource requirements (costs) and cost-effectiveness	☐ Large costs ☐ Moderate costs ☐ Negligible costs and savings ☐ Moderate savings ☐ Large savings ☒ Varies ☐ Uncertain	☐ Large costs ☐ Moderate costs ☐ Negligible costs and savings ☐ Moderate savings ☐ Large savings ☒ Varies ☐ Uncertain	☐ Large costs ☐ Moderate costs ☐ Negligible costs and savings ☐ Moderate savings ☐ Large savings ☒ Varies ☐ Uncertain	☐ Large costs ☐ Moderate costs ☐ Negligible costs and savings ☐ Moderate savings ☐ Large savings ☒ Varies ☐ Uncertain	☐ Large costs ☐ Moderate costs ☐ Negligible costs and savings ☐ Moderate savings ☐ Large savings ☒ Varies ☐ Uncertain
Equity	☐ Increased ☒ Probably increased ☐ Reduced ☐ Varies ☐ Uncertain	☐ Increased ☒ Probably increased ☐ Reduced ☐ Varies ☐ Uncertain	☐ Increased ☒ Probably increased ☐ Reduced ☐ Varies ☐ Uncertain	☐ Increased ☒ Probably increased ☐ Reduced ☐ Varies ☐ Uncertain	☐ Increased ☒ Probably increased ☐ Reduced ☐ Varies ☐ Uncertain

Table WA.3.1b Improving access to, uptake of, and continued use of contraception among adolescents (continued)

Recommendation	Recommendation 2.1a	Recommendation 2.1b	Recommendation 2.2	Recommendation 2.3	Recommendation 2.4
Intervention	Gender-transformative behaviour change interventions with adolescents to strengthen contraceptive decision-making	Interventions to shift gender and other social norms to support contraceptive decision-making and use among adolescents	Self-care interventions relevant to contraceptive use	Interventions to reduce financial barriers related to contraceptive use among adolescents	Accurate and safe digital health interventions for adolescents as part of sexual and reproductive health (SRH) programming
Acceptability	☐ Yes ☒ Probably yes ☐ No ☐ Varies ☐ Uncertain	☐ Yes ☒ Probably yes ☐ No ☐ Varies ☐ Uncertain	☐ Yes ☒ Probably yes ☐ No ☐ Varies ☐ Uncertain	☐ Yes ☒ Probably yes ☐ No ☐ Varies ☐ Uncertain	☐ Yes ☒ Probably yes ☐ No ☐ Varies ☐ Uncertain
Feasibility	☐ Yes ☒ Probably yes ☐ No ☐ Varies ☐ Uncertain	☐ Yes ☒ Probably yes ☐ No ☐ Varies ☐ Uncertain	☐ Yes ☒ Probably yes ☐ No ☐ Varies ☐ Uncertain	☐ Yes ☒ Probably yes ☐ No ☐ Varies ☐ Uncertain	☐ Yes ☒ Probably yes ☐ No ☐ Varies ☐ Uncertain
Enough evidence to make a recommendation?	☒ Yes ☐ No	☒ Yes ☐ No	☒ Yes ☐ No	☒ Yes ☐ No	☒ Yes ☐ No
Recommendation in favour or against the intervention	☒ In favour ☐ Against	☒ In favour ☐ Against	☒ In favour ☐ Against	☒ In favour ☐ Against	☒ In favour ☐ Against
Type of recommendation	☒ Strong recommendation ☐ Conditional recommendation ☐ Good practice statement	☒ Strong recommendation ☐ Conditional recommendation ☐ Good practice statement	☒ Strong recommendation ☐ Conditional recommendation ☐ Good practice statement	☐ Strong recommendation ☒ Conditional recommendation ☐ Good practice statement	☐ Strong recommendation ☒ Conditional recommendation ☐ Good practice statement

Table WA.3.1b (continued)

Recommendation	Good practice statement 2.1	Good practice statement 2.2	Good practice statement 2.3	Good practice statement 2.4
Intervention	Mobilization of political, governmental, religious, traditional and other influential leaders to support contraceptive use among adolescents	Interventions to improve the quality of health services, to improve contraceptive use among adolescents	Formulation and implementation of enabling laws and policies on age, marital status and consent procedures in relation to sexual activity, access to sexual and reproductive health (SRH) services and access to specific contraceptive methods	Meaningful engagement of adolescents in the design, implementation, monitoring and evaluation of efforts to address their contraceptive needs and rights
Problem a priority?	☒ Yes ☐ No ☐ Uncertain ☐ Varies	☒ Yes ☐ No ☐ Uncertain ☐ Varies	☒ Yes ☐ No ☐ Uncertain ☐ Varies	☒ Yes ☐ No ☐ Uncertain ☐ Varies
Certainty of the evidence	☐ High ☐ Moderate ☐ Low ☐ Very low ☒ Nil	☐ High ☐ Moderate ☐ Low ☒ Very low ☐ Nil	☐ High ☐ Moderate ☐ Low ☒ Very low ☐ Nil	☐ High ☐ Moderate ☐ Low ☐ Very low ☒ Nil
Balance of benefits and harms	☐ Favours intervention ☒ Probably favours intervention ☐ Against intervention ☐ Varies ☐ Uncertain	☐ Favours intervention ☒ Probably favours intervention ☐ Against intervention ☐ Varies ☐ Uncertain	☐ Favours intervention ☐ Probably favours intervention ☐ Against intervention ☐ Varies ☒ Uncertain	☒ Favours intervention ☐ Probably favours intervention ☐ Against intervention ☐ Varies ☐ Uncertain
Resource requirements (costs) and cost-effectiveness	☐ Large costs ☐ Moderate costs ☐ Negligible costs and savings ☐ Moderate savings ☐ Large savings ☒ Varies ☐ Uncertain	☐ Large costs ☐ Moderate costs ☒ Negligible costs and savings ☐ Moderate savings ☐ Large savings ☐ Varies ☐ Uncertain	☐ Large costs ☐ Moderate costs ☐ Negligible costs and savings ☐ Moderate savings ☐ Large savings ☒ Varies ☐ Uncertain	☐ Large costs ☐ Moderate costs ☐ Negligible costs and savings ☐ Moderate savings ☐ Large savings ☒ Varies ☐ Uncertain
Equity	☐ Increased ☒ Probably increased ☐ Reduced ☐ Varies ☐ Uncertain	☐ Increased ☒ Probably increased ☐ Reduced ☐ Varies ☐ Uncertain	☐ Increased ☐ Probably increased ☐ Reduced ☐ Varies ☒ Uncertain	☐ Increased ☒ Probably increased ☐ Reduced ☐ Varies ☐ Uncertain

Table WA.3.1b Improving access to, uptake of, and continued use of contraception among adolescents (continued)

Recommendation	Good practice statement 2.1	Good practice statement 2.2	Good practice statement 2.3	Good practice statement 2.4
Intervention	Mobilization of political, governmental, religious, traditional and other influential leaders to support contraceptive use among adolescents	Interventions to improve the quality of health services, to improve contraceptive use among adolescents	Formulation and implementation of enabling laws and policies on age, marital status and consent procedures in relation to sexual activity, access to sexual and reproductive health (SRH) services and access to specific contraceptive methods	Meaningful engagement of adolescents in the design, implementation, monitoring and evaluation of efforts to address their contraceptive needs and rights
Acceptability	☐ Yes ☒ Probably yes ☐ No ☐ Varies ☐ Uncertain	☐ Yes ☒ Probably yes ☐ No ☐ Varies ☐ Uncertain	☐ Yes ☐ Probably yes ☐ No ☒ Varies ☐ Uncertain	☐ Yes ☒ Probably yes ☐ No ☐ Varies ☐ Uncertain
Feasibility	☐ Yes ☒ Probably yes ☐ No ☐ Varies ☐ Uncertain	☐ Yes ☒ Probably yes ☐ No ☐ Varies ☐ Uncertain	☐ Yes ☐ Probably yes ☐ No ☒ Varies ☐ Uncertain	☐ Yes ☒ Probably yes ☐ No ☐ Varies ☐ Uncertain
Enough evidence to make a recommendation?	☒ Yes ☐ No	☒ Yes ☐ No	☒ Yes ☐ No	☒ Yes ☐ No
Recommendation in favour or against the intervention	☒ In favour ☐ Against	☒ In favour ☐ Against	☒ In favour ☐ Against	☒ In favour ☐ Against
Type of recommendation	☐ Strong recommendation ☐ Conditional recommendation ☒ Good practice statement	☐ Strong recommendation ☐ Conditional recommendation ☒ Good practice statement	☐ Strong recommendation ☐ Conditional recommendation ☒ Good practice statement	☐ Strong recommendation ☐ Conditional recommendation ☒ Good practice statement

4. Reviews conducted for the development of this guideline

4.1 Preventing child marriage and responding to the needs and rights of married girls

Search strategy: Given that multiple systematic reviews on interventions to prevent child marriage had been published in the preceding five years, the evidence team for the questions on child marriage began by appraising the literature included in these reviews. To identify studies that evaluated interventions which specified child marriage as an outcome of interest, the evidence team appraised studies included in the five most recent systematic reviews on child marriage interventions (1-5). To identify studies of interventions focused primarily on education but which also include child marriage as an outcome measure, the evidence team appraised studies included in a recent systematic review of interventions focused on eliminating gender-based barriers to education (6). To identify studies of interventions for married girls, which present rigorous results and disaggregate by marital status, the evidence team appraised studies included in two reviews of interventions targeting married girls (1, 7). An additional search was conducted for more recent publications focused on the

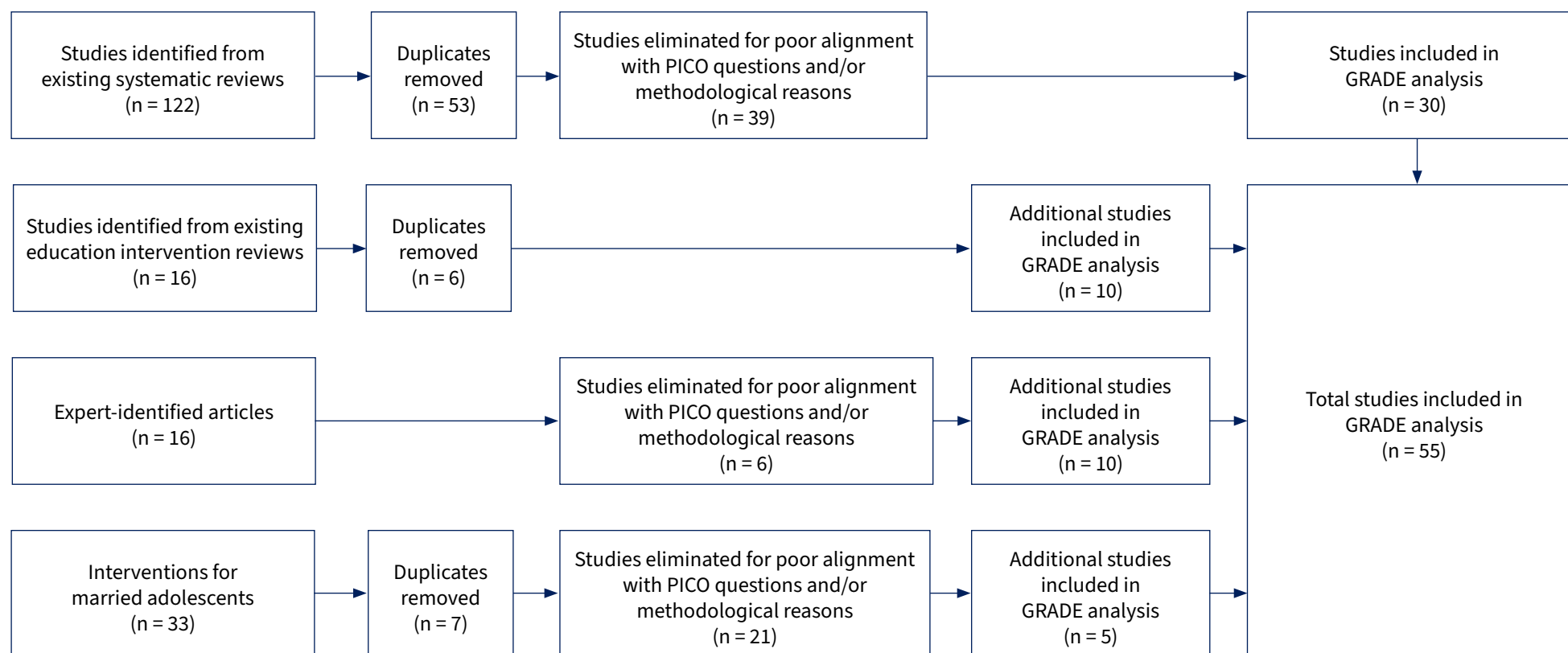
same interventions for which the studies included in previous reviews did not present final results. In such cases, the studies identified through the process above were replaced with more recent publications. Finally, experts were consulted about recent observational studies that might be considered for inclusion, particularly for PICO questions 4 (education) and 6 (child marriage laws).

Screening process: Studies were excluded based on non-alignment with PICO questions, concerns related to study design and/or concerns related to the method of analysis. Data from eligible articles were abstracted into standardized forms.

Extraction and synthesis process: Next, data were summarized in GRADE tables (see [section 1 of this web annex](#)) for presentation to the Guideline Development Group (GDG).

Search terms: No search terms are included due to the lack of a *de novo* search strategy (see “Search strategy” above).

Fig. WA.4.1 Child marriage PRISMA flowchart



GRADE: Grading of Recommendations Assessment, Development and Evaluation; PICO: population, intervention, comparator, outcomes; PRISMA: Preferred reporting items for systematic reviews and meta-analyses.

4.2 Improving access to, uptake of, and continued use of contraception among adolescents

Search strategy: A single systematic review was conducted for all PICO questions, using search terms for (i) contraception/family planning methods, (ii) adolescents aged 10–19, (iii) quantitative comparative study design and (iv) low- and middle-income countries (LMICs). The search included peer-reviewed journal publications in English in three databases: PubMed, Embase and Cumulative Index to Nursing and Allied Health Literature (CINAHL). Search dates were from 1 January 2010 to 22 December 2022. The specific search terms are provided in [section 4.3](#).

This was supplemented by a secondary search for articles to complement the database search, which included:

- reviewing articles included in other reviews (including the scoping review that was conducted to inform this guideline);
- reviewing reference lists of included articles; and
- asking GDG members for suggested literature.

Eligibility criteria: The following eligibility criteria were used for the systematic review.

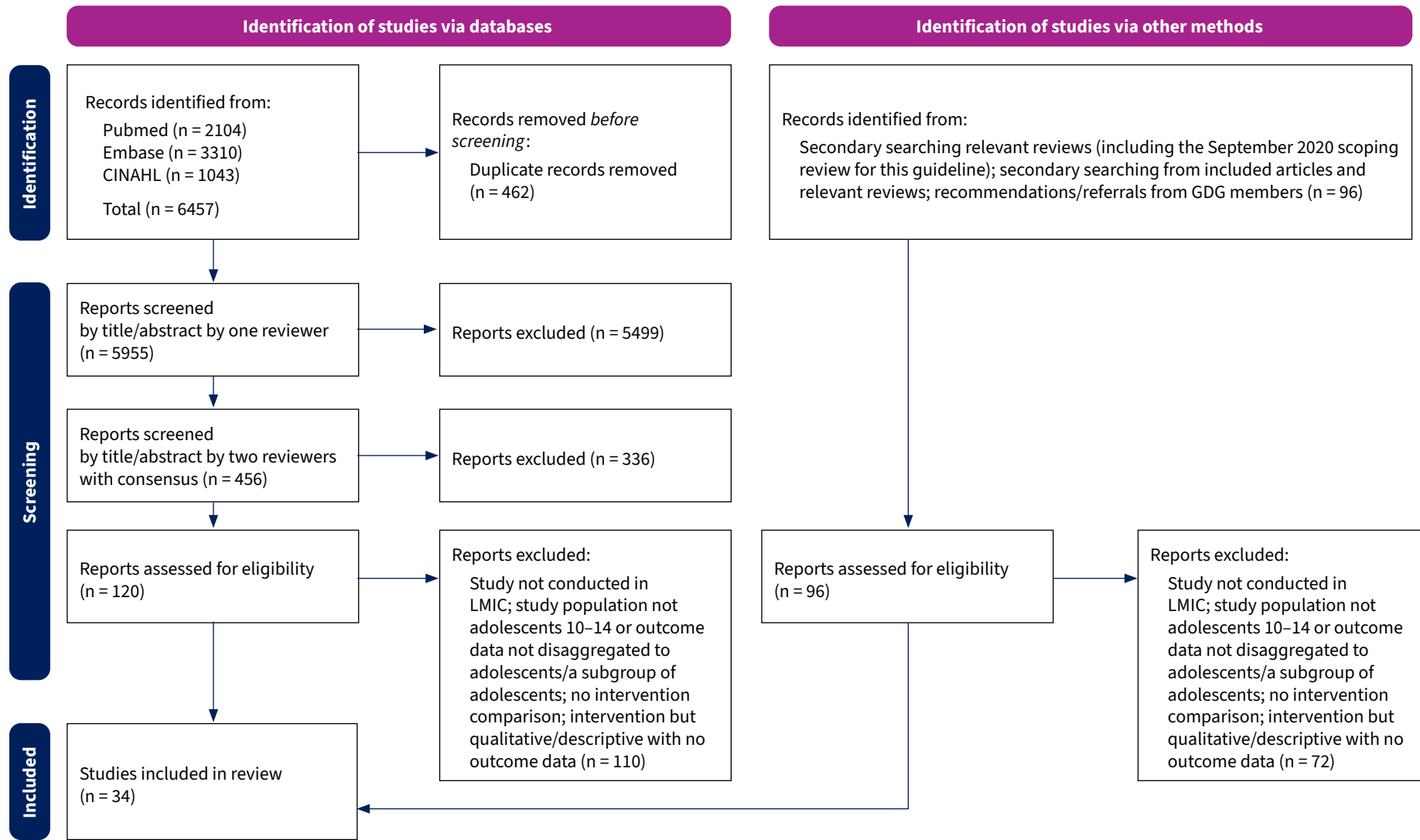
- Age range: adolescents, defined as being 10–19 years old; i.e. the study must have data disaggregated for adolescents aged 10–19 years (or at least one subset within that age range). Studies on “young people” more broadly (e.g. 15–24 years) were excluded if disaggregation was not possible.
- Sub-populations of interest (based on the PICO questions):
 - adolescents aged 10–14 years
 - married adolescents
 - unmarried adolescents
 - first-time parents
 - boys/young men
 - adolescents with disabilities

- adolescents in school
- adolescents out of school
- lesbian, gay, bisexual, transgender, queer/questioning, intersex or asexual (LGBTQIA+) adolescents
- adolescents in humanitarian crisis contexts
- adolescents living with HIV.
- Year range: 2010–search date; however, the evidence team for the questions on adolescent contraceptive use was open to including articles published before 2010 if they were included in the prior scoping review conducted to inform this guideline.
- Geographic restrictions: LMICs only.
- Type of literature: peer-reviewed literature (may include grey literature strategically via secondary searching through referral by the GDG members if needed for specific subtopics).
- Language: English.
- Must provide quantitative comparative data (pre–post or multi-arm study design) on the intervention versus comparator(s) for at least one of the outcomes in at least one of the PICO questions (see Annex 4 in the guideline document).

Screening process: Two rounds of title and abstract screening were conducted. The first round was done by one research assistant to remove search results that were clearly not relevant, and the second round was done in duplicate independently by two research assistants to remove search results that were not eligible. Full-text articles were then reviewed in duplicate. Data from eligible articles were abstracted into standardized forms.

Extraction and synthesis process: Data were then summarized in GRADE tables for presentation to the GDG. For more parsimonious presentation of the highest quality evidence, RCT data were prioritized for GRADE tables where available. Therefore, if an observational study presented the same outcome as an RCT for the same PICO question, the observational study’s data were not included in the GRADE table ([section 2 of this web annex](#)).

Fig. WA.4.2 Adolescent contraception PRISMA flowchart



CINAHL: Cumulative Index to Nursing and Allied Health Literature; GDG: Guideline Development Group; LMIC: low- and middle-income countries; PRISMA: Preferred reporting items for systematic reviews and meta-analyses.

4.3 Search terms

Database: PubMed

Search dates: 1 January 2010 to 22 December 2022

Concept 1: Contraception

("Contraceptive Devices"[Majr] OR "Contraceptive Agents"[Majr] OR "Birth Intervals"[Majr] OR "Family Planning Services"[Majr] OR "Contraception"[Majr] OR "Sexual Abstinence"[Majr] OR "Fertility"[Majr] OR "Gravidity"[Majr] OR "Pregnancy"[Majr] OR "Parity"[Majr]) OR (contraception[tiab] OR contraceptive[tiab] OR "female contraceptive devices"[tiab] OR "female contraceptive device"[tiab] OR "vaginal ring"[tiab] OR "vaginal rings"[tiab] OR "female contraceptive agents"[tiab] OR "female contraceptives"[tiab] OR "female contraceptives"[tiab] OR "female contraceptive agents"[tiab] OR "oral contraceptives"[tiab] OR "hormonal contraceptives"[tiab] OR "hormonal contraceptive agents"[tiab] OR "hormonal oral contraceptive agents"[tiab] OR "estrogen contraceptive agents"[tiab] OR "sequential oral contraceptives"[tiab] OR "sequential oral contraceptive agents"[tiab] OR "oral synthetic contraceptives"[tiab] OR "oral hormonal contraceptives"[tiab] OR "oral hormonal contraceptive agents"[tiab] OR "oral combined contraceptives"[tiab] OR "combined oral contraceptives"[tiab] OR "combined oral contraceptive agents"[tiab] OR "oral combined contraceptive agents"[tiab] OR "hormone-releasing intrauterine devices"[tiab] OR "hormone-releasing intrauterine device"[tiab] OR "hormonal intrauterine device"[tiab] OR "hormonal intrauterine devices"[tiab] OR "IUD"[tiab] OR "hormonal IUD"[tiab] OR "medicated intrauterine device"[tiab] OR "medicated intrauterine devices"[tiab] OR "progesterone-releasing intrauterine device"[tiab] OR "progesterone-releasing intrauterine devices"[tiab] OR "progesterone-releasing vaginal ring"[tiab] OR "contraceptive IUD"[tiab]

OR "contraceptive IUDs"[tiab] OR "intrauterine contraceptive device"[tiab] OR "intrauterine contraceptive devices"[tiab] OR "copper-bearing intrauterine devices"[tiab] OR "cyclofem"[tiab] OR "depo-provera"[tiab] OR "injectable contraceptive"[tiab] OR "injectable contraceptives"[tiab] OR "DMPA"[tiab] OR "depot medroxyprogesterone acetate"[tiab] OR "injectable progestogen-only contraceptives"[tiab] OR "norethisterone oenanthate"[tiab] OR "norethisterone enanthate"[tiab] OR "contraceptive implant"[tiab] OR "contraceptive implants"[tiab] OR "subdermal contraceptive implant"[tiab] OR "subdermal contraceptive

implants"[tiab] OR "single rod contraceptive implant"[tiab] OR "femplant"[tiab] OR "hormonal implant"[tiab] OR "contraceptive patch"[tiab] OR "transdermal contraceptive patch"[tiab] OR "evra transdermal contraceptive patch"[tiab] OR "ortho evra transdermal contraceptive patch"[tiab] OR "contraceptive ring"[tiab] OR "vaginal contraceptive ring"[tiab] OR "vaginal contraceptive rings"[tiab] OR "contraceptive rings"[tiab] OR "vaginal rings"[tiab] OR "vaginal ring"[tiab] OR "levonorgestrel intrauterine device"[tiab] OR "levonorgestrel-releasing intrauterine device"[tiab] OR "levonorgestrel implant"[tiab] OR "etonogestrel implant"[tiab] OR "hormonal intrauterine device"[tiab] OR "female contraceptive agents"[tiab] OR "hormonal oral contraceptive agents"[tiab] OR "hormonal contraception"[tiab] OR "hormonal contraception"[tiab] OR "hormonal contraceptives"[tiab] OR "birth spacing"[tiab] OR "family planning services"[tiab] OR "family planning"[tiab] OR "birth intervals"[tiab] OR "birth interval"[tiab] OR "birth spacings"[tiab] OR "pregnancy intervals"[tiab] OR "pregnancy interval"[tiab] OR "emergency contraceptive"[tiab] OR "emergency contraception"[tiab] OR "barrier method"[tiab] OR "fertility awareness"[tiab] OR "lactational amenorrhoea method"[tiab] OR "coitus interruptus"[tiab] OR "female sterilization"[tiab] OR "male sterilization"[tiab] OR abstinence[tiab] OR fertility[tiab] OR fecundity[tiab] OR gravidity[tiab] OR parity[tiab] OR pregnancy[tiab] OR pregnant[tiab] OR "sayana press"[tiab] OR "uniject"[tiab] OR "contraceptive self-injection"[tiab] OR "self-injectable contraception"[tiab] OR "self-injected contraceptive"[tiab] OR "calendar method"[tiab] OR "calendar-based method"[tiab] OR "calendar-based app"[tiab] OR "menstrual cycle app"[tiab] OR "menstrual tracking app"[tiab] OR fertility tracking app[tiab] OR "period tracker app"[tiab] OR "ovulation prediction app"[tiab] OR "calcuothermal app"[tiab] OR "symptothermal app"[tiab] OR "flo mobile app"[tiab] OR "clue menstruations"[tiab] OR "Period tracker deluxe"[tiab] OR "womanlog"[tiab] OR "ovy"[tiab] OR "natural cycles"[tiab] OR "myNFP"[tiab] OR "lady cycle"[tiab] OR "ovuvue"[tiab] OR "contraceptive accessibility"[tiab] OR "contraceptive discontinuation"[tiab] OR "rapid repeat pregnancy"[tiab] OR "coital based method"[tiab] OR "LARC"[tiab] OR "short-acting method"[tiab])

Concept 2: Adolescents 10–19 years old

(adolescent[Majr] OR young adult[Majr] OR students[Majr] OR child[Majr] OR youth[Majr] OR adolescence[tiab] OR adolescent[tiab] OR adolescents[tiab] OR teen[tiab] OR teenager[tiab] OR teenagers[tiab] OR "young adult"[tiab] OR "young adults"[tiab] OR "young male"[tiab] OR "young males"[tiab] OR "young female"[tiab] OR "young females"[tiab] OR "young couples"[tiab] OR "young parents"[tiab] OR

"first time parents"[tiab] OR youth[tiab] OR youths[tiab] OR "married girls"[tiab] OR "unmarried girls"[tiab] OR child[tiab] OR children[tiab] OR "school child"[tiab] OR schoolchild[tiab] OR "school children"[tiab] OR schoolchildren[tiab] OR pupil[tiab] OR pupils[tiab] OR student[tiab] OR students[tiab] OR "school girls"[tiab] OR "schoolgirls"[tiab] OR "school going girls"[tiab] OR "high school girls"[tiab] OR "high school boys"[tiab] OR "high school students"[tiab] OR "college students"[tiab] OR "university students"[tiab] OR "young migrants"[tiab] OR "young factory workers"[tiab])

Concept 3: LMICs

("afghanistan"[MeSH Terms] OR "albania"[MeSH Terms] OR "algeria"[MeSH Terms] OR "american samoa"[MeSH Terms] OR "angola"[MeSH Terms] OR "antigua and barbuda"[MeSH Terms] OR "argentina"[MeSH Terms] OR "armenia"[MeSH Terms] OR "aruba"[MeSH Terms] OR "azerbaijan"[MeSH Terms] OR "bahrain"[MeSH Terms] OR "bangladesh"[MeSH Terms] OR "barbados"[MeSH Terms] OR "republic of belarus"[MeSH Terms] OR "belize"[MeSH Terms] OR "benin"[MeSH Terms] OR "bhutan"[MeSH Terms] OR "bolivia"[MeSH Terms] OR "bosnia and herzegovina"[MeSH Terms] OR "botswana"[MeSH Terms] OR "brazil"[MeSH Terms] OR "bulgaria"[MeSH Terms] OR "burkina faso"[MeSH Terms] OR "burundi"[MeSH Terms] OR "cabo verde"[MeSH Terms] OR "cambodia"[MeSH Terms] OR "cameroon"[MeSH Terms] OR "central african republic"[MeSH Terms] OR "chad"[MeSH Terms] OR "chile"[MeSH Terms] OR "china"[MeSH Terms] OR "colombia"[MeSH Terms] OR "comoros"[MeSH Terms] OR "democratic republic of the congo"[MeSH Terms] OR "congo"[MeSH Terms] OR "costa rica"[MeSH Terms] OR "cote d ivoire"[MeSH Terms] OR "croatia"[MeSH Terms] OR "cuba"[MeSH Terms] OR "cyprus"[MeSH Terms] OR "czech republic"[MeSH Terms] OR "djibouti"[MeSH Terms] OR "dominica"[MeSH Terms] OR "dominican republic"[MeSH Terms] OR "ecuador"[MeSH Terms] OR "egypt"[MeSH Terms] OR "el salvador"[MeSH Terms] OR "equatorial guinea"[MeSH Terms] OR "eritrea"[MeSH Terms] OR "estonia"[MeSH Terms] OR "eswatini"[MeSH Terms] OR "eswatini"[MeSH Terms] OR "ethiopia"[MeSH Terms] OR "fiji"[MeSH Terms] OR "gabon"[MeSH Terms] OR "gambia"[MeSH Terms] OR "georgia republic"[MeSH Terms] OR "ghana"[MeSH Terms] OR "gibraltar"[MeSH Terms] OR "grenada"[MeSH Terms] OR "guam"[MeSH Terms] OR "guatemala"[MeSH Terms] OR "guinea"[MeSH Terms] OR "guinea bissau"[MeSH Terms] OR "guyana"[MeSH Terms] OR "haiti"[MeSH Terms] OR "honduras"[MeSH Terms] OR "india"[MeSH Terms] OR "indonesia"[MeSH Terms] OR "iran"[MeSH Terms] OR "iraq"[MeSH Terms] OR "jamaica"[MeSH Terms] OR "jordan"[MeSH Terms] OR

"kazakhstan"[MeSH Terms] OR "kenya"[MeSH Terms] OR "democratic people's republic of korea"[MeSH Terms] OR "republic of korea"[MeSH Terms] OR "kosovo"[MeSH Terms] OR "kyrgyzstan"[MeSH Terms] OR "laos"[MeSH Terms] OR "latvia"[MeSH Terms] OR "lebanon"[MeSH Terms] OR "lesotho"[MeSH Terms] OR "liberia"[MeSH Terms] OR "libya"[MeSH Terms] OR "lithuania"[MeSH Terms] OR "macau"[MeSH Terms] OR "republic of north macedonia"[MeSH Terms] OR "madagascar"[MeSH Terms] OR "malawi"[MeSH Terms] OR "malaysia"[MeSH Terms] OR "indian ocean islands"[MeSH Terms] OR "mali"[MeSH Terms] OR "malta"[MeSH Terms] OR "micronesia"[MeSH Terms] OR "palau"[MeSH Terms] OR "mauritania"[MeSH Terms] OR "mauritius"[MeSH Terms] OR "mexico"[MeSH Terms] OR "moldova"[MeSH Terms] OR "mongolia"[MeSH Terms] OR "montenegro"[MeSH Terms] OR "morocco"[MeSH Terms] OR "mozambique"[MeSH Terms] OR "myanmar"[MeSH Terms] OR "namibia"[MeSH Terms] OR "nepal"[MeSH Terms] OR "netherlands antilles"[MeSH Terms] OR "nicaragua"[MeSH Terms] OR "niger"[MeSH Terms] OR "nigeria"[MeSH Terms] OR "oman"[MeSH Terms] OR "pakistan"[MeSH Terms] OR "panama"[MeSH Terms] OR "papua new guinea"[MeSH Terms] OR "paraguay"[MeSH Terms] OR "peru"[MeSH Terms] OR "philippines"[MeSH Terms] OR "poland"[MeSH Terms] OR "portugal"[MeSH Terms] OR "puerto rico"[MeSH Terms] OR "romania"[MeSH Terms] OR "russia"[MeSH Terms] OR "rwanda"[MeSH Terms] OR "samoa"[MeSH Terms] OR "sao tome and principe"[MeSH Terms] OR "saudi arabia"[MeSH Terms] OR "senegal"[MeSH Terms] OR "serbia"[MeSH Terms] OR "seychelles"[MeSH Terms] OR "sierra leone"[MeSH Terms] OR "slovakia"[MeSH Terms] OR "slovenia"[MeSH Terms] OR "melanesia"[MeSH Terms] OR "somalia"[MeSH Terms] OR "south africa"[MeSH Terms] OR "south sudan"[MeSH Terms] OR "sri lanka"[MeSH Terms] OR "saint kitts and nevis"[MeSH Terms] OR "saint lucia"[MeSH Terms] OR "saint vincent and the grenadines"[MeSH Terms] OR "sudan"[MeSH Terms] OR "suriname"[MeSH Terms] OR "syria"[MeSH Terms] OR "tajikistan"[MeSH Terms] OR "tanzania"[MeSH Terms] OR "thailand"[MeSH Terms] OR "timor leste"[MeSH Terms] OR "togo"[MeSH Terms] OR "tonga"[MeSH Terms] OR "trinidad and tobago"[MeSH Terms] OR "tunisia"[MeSH Terms] OR "turkey"[MeSH Terms] OR "turkmenistan"[MeSH Terms] OR "uganda"[MeSH Terms] OR "ukraine"[MeSH Terms] OR "uruguay"[MeSH Terms] OR "uzbekistan"[MeSH Terms] OR "vanuatu"[MeSH Terms] OR "venezuela"[MeSH Terms] OR "vietnam"[MeSH Terms] OR "middle east"[MeSH Terms] OR "yemen"[MeSH Terms] OR "yugoslavia"[MeSH Terms] OR "zambia"[MeSH Terms] OR "zimbabwe"[MeSH Terms] OR "africa south of the sahara"[MeSH Terms] OR "africa, central"[MeSH Terms] OR "africa, northern"[MeSH Terms] OR "africa, southern"[MeSH Terms] OR "africa, eastern"[MeSH Terms] OR

"africa, western"[MeSH Terms] OR "west indies"[MeSH Terms] OR "indian ocean islands"[MeSH Terms] OR "caribbean region"[MeSH Terms] OR "central america"[MeSH Terms] OR "latin america"[MeSH Terms] OR "south america"[MeSH Terms] OR "asia, central"[MeSH Terms] OR "asia, northern"[MeSH Terms] OR "asia, southeastern"[MeSH Terms] OR "asia, western"[MeSH Terms] OR "europe, eastern"[MeSH Terms] OR "developing countries"[MeSH Terms] OR "afghanistan"[Text Word] OR "albania"[Text Word] OR "algeria"[Text Word] OR "american samoa"[Text Word] OR "angola"[Text Word] OR "antigua"[Text Word] OR "barbuda"[Text Word] OR "argentina"[Text Word] OR "armenia"[Text Word] OR "armenian"[Text Word] OR "aruba"[Text Word] OR "azerbaijan"[Text Word] OR "bahrain"[Text Word] OR "bangladesh"[Text Word] OR "barbados"[Text Word] OR "belarus"[Text Word] OR "byelarus"[Text Word] OR "belorussia"[Text Word] OR "byelorussian"[Text Word] OR "belize"[Text Word] OR "british honduras"[Text Word] OR "benin"[Text Word] OR "dahomey"[Text Word] OR "bhutan"[Text Word] OR "bolivia"[Text Word] OR "bosnia"[Text Word] OR "herzegovina"[Text Word] OR "botswana"[Text Word] OR "bechuanaland"[Text Word] OR "brazil"[Text Word] OR "brasil"[Text Word] OR "bulgaria"[Text Word] OR "burkina faso"[Text Word] OR "burkina fasso"[Text Word] OR "upper volta"[Text Word] OR "burundi"[Text Word] OR "urundi"[Text Word] OR "cabo verde"[Text Word] OR "cape verde"[Text Word] OR "cambodia"[Text Word] OR "kampuchea"[Text Word] OR "khmer republic"[Text Word] OR "cameroon"[Text Word] OR "cameron"[Text Word] OR "cameroun"[Text Word] OR "central african republic"[Text Word] OR "ubangi shari"[Text Word] OR "chad"[Text Word] OR "chile"[Text Word] OR "china"[Text Word] OR "colombia"[Text Word] OR "comoros"[Text Word] OR "comoro islands"[Text Word] OR "mayotte"[Text Word] OR "congo"[Text Word] OR "zaire"[Text Word] OR "costa rica"[Text Word] OR "cote d ivoire"[Text Word] OR "cote d ivoire"[Text Word] OR "cote d ivoire"[Text Word] OR "ivory coast"[Text Word] OR "croatia"[Text Word] OR "cuba"[Text Word] OR "cyprus"[Text Word] OR "czech republic"[Text Word] OR "czechoslovakia"[Text Word] OR "djibouti"[Text Word] OR "french somaliland"[Text Word] OR "dominica"[Text Word] OR "dominican republic"[Text Word] OR "ecuador"[Text Word] OR "egypt"[Text Word] OR "united arab republic"[Text Word] OR "el salvador"[Text Word] OR "equatorial guinea"[Text Word] OR "spanish guinea"[Text Word] OR "eritrea"[Text Word] OR "estonia"[Text Word] OR "eswatini"[Text Word] OR "swaziland"[Text Word] OR "ethiopia"[Text Word] OR "fiji"[Text Word] OR "gabon"[Text Word] OR "gabonese republic"[Text Word] OR "gambia"[Text Word] OR "georgia"[Text Word] OR "georgian"[Text Word] OR "ghana"[Text Word] OR "gold coast"[Text Word] OR "gibraltar"[Text Word] OR "grenada"[Text Word] OR "guam"[Text Word] OR

"guatemala"[Text Word] OR "guinea"[Text Word] OR "guyana"[Text Word] OR "guiana"[Text Word] OR "haiti"[Text Word] OR "hispaniola"[Text Word] OR "honduras"[Text Word] OR "india"[Text Word] OR "indonesia"[Text Word] OR "timor"[Text Word] OR "iran"[Text Word] OR "iraq"[Text Word] OR "isle of man"[Text Word] OR "jamaica"[Text Word] OR "jordan"[Text Word] OR "kazakhstan"[Text Word] OR "kazakh"[Text Word] OR "kenya"[Text Word] OR "korea"[Text Word] OR "kosovo"[Text Word] OR "kyrgyzstan"[Text Word] OR "kirghizia"[Text Word] OR "kirgizstan"[Text Word] OR "kyrgyz republic"[Text Word] OR "kirghiz"[Text Word] OR "laos"[Text Word] OR "lao pdr"[Text Word] OR "lao people s democratic republic"[Text Word] OR "latvia"[Text Word] OR "lebanon"[Text Word] OR "lesotho"[Text Word] OR "basutoland"[Text Word] OR "liberia"[Text Word] OR "libya"[Text Word] OR "libyan arab jamahiriya"[Text Word] OR "lithuania"[Text Word] OR "macau"[Text Word] OR "macao"[Text Word] OR "macedonia"[Text Word] OR "madagascar"[Text Word] OR "malagasy republic"[Text Word] OR "malawi"[Text Word] OR "nyasaland"[Text Word] OR "malaysia"[Text Word] OR "maldives"[Text Word] OR "indian ocean"[Text Word] OR "mali"[Text Word] OR "malta"[Text Word] OR "micronesia"[Text Word] OR "kiribati"[Text Word] OR "marshall islands"[Text Word] OR "nauru"[Text Word] OR "northern mariana islands"[Text Word] OR "palau"[Text Word] OR "tuvalu"[Text Word] OR "mauritania"[Text Word] OR "mauritius"[Text Word] OR "mexico"[Text Word] OR "moldova"[Text Word] OR "moldovian"[Text Word] OR "mongolia"[Text Word] OR "montenegro"[Text Word] OR "morocco"[Text Word] OR "ifni"[Text Word] OR "mozambique"[Text Word] OR "portuguese east africa"[Text Word] OR "myanmar"[Text Word] OR "burma"[Text Word] OR "namibia"[Text Word] OR "nepal"[Text Word] OR "netherlands antilles"[Text Word] OR "nicaragua"[Text Word] OR "niger"[Text Word] OR "nigeria"[Text Word] OR "oman"[Text Word] OR "muscat"[Text Word] OR "pakistan"[Text Word] OR "panama"[Text Word] OR "papua new guinea"[Text Word] OR "paraguay"[Text Word] OR "peru"[Text Word] OR "philippines"[Text Word] OR "philipines"[Text Word] OR "phillipines"[Text Word] OR "phillippines"[Text Word] OR "poland"[Text Word] OR "polish people s republic"[Text Word] OR "portugal"[Text Word] OR "portuguese republic"[Text Word] OR "puerto rico"[Text Word] OR "romania"[Text Word] OR "russia"[Text Word] OR "russian federation"[Text Word] OR "ussr"[Text Word] OR "soviet union"[Text Word] OR "union of soviet socialist republics"[Text Word] OR "rwanda"[Text Word] OR "ruanda"[Text Word] OR "samoa"[Text Word] OR "pacific islands"[Text Word] OR "polynesia"[Text Word] OR "samoan islands"[Text Word] OR "sao tome and principe"[Text Word] OR "saudi arabia"[Text Word] OR "senegal"[Text Word] OR "serbia"[Text Word] OR

"seychelles"[Text Word] OR "sierra leone"[Text Word] OR "slovakia"[Text Word] OR "slovak republic"[Text Word] OR "slovenia"[Text Word] OR "melanesia"[Text Word] OR "solomon island"[Text Word] OR "solomon islands"[Text Word] OR "norfolk island"[Text Word] OR "somalia"[Text Word] OR "south africa"[Text Word] OR "south sudan"[Text Word] OR "sri lanka"[Text Word] OR "ceylon"[Text Word] OR "saint kitts and nevis"[Text Word] OR "st kitts and nevis"[Text Word] OR "saint lucia"[Text Word] OR "st lucia"[Text Word] OR "saint vincent"[Text Word] OR "st vincent"[Text Word] OR "grenadines"[Text Word] OR "sudan"[Text Word] OR "suriname"[Text Word] OR "surinam"[Text Word] OR "syria"[Text Word] OR "syrian arab republic"[Text Word] OR "tajikistan"[Text Word] OR "tadjikistan"[Text Word] OR "tadzhikistan"[Text Word] OR "tadzhik"[Text Word] OR "tanzania"[Text Word] OR "tanganyika"[Text Word] OR "thailand"[Text Word] OR "siam"[Text Word] OR "timor leste"[Text Word] OR "east timor"[Text Word] OR "togo"[Text Word] OR "togolese republic"[Text Word] OR "tonga"[Text Word] OR "trinidad"[Text Word] OR "tobago"[Text Word] OR "tunisia"[Text Word] OR "turkey"[Text Word] OR "turkmenistan"[Text Word] OR "turkmen"[Text Word] OR "uganda"[Text Word] OR "ukraine"[Text Word] OR "uruguay"[Text Word] OR "uzbekistan"[Text Word] OR "uzbek"[Text Word] OR "vanuatu"[Text Word] OR "new hebrides"[Text Word] OR "venezuela"[Text Word] OR "vietnam"[Text Word] OR "viet nam"[Text Word] OR "middle east"[Text Word] OR "west bank"[Text Word] OR "gaza"[Text Word] OR "palestine"[Text Word] OR "yemen"[Text Word] OR "yugoslavia"[Text Word] OR "zambia"[Text Word] OR "zimbabwe"[Text Word] OR "northern rhodesia"[Text Word] OR "global south"[Text Word] OR "africa south of the sahara"[Text Word] OR "sub saharan africa"[Text Word] OR "subsaharan africa"[Text Word] OR "central africa"[Text Word] OR "north africa"[Text Word] OR "northern africa"[Text Word] OR "magreb"[Text Word] OR "maghrib"[Text Word] OR "sahara"[Text Word] OR "southern africa"[Text Word] OR "east africa"[Text Word] OR "eastern africa"[Text Word] OR "west africa"[Text Word] OR "western africa"[Text Word] OR "west indies"[Text Word] OR "indian ocean islands"[Text Word] OR "caribbean"[Text Word] OR "central america"[Text Word] OR "latin america"[Text Word] OR "south america"[Text Word] OR "central asia"[Text Word] OR "north asia"[Text Word] OR "northern asia"[Text Word] OR "southeastern asia"[Text Word] OR "south eastern asia"[Text Word] OR "southeast asia"[Text Word] OR "south east asia"[Text Word] OR "western asia"[Text Word] OR "east europe"[Text Word] OR "eastern europe"[Text Word] OR "developing country"[Text Word] OR "developing countries"[Text Word] OR "developing nation"[Text Word] OR "developing nations"[Text Word] OR "developing population"[Text Word] OR "developing populations"[Text Word] OR "developing

world"[Text Word] OR "less developed country"[Text Word] OR "less developed countries"[Text Word] OR "less developed nation"[Text Word] OR "less developed nations"[Text Word] OR "less developed world"[Text Word] OR "lesser developed countries"[Text Word] OR "lesser developed nations"[Text Word] OR "under developed country"[Text Word] OR "under developed countries"[Text Word] OR "under developed nations"[Text Word] OR "under developed world"[Text Word] OR "underdeveloped country"[Text Word] OR "underdeveloped countries"[Text Word] OR "underdeveloped nation"[Text Word] OR "underdeveloped nations"[Text Word] OR "underdeveloped population"[Text Word] OR "underdeveloped populations"[Text Word] OR "underdeveloped world"[Text Word] OR "middle income country"[Text Word] OR "middle income countries"[Text Word] OR "middle income nation"[Text Word] OR "middle income nations"[Text Word] OR "middle income population"[Text Word] OR "middle income populations"[Text Word] OR "low income country"[Text Word] OR "low income countries"[Text Word] OR "low income nation"[Text Word] OR "low income nations"[Text Word] OR "low income population"[Text Word] OR "low income populations"[Text Word] OR "lower income country"[Text Word] OR "lower income countries"[Text Word] OR "lower income nations"[Text Word] OR "lower income population"[Text Word] OR "lower income populations"[Text Word] OR "underserved countries"[Text Word] OR "underserved nations"[Text Word] OR "underserved population"[Text Word] OR "underserved populations"[Text Word] OR "under served population"[Text Word] OR "under served populations"[Text Word] OR "deprived countries"[Text Word] OR "deprived population"[Text Word] OR "deprived populations"[Text Word] OR "poor country"[Text Word] OR "poor countries"[Text Word] OR "poor nation"[Text Word] OR "poor nations"[Text Word] OR "poor population"[Text Word] OR "poor populations"[Text Word] OR "poor world"[Text Word] OR "poorer countries"[Text Word] OR "poorer nations"[Text Word] OR "poorer population"[Text Word] OR "poorer populations"[Text Word] OR "developing economy"[Text Word] OR "developing economies"[Text Word] OR "less developed economy"[Text Word] OR "less developed economies"[Text Word] OR "underdeveloped economies"[Text Word] OR "middle income economy"[Text Word] OR "middle income economies"[Text Word] OR "low income economy"[Text Word] OR "low income economies"[Text Word] OR "lower income economies"[Text Word] OR "low gdp"[Text Word] OR "low gnp"[Text Word] OR "low gross domestic"[Text Word] OR "low gross national"[Text Word] OR "lower gdp"[Text Word] OR "lower gross domestic"[Text Word] OR "lmic"[Text Word] OR "lmics"[Text Word] OR "third world"[Text Word] OR "lami country"[Text Word] OR "lami countries"[Text Word] OR "transitional

country"[Text Word] OR "transitional countries"[Text Word] OR "emerging economies"[Text Word] OR "emerging nation"[Text Word] OR "emerging nations"[Text Word])

Concept 4: Quantitative comparative study design

("Randomized Controlled Trials as Topic" [mh] OR "randomized controlled trial" [pt] OR "random allocation" [mh] OR "double blind method" [mh] OR "single blind method" [mh] OR "clinical trial" [pt] OR "controlled clinical trial" [pt] OR "multicenter study" [pt] OR "multicenter studies as topic" [mh] OR "clinical trial" [tw] OR "clinical trials" [tw] OR "single blind" [tw] OR "double blind" [tw] OR "triple blind" [tw] OR "single mask" [tw] OR "double mask" [tw] OR placebos [mh] OR placebo [tw] OR "randomly allocated" [tw] OR "epidemiologic studies" [mh] OR "case-control studies" [mh] OR "cohort studies" [mh] OR "case-control" [tiab] OR "cohort study" [tiab] OR "cohort studies" [tiab] OR "cohort analysis" [tiab] OR "cohort analyses" [tiab] OR "follow up study" [tiab] OR "follow up studies" [tiab] OR "observational study" [tiab] OR "observational studies" [tiab] OR "longitudinal" [tiab] OR "retrospective" [tiab] OR "prospective" [tiab] OR "cross-sectional" [tiab] OR "cross-sectional studies" [mh] NOT ("case report" [tw] OR letter [pt] OR "correspondence as topic" [mh] OR "historical article" [pt] OR comment [pt] OR editorial [pt] OR "practice guideline" [pt] OR (animals [mh] NOT humans [mh]) OR (("random sample" [tiab] OR "random digit" [tiab] OR "random effect" [tiab] OR "random effects" [tiab] OR random survey [tiab] OR "random regression" [tiab]) NOT "randomized controlled trial" [pt])

Database: Embase

*Add these at the end: AND [humans]/lim AND [2010-2023]/py AND ([article]/lim OR [article in press]/lim) AND ([school]/lim OR [adolescent]/lim OR [young adult]/lim)

Concept 1: Contraception

((('birth control'/exp OR 'contraception'/exp OR 'contraceptive agent'/exp OR 'fertility'/exp OR 'sexual abstinence'/exp OR 'adolescent pregnancy'/exp) OR ('contraception':ab,ti OR 'contraceptive':ab,ti OR 'female contraceptive devices':ab,ti OR 'female contraceptive device':ab,ti OR 'vaginal ring':ab,ti OR 'vaginal rings':ab,ti OR 'female contraceptive agents':ab,ti OR 'female contraceptives':ab,ti OR 'female contraceptives':ab,ti OR 'female contraceptive agents':ab,ti OR 'oral contraceptives':ab,ti OR 'hormonal contraceptives':ab,ti OR 'hormonal contraceptive agents':ab,ti OR 'hormonal oral contraceptive agents':ab,ti OR 'estrogen

contraceptive agents':ab,ti OR 'sequential oral contraceptives':ab,ti OR 'sequential oral contraceptive agents':ab,ti OR 'oral synthetic contraceptives':ab,ti OR 'oral hormonal contraceptives':ab,ti OR 'oral hormonal contraceptive agents':ab,ti OR 'oral combined contraceptives':ab,ti OR 'combined oral contraceptives':ab,ti OR 'combined oral contraceptive agents':ab,ti OR 'oral combined contraceptive agents':ab,ti OR 'hormone-releasing intrauterine devices':ab,ti OR 'hormone-releasing intrauterine device':ab,ti OR 'hormonal intrauterine device':ab,ti OR 'hormonal intrauterine devices':ab,ti OR 'IUD':ab,ti OR 'hormonal IUD':ab,ti OR 'medicated intrauterine device':ab,ti OR 'medicated intrauterine devices':ab,ti OR 'progesterone-releasing intrauterine device':ab,ti OR 'progesterone-releasing intrauterine devices':ab,ti OR 'progesterone-releasing vaginal ring':ab,ti OR 'contraceptive IUD':ab,ti OR 'contraceptive IUDs':ab,ti OR 'intrauterine contraceptive device':ab,ti OR 'intrauterine contraceptive devices':ab,ti OR 'copper-bearing intrauterine devices':ab,ti OR 'cyclofem':ab,ti OR 'depo-provera':ab,ti OR 'injectable contraceptive':ab,ti OR 'injectable contraceptives':ab,ti OR 'DMPA':ab,ti OR 'depot medroxyprogesterone acetate':ab,ti OR 'injectable progestogen-only contraceptives':ab,ti OR 'norethisterone oenanthate':ab,ti OR 'norethisterone enanthate':ab,ti OR 'contraceptive implant':ab,ti OR 'contraceptive implants':ab,ti OR 'subdermal contraceptive implant':ab,ti OR 'subdermal contraceptive implants':ab,ti OR 'single rod contraceptive implant':ab,ti OR 'femplant':ab,ti OR 'hormonal implant':ab,ti OR 'contraceptive patch':ab,ti OR 'transdermal contraceptive patch':ab,ti OR 'evra transdermal contraceptive patch':ab,ti OR 'ortho evra transdermal contraceptive patch':ab,ti OR 'contraceptive ring':ab,ti OR 'vaginal contraceptive ring':ab,ti OR 'vaginal contraceptive rings':ab,ti OR 'contraceptive rings':ab,ti OR 'vaginal rings':ab,ti OR 'vaginal ring':ab,ti OR 'levonorgestrel intrauterine device':ab,ti OR 'levonorgestrel-releasing intrauterine device':ab,ti OR 'levonorgestrel implant':ab,ti OR 'etonogestrel implant':ab,ti OR 'hormonal intrauterine device':ab,ti OR 'female contraceptive agents':ab,ti OR 'hormonal oral contraceptive agents':ab,ti OR 'hormonal contraception':ab,ti OR 'hormonal contraception':ab,ti OR 'hormonal contraceptives':ab,ti OR 'birth spacing':ab,ti OR 'family planning services':ab,ti OR 'family planning':ab,ti OR 'birth intervals':ab,ti OR 'birth interval':ab,ti OR 'birth spacings':ab,ti OR 'pregnancy intervals':ab,ti OR 'pregnancy interval':ab,ti OR 'emergency contraceptive':ab,ti OR 'emergency contraception':ab,ti OR 'barrier method':ab,ti OR 'fertility awareness':ab,ti OR 'lactational amenorrhoea method':ab,ti OR 'coitus interruptus':ab,ti OR 'female sterilization':ab,ti OR 'male sterilization':ab,ti OR 'abstinence':ab,ti OR 'fertility':ab,ti OR 'fecundity':ab,ti OR 'gravidity':ab,ti OR 'parity':ab,ti OR 'pregnancy':ab,ti

OR 'pregnant':ab,ti OR 'sayana press':ab,ti OR 'uniject':ab,ti OR 'contraceptive self-injection':ab,ti OR 'self-injectable contraception':ab,ti OR 'self-injected contraceptive':ab,ti OR 'calendar method':ab,ti OR 'calendar-based method':ab,ti OR 'calendar-based app':ab,ti OR 'menstrual cycle app':ab,ti OR 'menstrual tracking app':ab,ti OR 'fertility tracking app':ab,ti OR 'period tracker app':ab,ti OR 'ovulation prediction app':ab,ti OR 'calcuothermal app':ab,ti OR 'symptothermal app':ab,ti OR 'flo mobile app':ab,ti OR 'clue menstruations':ab,ti OR 'Period tracker deluxe':ab,ti OR 'womanlog':ab,ti OR 'ovv':ab,ti OR 'natural cycles':ab,ti OR 'myNFP':ab,ti OR 'lady cycle':ab,ti OR 'ovuvue':ab,ti OR 'contraceptive accessibility':ab,ti OR 'contraceptive discontinuation':ab,ti OR 'rapid repeat pregnancy':ab,ti OR 'coital based method':ab,ti OR 'LARC':ab,ti OR 'short-acting method':ab,ti))

Concept 2: Adolescents 10–19 years old

('adolescence':ab,ti OR 'adolescent':ab,ti OR 'adolescents':ab,ti OR 'teen':ab,ti OR 'teenager':ab,ti OR 'teenagers':ab,ti OR 'young adult':ab,ti OR 'young adults':ab,ti OR 'young male':ab,ti OR 'young males':ab,ti OR 'young female':ab,ti OR 'young females':ab,ti OR 'young couples':ab,ti OR 'young parents':ab,ti OR 'first time parents':ab,ti OR 'youth':ab,ti OR 'youths':ab,ti OR 'married girls':ab,ti OR 'unmarried girls':ab,ti OR 'child':ab,ti OR 'children':ab,ti OR 'school child':ab,ti OR 'schoolchild':ab,ti OR 'school children':ab,ti OR 'schoolchildren':ab,ti OR 'pupil':ab,ti OR 'pupils':ab,ti OR 'student':ab,ti OR 'students':ab,ti OR 'school girls':ab,ti OR 'schoolgirls':ab,ti OR 'school going girls':ab,ti OR 'high school girls':ab,ti OR 'high school boys':ab,ti OR 'high school students':ab,ti OR 'college students':ab,ti OR 'university students':ab,ti OR 'young migrants':ab,ti OR 'young factory workers':ab,ti)

Concept 3: LMICs

('developing country':ab,ti OR 'developing countries':ab,ti OR 'developing nation':ab,ti OR 'developing nations':ab,ti OR 'developing population':ab,ti OR 'developing populations':ab,ti OR 'developing world':ab,ti OR 'less developed country':ab,ti OR 'less developed countries':ab,ti OR 'less developed nation':ab,ti OR 'less developed nations':ab,ti OR 'less developed population':ab,ti OR 'less developed populations':ab,ti OR 'less developed world':ab,ti OR 'lesser developed country':ab,ti OR 'lesser developed countries':ab,ti OR 'lesser developed nation':ab,ti OR 'lesser developed nations':ab,ti OR 'lesser developed population':ab,ti OR 'lesser developed populations':ab,ti OR 'lesser developed world':ab,ti OR 'under developed country':ab,ti OR 'under developed countries':ab,ti OR 'under developed nation':ab,ti OR 'under

developed nations':ab,ti OR 'under developed population':ab,ti OR 'under developed populations':ab,ti OR 'under developed world':ab,ti OR 'underdeveloped country':ab,ti OR 'underdeveloped countries':ab,ti OR 'underdeveloped nation':ab,ti OR 'underdeveloped nations':ab,ti OR 'underdeveloped population':ab,ti OR 'underdeveloped populations':ab,ti OR 'underdeveloped world':ab,ti OR 'middle income country':ab,ti OR 'middle income countries':ab,ti OR 'middle income nation':ab,ti OR 'middle income nations':ab,ti OR 'middle income population':ab,ti OR 'middle income populations':ab,ti OR 'low income country':ab,ti OR 'low income countries':ab,ti OR 'low income nation':ab,ti OR 'low income nations':ab,ti OR 'low income population':ab,ti OR 'low income populations':ab,ti OR 'lower income country':ab,ti OR 'lower income countries':ab,ti OR 'lower income nation':ab,ti OR 'lower income nations':ab,ti OR 'lower income population':ab,ti OR 'lower income populations':ab,ti OR 'underserved country':ab,ti OR 'underserved countries':ab,ti OR 'underserved nation':ab,ti OR 'underserved nations':ab,ti OR 'underserved population':ab,ti OR 'underserved populations':ab,ti OR 'underserved world':ab,ti OR 'under served country':ab,ti OR 'under served countries':ab,ti OR 'under served nation':ab,ti OR 'under served nations':ab,ti OR 'under served population':ab,ti OR 'under served populations':ab,ti OR 'under served world':ab,ti OR 'deprived country':ab,ti OR 'deprived countries':ab,ti OR 'deprived nation':ab,ti OR 'deprived nations':ab,ti OR 'deprived population':ab,ti OR 'deprived populations':ab,ti OR 'deprived world':ab,ti OR 'poor country':ab,ti OR 'poor countries':ab,ti OR 'poor nation':ab,ti OR 'poor nations':ab,ti OR 'poor population':ab,ti OR 'poor populations':ab,ti OR 'poor world':ab,ti OR 'poorer country':ab,ti OR 'poorer countries':ab,ti OR 'poorer nation':ab,ti OR 'poorer nations':ab,ti OR 'poorer population':ab,ti OR 'poorer populations':ab,ti OR 'poorer world':ab,ti OR 'developing economy':ab,ti OR 'developing economies':ab,ti OR 'less developed economy':ab,ti OR 'less developed economies':ab,ti OR 'lesser developed economy':ab,ti OR 'lesser developed economies':ab,ti OR 'under developed economy':ab,ti OR 'under developed economies':ab,ti OR 'underdeveloped economy':ab,ti OR 'underdeveloped economies':ab,ti OR 'middle income economy':ab,ti OR 'middle income economies':ab,ti OR 'low income economy':ab,ti OR 'low income economies':ab,ti OR 'lower income economy':ab,ti OR 'lower income economies':ab,ti OR 'low gdp':ab,ti OR 'low gnp':ab,ti OR 'low gross domestic':ab,ti OR 'low gross national':ab,ti OR 'lower gdp':ab,ti OR 'lower gnp':ab,ti OR 'lower gross domestic':ab,ti OR 'lower gross national':ab,ti OR 'lami':ab,ti OR 'lamic':ab,ti OR 'lamic':ab,ti OR 'third world':ab,ti OR 'lami country':ab,ti OR 'lami countries':ab,ti OR 'transitional country':ab,ti OR 'transitional countries':ab,ti

OR africa:ti,ab OR asia:ti,ab OR caribbean:ti,ab OR 'west indies':ti,ab OR 'south america':ti,ab OR 'latin america':ti,ab OR 'central america':ti,ab OR 'atlantic islands':ab,ti OR 'commonwealth of independent states':ab,ti OR 'pacific islands':ab,ti OR 'indian ocean islands':ab,ti OR 'eastern europe':ab,ti OR afghanistan:ti,ab OR albania:ti,ab OR algeria:ti,ab OR 'american samoa':ti,ab OR angola:ti,ab OR antigua:ti,ab OR barbuda:ti,ab OR argentina:ti,ab OR armenia:ti,ab OR armenian:ti,ab OR aruba:ti,ab OR azerbaijan:ti,ab OR bahrain:ti,ab OR bangladesh:ti,ab OR barbados:ti,ab OR benin:ti,ab OR byelarus:ti,ab OR byelorussian:ti,ab OR belarus:ti,ab OR belorussian:ti,ab OR belorussia:ti,ab OR belize:ti,ab OR bhutan:ti,ab OR bolivia:ti,ab OR bosnia:ti,ab OR herzegovina:ti,ab OR hercegovina:ti,ab OR botswana:ti,ab OR brasil:ti,ab OR brazil:ti,ab OR bulgaria:ti,ab OR 'burkina faso':ti,ab OR 'burkina fasso':ti,ab OR 'upper volta':ti,ab OR burundi:ti,ab OR urundi:ti,ab OR cambodia:ti,ab OR 'khmer republic':ti,ab OR kampuchea:ti,ab OR cameroon:ti,ab OR cameroons:ti,ab OR cameron:ti,ab OR camérons:ti,ab OR 'cape verde':ti,ab OR 'cabo verde':ti,ab OR 'central african republic':ti,ab OR chad:ti,ab OR chile:ti,ab OR china:ti,ab OR colombia:ti,ab OR comoros:ti,ab OR 'comoro islands':ti,ab OR comores:ti,ab OR mayotte:ti,ab OR congo:ti,ab OR zaire:ti,ab OR 'costa rica':ti,ab OR 'cote d`ivoire' OR 'ivory coast':ti,ab OR croatia:ti,ab OR cuba:ti,ab OR cyprus:ti,ab OR czechoslovakia:ti,ab OR 'czech republic':ti,ab OR slovakia:ti,ab OR 'slovak republic':ti,ab OR djibouti:ti,ab OR 'french somaliland':ti,ab OR dominica:ti,ab OR 'dominican republic':ti,ab OR 'east timor':ti,ab OR 'east timur':ti,ab OR 'timor leste':ti,ab OR ecuador:ti,ab OR egypt:ti,ab OR 'united arab republic':ti,ab OR 'el salvador':ti,ab OR eritrea:ti,ab OR estonia:ti,ab OR eswatini:ti,ab OR ethiopia:ti,ab OR fiji:ti,ab OR gabon:ti,ab OR 'gabonese republic':ti,ab OR gambia:ti,ab OR gaza:ti,ab OR 'georgia republic':ti,ab OR 'georgian republic':ti,ab OR ghana:ti,ab OR 'gold coast':ti,ab OR grenada:ti,ab OR guatemala:ti,ab OR guinea:ti,ab OR guam:ti,ab OR guiana:ti,ab OR guyana:ti,ab OR haiti:ti,ab OR honduras:ti,ab OR india:ti,ab OR maldives:ti,ab OR indonesia:ti,ab OR iran:ti,ab OR iraq:ti,ab OR 'isle of man':ti,ab OR jamaica:ti,ab OR jordan:ti,ab OR kazakhstan:ti,ab OR kazakh:ti,ab OR kenya:ti,ab OR kiribati:ti,ab OR korea:ti,ab OR kosovo:ti,ab OR kyrgyzstan:ti,ab OR kirghizia:ti,ab OR 'kyrgyz republic':ti,ab OR kirghiz:ti,ab OR kirgizstan:ti,ab OR 'lao pdr':ti,ab OR laos:ti,ab OR latvia:ti,ab OR lebanon:ti,ab OR lesotho:ti,ab OR basutoland:ti,ab OR liberia:ti,ab OR libya:ti,ab OR lithuania:ti,ab OR macedonia:ti,ab OR madagascar:ti,ab OR 'malagasy republic':ti,ab OR malaysia:ti,ab OR malaya:ti,ab OR malay:ti,ab OR sabah:ti,ab OR sarawak:ti,ab OR malawi:ti,ab OR nysaland:ti,ab OR mali:ti,ab OR malta:ti,ab OR

'marshall islands':ti,ab OR mauritania:ti,ab OR mauritius:ti,ab OR melanesia:ab,ti OR 'agalega islands':ti,ab OR mexico:ti,ab OR micronesia:ti,ab OR 'middle east':ti,ab OR moldova:ti,ab OR moldovia:ti,ab OR moldovian:ti,ab OR mongolia:ti,ab OR morocco:ti,ab OR ifni:ti,ab OR mozambique:ti,ab OR myanmar:ti,ab OR myanma:ti,ab OR burma:ti,ab OR namibia:ti,ab OR nepal:ti,ab OR 'netherlands antilles':ti,ab OR 'new caledonia':ti,ab OR nicaragua:ti,ab OR niger:ti,ab OR nigeria:ti,ab OR 'northern mariana islands':ti,ab OR oman:ti,ab OR muscat:ti,ab OR pakistan:ti,ab OR palau:ti,ab OR palestine:ti,ab OR panama:ti,ab OR paraguay:ti,ab OR peru:ti,ab OR philippines:ti,ab OR philipines:ti,ab OR phillipines:ti,ab OR phillippines:ti,ab OR poland:ti,ab OR portugal:ti,ab OR 'puerto rico':ti,ab OR romania:ti,ab OR rumania:ti,ab OR roumania:ti,ab OR russia:ti,ab OR russian:ti,ab OR rwanda:ti,ab OR ruanda:ti,ab OR 'saint kitts':ti,ab OR 'st kitts':ti,ab OR nevis:ti,ab OR 'saint lucia':ti,ab OR 'st lucia':ti,ab OR 'saint vincent':ti,ab OR 'st vincent':ti,ab OR grenadines:ti,ab OR samoa:ti,ab OR 'samoan islands':ti,ab OR 'navigator island':ti,ab OR 'navigator islands':ti,ab OR 'sao tome':ti,ab OR 'saudi arabia':ti,ab OR senegal:ti,ab OR serbia:ti,ab OR montenegro:ti,ab OR seychelles:ti,ab OR 'sierra leone':ti,ab OR slovenia:ti,ab OR 'sri lanka':ti,ab OR ceylon:ti,ab OR 'solomon islands':ti,ab OR somalia:ti,ab OR sudan:ti,ab OR suriname:ti,ab OR surinam:ti,ab OR swaziland:ti,ab OR syria:ti,ab OR syrian:ti,ab OR tajikistan:ti,ab OR tadjhikistan:ti,ab OR tadjikistan:ti,ab OR tadjhik:ti,ab OR tanzania:ti,ab OR thailand:ti,ab OR togo:ti,ab OR 'togolese republic':ti,ab OR tonga:ti,ab OR trinidad:ti,ab OR tobago:ti,ab OR tunisia:ti,ab OR turkey:ti,ab OR turkmenistan:ti,ab OR turkmen:ti,ab OR tuvalu:ti,ab OR uganda:ti,ab OR ukraine:ti,ab OR uruguay:ti,ab OR ussr:ti,ab OR 'soviet union':ti,ab OR 'union of soviet socialist republics':ti,ab OR uzbekistan:ti,ab OR uzbek OR vanuatu:ti,ab OR 'new hebrides':ti,ab OR venezuela:ti,ab OR vietnam:ti,ab OR 'viet nam':ti,ab OR 'west bank':ti,ab OR yemen:ti,ab OR yugoslavia:ti,ab OR zambia:ti,ab OR zimbabwe:ti,ab OR rhodesia:ti,ab OR 'developing country'/exp OR 'africa'/de OR 'africa south of the sahara'/de OR 'north africa'/de OR 'central africa'/de OR 'asia'/de OR 'south asia'/de OR 'southeast asia'/de OR 'south america'/de OR 'central america'/de OR 'south and central america'/de OR 'atlantic islands'/de OR 'caribbean islands'/de OR 'pacific islands'/de OR 'indian ocean'/de OR 'eastern europe'/de OR 'afghanistan'/exp OR 'albania'/exp OR 'algeria'/exp OR 'american samoa'/exp OR 'angola'/exp OR 'antigua and barbuda'/exp OR 'argentina'/exp OR 'armenia'/exp OR 'azerbaijan'/exp OR 'bahrain'/exp OR 'bangladesh'/exp OR 'barbados'/exp OR 'benin'/exp OR 'belarus'/exp OR 'baltic states'/exp OR 'belize'/exp OR 'bhutan'/exp OR 'bolivia'/exp OR 'bosnia and herzegovina'/exp

OR 'botswana'/exp OR 'brazil'/exp OR 'bulgaria'/exp OR 'burkina faso'/exp OR 'burundi'/exp OR 'cambodia'/exp OR 'cameroon'/exp OR 'cape verde'/exp OR 'central African republic'/exp OR 'chad'/exp OR 'chile'/exp OR 'china'/exp OR 'colombia'/exp OR 'comoros'/exp OR 'congo'/exp OR 'costa rica'/exp OR 'cote d`ivoire'/exp OR 'croatia'/exp OR 'cuba'/exp OR 'cyprus'/exp OR 'czechoslovakia'/exp OR 'czech republic'/exp OR 'slovakia'/exp OR 'djibouti'/exp OR 'democratic republic congo'/exp OR 'dominica'/exp OR 'dominican republic'/exp OR 'timor-leste'/exp OR 'ecuador'/exp OR 'egypt'/exp OR 'el salvador'/exp OR 'eritrea'/exp OR 'estonia'/exp OR 'eswatini'/exp OR 'ethiopia'/exp OR 'french guiana'/exp OR 'fiji'/exp OR 'gabon'/exp OR 'gambia'/exp OR 'georgia (republic)'/exp OR 'ghana'/exp OR 'grenada'/exp OR 'guatemala'/exp OR 'guinea'/exp OR 'guinea bissau'/exp OR 'guam'/exp OR 'guyana'/exp OR 'haiti'/exp OR 'honduras'/exp OR 'india'/exp OR 'indonesia'/exp OR 'iran'/exp OR 'iraq'/exp OR 'jamaica'/exp OR 'jordan'/exp OR 'kazakhstan'/exp OR 'kenya'/exp OR 'korea'/exp OR 'kyrgyzstan'/exp OR 'laos'/exp OR 'latvia'/exp OR 'lebanon'/exp OR 'lesotho'/exp OR 'liberia'/exp OR 'libyan arab jamahiriya'/exp OR 'lithuania'/exp OR 'macedonia (republic)'/exp OR 'madagascar'/exp OR 'malaysia'/exp OR 'malawi'/exp OR 'mali'/exp OR 'malta'/exp OR 'mauritania'/exp OR 'mauritius'/exp OR 'melanesia'/exp OR 'mexico'/exp OR 'federated states of micronesia'/exp OR 'middle east'/de OR 'moldova'/exp OR 'mongolia'/exp OR 'montenegro'/exp OR 'morocco'/exp OR 'mozambique'/exp OR 'myanmar'/exp OR 'namibia'/exp OR 'nepal'/exp OR 'netherlands antilles'/exp OR 'new caledonia'/exp OR 'nicaragua'/exp OR 'niger'/exp OR 'nigeria'/exp OR 'north korea'/exp OR 'oman'/exp OR 'pakistan'/exp OR 'palau'/exp OR 'panama'/exp OR 'papua new guinea'/exp OR 'paraguay'/exp OR 'peru'/exp OR 'philippines'/exp OR 'poland'/exp OR 'portugal'/exp OR 'puerto rico'/exp OR 'romania'/exp OR 'russian federation'/exp OR 'rwanda'/exp OR 'saint kitts and nevis'/exp OR 'saint lucia'/exp OR 'saint vincent and the grenadines'/exp OR 'samoan islands'/exp OR 'samoa'/exp OR 'saudi arabia'/exp OR 'senegal'/exp OR 'serbia'/exp OR 'montenegro (republic)'/exp OR 'seychelles'/exp OR 'sierra leone'/exp OR 'slovenia'/exp OR 'sri lanka'/exp OR 'somalia'/exp OR 'south korea'/exp OR 'south africa'/exp OR 'sudan'/exp OR 'suriname'/exp OR 'swaziland'/exp OR 'syrian arab republic'/exp OR 'tajikistan'/exp OR 'tanzania'/exp OR 'thailand'/exp OR 'togo'/exp OR 'tonga'/exp OR 'trinidad and tobago'/exp OR 'tunisia'/exp OR 'turkey (republic)'/exp OR 'turkmenistan'/exp OR 'uganda'/exp OR 'ukraine'/exp OR 'uruguay'/exp OR 'ussr'/exp OR 'uzbekistan'/exp OR 'vanuatu'/exp OR 'venezuela'/exp OR 'viet nam'/exp OR 'yemen'/exp OR 'yugoslavia'/exp OR 'yugoslavia (pre-1992)'/exp OR 'zambia'/exp OR 'zimbabwe'/exp

Concept 4: Quantitative comparative study design

('controlled study'/exp OR 'comparative effectiveness'/exp OR 'experimental study' OR 'feasibility study' OR 'field study' OR 'observational study')

Database: CINAHL

*Check off: Peer-reviewed, Publication date Jan 2010–any, Human, Research Article, Age groups: Child (6–12), Adolescent (13–18), Adult (19–44)

Concept 1: Contraception

((MW "Contraception") OR (MW "Hormonal Contraception") OR (MW "Contraceptives, Postcoital") OR (MW "Contraceptives, Oral Combined") OR (MW "Contraceptive Agents") OR (MW "Diaphragms, Contraceptive") OR (MW "Birth Intervals") OR (MW "Family Planning") OR (MW "Pregnancy") OR (MW "Sexual abstinence") OR (MW "fertility") OR (MW "parity") OR (MW "intrauterine devices") OR (MW "condoms") OR (MW "female condoms")) OR ((AB "contraception") OR (AB "contraceptive") OR (AB "female contraceptive devices") OR (AB "female contraceptive device") OR (AB "vaginal ring") OR (AB "vaginal rings") OR (AB "female contraceptive agents") OR (AB "female contraceptives") OR (AB "female contraceptive agents") OR (AB "oral contraceptives") OR (AB "hormonal contraceptives") OR (AB "hormonal contraceptive agents") OR (AB "hormonal oral contraceptive agents") OR (AB "estrogen contraceptive agents") OR (AB "sequential oral contraceptives") OR (AB "sequential oral contraceptive agents") OR (AB "oral synthetic contraceptives") OR (AB "oral hormonal contraceptives") OR (AB "oral hormonal contraceptive agents") OR (AB "oral combined contraceptives") OR (AB "combined oral contraceptives") OR (AB "combined oral contraceptive agents") OR (AB "oral combined contraceptive agents") OR (AB "hormone-releasing intrauterine devices") OR (AB "hormone-releasing intrauterine device") OR (AB "hormonal intrauterine device") OR (AB "hormonal intrauterine devices") OR (AB "IUD") OR (AB "hormonal IUD") OR (AB "medicated intrauterine device") OR (AB "medicated intrauterine devices") OR (AB "progesterone-releasing intrauterine device") OR (AB "progesterone-releasing intrauterine devices") OR (AB "progesterone-releasing vaginal ring") OR (AB "contraceptive IUD") OR (AB "contraceptive IUDs") OR (AB "intrauterine contraceptive device") OR (AB "intrauterine contraceptive devices") OR (AB "copper- bearing intrauterine devices") OR (AB "cyclofem") OR (AB "depo-provera") OR (AB "injectable contraceptive") OR (AB "injectable contraceptives") OR (AB "DMPA") OR (AB "depot medroxyprogesterone acetate") OR (AB "injectable progestogen-only contraceptives") OR (AB "norethisterone

oenanthate") OR (AB "norethisterone enanthate") OR (AB "contraceptive implant")
 OR (AB "contraceptive implants") OR (AB "subdermal contraceptive implant") OR
 (AB "subdermal contraceptive implants") OR (AB "single rod contraceptive implant")
 OR (AB "femplant") OR (AB "hormonal implant") OR (AB "contraceptive patch") OR
 (AB "transdermal contraceptive patch") OR (AB "evra transdermal contraceptive
 patch") OR (AB "ortho evra transdermal contraceptive patch") OR (AB "contraceptive
 ring") OR (AB "vaginal contraceptive ring") OR (AB "vaginal contraceptive rings")
 OR (AB "contraceptive rings") OR (AB "vaginal rings") OR (AB "vaginal ring") OR (AB
 "levonorgestrel intrauterine device") OR (AB "levonorgestrel-releasing intrauterine
 device") OR (AB "levonorgestrel implant") OR (AB "etonogestrel implant") OR (AB
 "hormonal intrauterine device") OR (AB "female contraceptive agents") OR (AB
 "hormonal oral contraceptive agents") OR (AB "hormonal contraception") OR (AB
 "hormonal contraception") OR (AB "hormonal contraceptives") OR (AB "birth spacing")
 OR (AB "family planning services") OR (AB "family planning") OR (AB "birth intervals")
 OR (AB "birth interval") OR (AB "birth spacings") OR (AB "pregnancy intervals") OR
 (AB "pregnancy interval") OR (AB "emergency contraceptive") OR (AB "emergency
 contraception") OR (AB "barrier method") OR (AB "fertility awareness") OR (AB
 "lactational amenorrhoea method") OR (AB "coitus interruptus") OR (AB "female
 sterilization") OR (AB "male sterilization") OR (AB "abstinence") OR (AB "fertility")
 OR (AB "fecundity") OR (AB "gravidity") OR (AB "parity") OR (AB "pregnancy") OR
 (AB "pregnant") OR (AB "sayana press") OR (AB "uniject") OR (AB "contraceptive
 self-injection") OR (AB "self-injectable contraception") OR (AB "self- injected
 contraceptive") OR (AB "calendar method") OR (AB "calendar-based method") OR (AB
 "calendar-based app") OR (AB "menstrual cycle app") OR (AB "menstrual tracking
 app"[tiab] OR fertility tracking app") OR (AB "period tracker app") OR (AB "ovulation
 prediction app") OR (AB "calcuothermal app") OR (AB "symptothermal app") OR (AB
 "flo mobile app") OR (AB "clue menstruations") OR (AB "Period tracker deluxe") OR (AB
 "womanlog") OR (AB "ovy") OR (AB "natural cycles") OR (AB "myNFP") OR (AB "lady
 cycle") OR (AB "ovuvue") OR (AB "contraceptive accessibility") OR (AB "contraceptive
 discontinuation") OR (AB "rapid repeat pregnancy") OR (AB "coital based method") OR
 (AB "LARC") OR (AB "short-acting method"))

Concept 2: Adolescents 10–19 years old

((MW "adolescence") OR (MW "young adult")) OR ((AB "adolescence") OR (AB
 "adolescent") OR (AB "adolescents") OR (AB "teen") OR (AB "teenager") OR (AB
 "teenagers") OR (AB "young adult") OR (AB "young adults") OR (AB "young male") OR

(AB "young males") OR (AB "young female") OR (AB "young females") OR (AB "young
 couples") OR (AB "young parents") OR (AB "first time parents") OR (AB "youth") OR
 (AB "youths") OR (AB "married girls") OR (AB "unmarried girls") OR (AB "child") OR
 (AB "children") OR (AB "school child") OR (AB "schoolchild") OR (AB "school children")
 OR (AB "schoolchildren") OR (AB "pupil") OR (AB "pupils") OR (AB "student") OR (AB
 "students") OR (AB "school girls") OR (AB "schoolgirls") OR (AB "school going girls")
 OR (AB "high school girls") OR (AB "high school boys") OR (AB "high school students")
 OR (AB "college students") OR (AB "university students") OR (AB "young migrants") OR
 (AB "young factory workers"))

Concept 3: LMICs

("Antigua and Barbuda" OR "Atlantic Islands" OR "Baltic States" OR "Commonwealth
 of Independent States" OR "Democratic People's Republic of Korea" OR "Democratic
 Republic of the Congo" OR "deprived countries" OR "deprived population" OR
 "deprived populations" OR "developing countries" OR "developing country" OR
 "developing economies" OR "developing economy" OR "developing nation" OR
 "developing nations" OR "developing population" OR "developing populations"
 OR "developing world" OR "Equatorial Guinea" OR "French Guiana" OR "Georgia
 Republic" OR "Independent State of Samoa" OR "Indian Ocean Islands" OR "lami
 countries" OR "lami country" OR "less developed countries" OR "less developed
 country" OR "less developed economies" OR "less developed economy" OR "less
 developed nation" OR "less developed nations" OR "less developed world" OR "lesser
 developed countries" OR "lesser developed nations" OR "low gdp" OR "low gnp" OR
 "low gross domestic" OR "low gross national" OR "low income countries" OR "low
 income country" OR "low income economies" OR "low income economy" OR "low
 income nations" OR "low income population" OR "low income populations" OR "lower
 gdp" OR "lower gross domestic" OR "lower income countries" OR "lower income
 country" OR "lower income nations" OR "lower income population" OR "lower income
 populations" OR "Macedonia Republic" OR "Melanesia" OR "middle income countries"
 OR "middle income country" OR "middle income economies" OR "middle income
 nation" OR "middle income nations" OR "middle income population" OR "middle
 income populations" OR "Pacific Islands" OR "poor countries" OR "poor country" OR
 "poor nation" OR "poor nations" OR "poor population" OR "poor populations" OR
 "poor world" OR "poorer countries" OR "poorer nations" OR "poorer population" OR
 "poorer populations" OR "Republic of Belarus" OR "Saint Kitts and Nevis" OR "Saint
 Vincent and the Grenadine" OR "South Sudan" OR "third world" OR "transitional

countries" OR "transitional country" OR "Trinidad and Tobago" OR "under developed countries" OR "under developed country" OR "under developed nations" OR "under developed world" OR "under served population" OR "under served populations" OR "underdeveloped countries" OR "underdeveloped country" OR "underdeveloped economies" OR "underdeveloped nations" OR "underdeveloped population" OR "underdeveloped world" OR "underserved countries" OR "underserved nations" OR "underserved population" OR "underserved populations" OR "Afghanistan" OR "Africa" OR "Albania" OR "Algeria" OR "American Samoa" OR "Angola" OR "Argentina" OR "Armenia" OR "Asia" OR "Azerbaijan" OR "Bahrain" OR "Bangladesh" OR "Barbados" OR "Belize" OR "Benin" OR "Bhutan" OR "Bolivia" OR "Bosnia-Herzegovina" OR "Botswana" OR "Brazil" OR "Bulgaria" OR "Burkina Faso" OR "Burundi" OR "Cambodia" OR "Cameroon" OR "Cape Verde" OR "Caribbean Region" OR "Central African Republic" OR "Central America" OR "Chad" OR "Chile" OR "China" OR "Colombia" OR "Comoros" OR "Congo" OR "Costa Rica" OR "Cote d'Ivoire" OR "Croatia" OR "Cuba" OR "Cyprus" OR "Czech Republic" OR "Czechoslovakia" OR "Developing Countries" OR "Djibouti" OR "Dominica" OR "Dominican Republic" OR "East Timor" OR "Ecuador" OR "Egypt" OR "El Salvador" OR "Eritrea" OR "Estonia" OR "Eswatini" OR "Ethiopia" OR "Fiji" OR "Gabon" OR "Gambia" OR "Ghana" OR "Grenada" OR "Guam" OR "Guatemala" OR "Guinea" OR "Guinea-Bissau" OR "Guyana" OR "Haiti" OR "Honduras" OR "India" OR "Indonesia" OR "Iran" OR "Iraq" OR "Jamaica" OR "Jordan" OR "Kazakhstan" OR "Kenya" OR "Korea" OR "Kyrgyzstan" OR "Laos" OR "Latin America" OR "Latvia" OR "Lebanon" OR "Lesotho" OR "Liberia" OR "Libya" OR "Lithuania" OR "Lmic" OR "Imics" OR "Madagascar" OR "Malawi" OR "Malaysia" OR "Mali" OR "Malta" OR "Mauritania" OR "Mauritius" OR "Mexico" OR "Micronesia" OR "Middle East" OR "Moldova" OR "Mongolia" OR "Montenegro" OR "Montenegro" OR "Morocco" OR "Mozambique" OR "Myanmar" OR "Namibia" OR "Nepal" OR "Netherlands Antilles" OR "New Caledonia" OR "Nicaragua" OR "Niger" OR "Nigeria" OR "Oman" OR "Pakistan" OR "Palau" OR "Panama" OR "Papua New Guinea" OR "Paraguay" OR "Peru" OR "Philippines" OR "Poland" OR "Portugal" OR "Puerto Rico" OR "Romania" OR "Russia" OR "Rwanda" OR "Saint Lucia" OR "Samoa" OR "Saudi Arabia" OR "Senegal" OR "Serbia" OR "Seychelles" OR "Sierra Leone" OR "Slovakia" OR "Slovenia" OR "Somalia" OR "South Africa" OR "South America" OR "Sri Lanka" OR "Sudan" OR "Suriname" OR "Swaziland" OR "Syria" OR "Tajikistan" OR "Tanzania" OR "Thailand" OR "Togo" OR "Tonga" OR "Tunisia" OR "Turkey" OR "Turkmenistan" OR "Uganda" OR "Ukraine" OR "Uruguay" OR "USSR" OR "Uzbekistan" OR "Vanuatu" OR "Venezuela" OR "Vietnam" OR "West Indies" OR "Yemen" OR "Yugoslavia" OR "Zambia" OR "Zimbabwe")

Concept 4: Quantitative comparative study design

((MH "Experimental studies+") OR (MM "Crossover Design") OR (MM "Empirical Research") OR (MM "Nonexperimental Studies+") OR (MM "Quasi-Experimental Studies+") OR (MM "Retrospective Design"))

References

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